The Symmetry-Quotient Attack on Odd-Characteristic SNOVA

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Abstract

We identify a deterministic loss of quadratic dimension in the proposed odd-characteristic ($q = 19$) redesign of SNOVA Version 2.3. If $A = \mathbb{F}_q[S]$ has dimension d, symmetry identifies the verifier's power-pair features labeled by (ξ, η) and (η, ξ). Hence the d^2 ordered features per public base form factor through $\mathrm{Sym}^2(A)$ and have dimension at most $\binom{d+1}{2}$: 16 becomes 10 for $\ell = 4$, and 4 becomes 3 for $\ell = 2$. This identity holds for every symmetric public key, before the public outer map. We turn it into exact public-key forgery reductions for all nine published $q = 19$ parameter sets, retain every verifier output, and defeat the proposed symmetric-signature rejection rule by fixed-skew or exact accepted-density arguments.

The root analysis no longer assumes a sampled global minimum rank. A reserved public vinegar line selects the relation, offsets, and attack slices from data disjoint from fresh internal public coefficients. In the random-XOF idealization of the official expansion, tensor injectivity and the exact modulo-19 maximum atom 14/256 bound the complete aggregate collision spectrum. The same spectrum also bounds Jacobian singularity: if $\bar{\eta}_L = \sum_{y \neq 0} 19^{-\operatorname{rank} T_y}$ and the verifier accepts density α, then a random target–coset trial contains an accepted root at which the full restricted Jacobian has rank $h = \dim L$ with probability at least

$$19^{h-K} \frac{(\alpha - \bar{\eta}_L/18)^2}{1 + \bar{\eta}_L}.$$

No uniqueness assumption and no independence between rejection, singularity, and the MQ equations is used.

This yields a selected-degree-free, fixed-core-free attack branch. After the four-eigenblock change, the 55 near-square family-count patterns form only 16 orbits under 19-Frobenius. Solving one representative per orbit and then checking every omitted residual equation, descent condition, rejection condition, reconstruction equation, and the original verifier gives robust per-certified-key AXN exponents 138.324, 179.758, and 220.832. A separate exact random-XOF structural-preflight theorem proves simultaneous success probability greater than 0.9423222679 for all six $\ell = 4$ shapes. After charging its inverse probability, the random-key-normalized exponents are 138.410, 179.844, and 220.918, leaving 4.590, 27.156, and 51.082 bits below the numerical references. This branch uses neither a selected Macaulay degree nor a preferred core-Jacobian premise. Its direct parametrizations can be enormous, so it is an operation-count theorem rather than a low-memory claim. An independent original-equation eigenblock-core route gives much smaller output ceilings

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and robust per-certified-key exponents 134.077, 196.246, and 247.432, conditioned only on an exact public nonzero-Jacobian preflight.

The complementary streamed-XL branch supplies the low-state frontier: state-minimizing four-block profiles cost $2^{141.70}$, $2^{203.63}$, and $2^{269.35}$ AXN operations with packed eight-array envelopes 17.7 GiB, 2.21 GiB, and 1.01 GiB, while an exact three-plus-one decomposition gives a robust Level-V point $2^{263.40}$ with 9.55 GiB. Selected-degree behavior remains an empirical input only for this low-state branch. Independently of every solver estimate, merely restoring the quadratic-image cap intended by the old ordered-label analysis while retaining symmetry and Version 2.3's parameter coupling requires 31.25%–60.71% more verifier outputs; and eliminating the new full-dimensional fresh-slice route by dimensions alone requires 38.89%–52.00% more outputs on the six $\ell = 4$ rows.

1 Introduction

The cryptanalytic task. A multivariate signature scheme publishes a polynomial map of total degree at most two,

$$\mathcal{P} : \mathbb{F}_q^N \longrightarrow \mathbb{F}_q^M.$$

For a message with hash-derived target $t \in \mathbb{F}_q^M$, the algebraic core of a forgery is to find x satisfying $\mathcal{P}(x) = t$; any additional format or rejection checks must also pass. This is an MQ problem: M equations of degree at most two in the coordinates of x. Linear and constant terms are allowed; throughout the paper, "quadratic" refers to the maximum degree. The apparent equation and variable counts are only a first approximation to security. Public algebraic structure can make different-looking quadratic features identical, can expose linear equations, or can split the remaining variables into blocks that a specialized solver can exploit.

SNOVA is one of the nine digital-signature candidates that NIST advanced to the third round of its additional post-quantum signature process in May 2026 [20, 1]. The 2026 wedge attack breaks the updated Round-2 construction in characteristic two [4]. NIST describes the odd-characteristic system studied here as a proposed response: it moves to $q = 19$ and replaces the asymmetric bilinear public forms by symmetric quadratic forms. NIST also states that SNOVA has not reached a stable form and that more substantial changes may be considered than for more mature candidates [1]. The object of this paper is therefore both concrete and time-sensitive: the public $q = 19$ Version 2.3 construction, not the broken characteristic-two submission and not every future third-round revision.

SNOVA stores the signature variables in small matrices and builds its public quadratics from pairs of powers of a public matrix S. Beullens showed that an attacker can impose an affine relation among the signature columns and thereby reduce forgery to a smaller underdetermined MQ system [3]. We find an additional, deterministic reduction inside the quadratic features themselves.

The hidden duplication. Let P_i be a public symmetric base matrix, let $A = \mathbb{F}_q[S]$, and write $\Sigma_\xi = I_n \otimes \xi$ for $\xi \in A$. The feature with left label ξ and right label η is

$$q_{i,\xi,\eta}(u) = u^\mathsf{T} \Sigma_\xi P_i \Sigma_\eta u.$$

Because the matrices are symmetric and the result is a scalar,

$$q_{i,\xi,\eta}(u) = q_{i,\eta,\xi}(u). \tag{1}$$

The verifier begins from ordered labels, but the polynomial depends only on the corresponding unordered pair. We call the operation of identifying these duplicate labels the *symmetry quotient*.

For a two-dimensional toy algebra with basis $\{1, S\}$, the four labels

$$(1,1), \qquad (1,S), \qquad (S,1), \qquad (S,S)$$

produce only three polynomials because the middle two are equal. More generally, if $\dim_{\mathbb{F}_q} A = d$, then at most

$$K = m_1 \binom{d+1}{2} \tag{2}$$

independent power-pair features remain across the m_1 base forms, rather than $m_1 d^2$. Thus each $\ell = 4$ base form drops from 16 ordered labels to 10 unordered labels; each $\ell = 2$ base form drops from 4 to 3. The loss occurs before the public outer linear map—the final public mixing of these features into verifier outputs—so later randomization cannot recreate the discarded directions. It holds for every symmetric public key and every attacker-chosen column relation.

Core idea. The verifier assigns two names to the same quadratic polynomial whenever it reverses a pair of power labels. The attack first removes those duplicate features, then uses ordinary linear algebra to retain all remaining verifier equations, and finally solves only the smaller quadratic core.

What is exact, and what is modeled. The paper deliberately separates algebraic statements from concrete-cost inputs. The distinction is summarized here and used consistently below.

The attack as a data-flow. The proof is easiest to follow as a sequence of exact transformations. The word *lift* below means the attacker-chosen affine rule that turns one column vector into a full signature candidate.

Full public verifier	$\longrightarrow$	impose one affine common-column relation; quadratic, linear, and constant terms become explicit
	$\downarrow$	
Symmetry quotient	$\longrightarrow$	identify (ξ, η) with (η, ξ) and keep only the distinct quadratic features
	$\downarrow$	
Exact affine elimination	$\longrightarrow$	use a left-kernel projection to expose every output direction unreachable by the quadratics as an ordinary linear constraint
	$\downarrow$	
Structured residual MQ	$\longrightarrow$	choose either a stable A-linear slice or a full-dimensional ordinary slice for $\ell = 4$, with exact rejection-density control, or a rejection-compatible relation for $\ell = 2$
	$\downarrow$	
Complementary solvers	$\longrightarrow$	stream the multigraded Macaulay operator; solve a Frobenius-channel core; or sweep every square multidegree-family projection needed to cover a full-Jacobian root
	$\downarrow$	
Candidate signature	$\longrightarrow$	map back to the original coordinates and run the unmodified public verifier

Table 1: Claim ledger. A public certificate can be checked from the public key and a solver transcript; "model input" means that the displayed exponent assumes the cited estimator continues to describe the production instance.

Statement	Status	What a verifier checks or assumes
Symmetry quotient	Unconditional theorem	Symmetry of S and the public base matrices; no random-key hypothesis.
Frobenius-channel normal form	Unconditional theorem and exact certificate	$\mathrm{Sym}^2_{\mathbb{F}_q}(\mathbb{F}_{q^4}) \cong \mathbb{F}_{q^4} \oplus \mathbb{F}_{q^4} \oplus \mathbb{F}_{q^2}$; the released 10×10 matrix has rank ten over $\mathbb{F}_{19}$.
Ordered-cap restoration floor	Unconditional necessary condition	To restore the ordered-label quadratic-image cap while retaining symmetry, a design needs both enough outputs and enough symmetric-square features. Under the current coupling this forces a 31.25%–60.71% output increase across the nine rows, before any security margin.
Affine reduction and rejection handling	Exact public certificate/theorem	An invertible target-coordinate split, matrix ranks, fixed-skew identities, and an accepted-density singleton bound for the square $\ell = 4$ rows.
Root and retry bounds	Exact theorem; seeded-model theorem or fixed-key certificate	A reserved-line construction leaves a fresh coefficient block and bounds the bad aggregate spectrum in the random-XOF idealization. For a fixed key, a finite projective rank histogram certifies the same sum exactly.
Special-XL extraction	Exact correctness once certified	Macaulay row combinations, a complete border prebasis, commuting multiplication matrices, and the original verifier. Quotient dimension $D > 1$ is sound but may require additional block linear algebra.
Implicit Macaulay state	Exact operator; standard black-box solver	Rows and transpose products are generated on demand. Random normal preconditioning uses $O(M)$ or $O(bM)$ field-vector state. Both the compact $\mathbb{F}_{19^8}$ and Level-I robust $\mathbb{F}_{19^{12}}$ scalar realizations are priced from exact row counts.
Displayed special-XL exponent	Model input with explicit sensitivity	The compact rows retain the published $D = 1$ prediction. The robust rows instead charge a finite-quotient certificate probability only $\sigma \geq 1/8$; any larger block/kernel work still has to fit the displayed ledger.
Channel-homotopy branch	Theorem-backed nonsingular-root recovery; explicit public preflight and sensitivity factor	A nonzero leading-Jacobian witness, the finite-cardinality homotopy theorem, and exact filtering remove the selected-degree assumption. The displayed leading exponent leaves κ_{hom} explicit and is not a production timing or a low-memory claim.
Basis-complete/square-slice homotopy	Theorem-backed recovery without a fixed core or selected degree	The aggregate spectrum bounds full-Jacobian singularity. Every possible family-count pattern is swept at Levels I/V; the complete square residual system is used at Level III. All soft-O and implementation overhead remains the explicit multiplier κ_{hom}.
Generic-MQ fallback	Model input	The same exact quotient system, but the pinned generic semi-regularity and linear-algebra estimator.
Circuit-unit conversion	Sensitivity calculation	Only the field arithmetic charged by the estimator; not a complete attack circuit or category claim.

Every transformation through the residual MQ system is exact and uses only public algebra and linear algebra. The solver model enters only when we estimate the cost of solving that residual system. The final public-verifier check makes the attack Las Vegas: a trial may be discarded, but an invalid signature is never returned.

Four jobs that a complete attack must perform. The equality (1) is only the first job. The construction also has to preserve all public outputs, ensure that targets have roots, and make the residual equations compatible with a practical solver:

1. **Constrain the signature columns.** Every signature column is written as an affine function of one common vector. Substitution separates the verifier into quotient quadratic features, offset-linear terms, and constants.
2. **Eliminate the affine part without dropping equations.** Output directions outside the quadratic image become linear constraints. For each target satisfying those affine consistency conditions, Gaussian elimination leaves a smaller MQ system with exactly the same solutions as the constrained verifier.
3. **Choose a solver-friendly search space.** For $\ell = 4$, a structured choice of offsets leaves a large A-linear kernel and an injective local skew map makes rejection cost less than 0.0014 bits. Over a splitting field this kernel becomes four variable blocks, preserving the multidegrees used by special XL. The same slice also has an exact Frobenius-channel form with two A-valued channels of degrees two and twenty. For $\ell = 2$, fixed-skew and filtered relations handle the verifier's rejection rule directly.
4. **Retry, solve, and verify.** The low-state branch uses streamed multigraded linear algebra. Two independent branches use symbolic homotopy: one on a Frobenius-channel core, and one on a complete family of square projections that is guaranteed to contain a nonsingular core whenever the full restricted Jacobian has rank h. Exact moment bounds control accepted nonsingular roots directly, without requiring a unique root. Every recovered candidate is transformed back and checked by the original verifier.

A complete dimension walk-through. The Level-I parameter set $(v, o, \ell, r) = (28, 5, 4, 4)$ illustrates where the reduction comes from: it has 28 vinegar blocks, 5 oil blocks, and blocks with 4 rows and 4 columns. Here $n = v + o = 33$, one stacked signature column has $N_0 = n\ell = 132$ coordinates, the verifier has $M = or\ell = 80$ scalar outputs, and there are $m_1 = 5$ public base-form groups.

Table 2: The dimensions in the $(28, 5, 4, 4)$ attack. This table is only a worked example; the general formulas appear in later sections.

Stage	Dimension	Meaning
Raw public feature vector	$N_{\text{raw}} = 1{,}280$	$m_1\ell^2r^2$ labeled slots before the column relation
After the common-column relation	80 ordered quadratics	one feature for each of $m_1\ell^2 = 5 \cdot 16$ ordered power pairs
After the symmetry quotient	$K = 50$ distinct quadratics	$m_1\binom{\ell+1}{2} = 5 \cdot 10$ unordered power pairs
Affine output directions	$M - K = 30$	linear equations that determine an affine search space
Full residual domain	102 coordinates	$N_0 - (M - K) = 132 - 30$ after affine elimination
Stable kernel	52 base-field dimensions	an A-linear space of A-dimension 13 on which offset-linear motion vanishes
Selected special-XL slice	$h = 40$	four blocks of $s = 10$ variables; hence $\tau = K - h = 10$

The final slice deliberately has fewer variables than equations. Full cross-polar rank is not needed. The exact load-bearing condition is the aggregate collision mass $\bar{\eta}_L \leq 19^{-1}$. The simple minimum-rank certificate $\operatorname{rank} T_y \geq h + 1 = 41$ for every nonzero slice direction implies that bound. Either certificate makes a random target–slice trial contain a unique base-field root with probability at least $19^{-10}(1 - 19^{-1})$, so the expected retry count is at most $19^{10} \cdot 19/18$. The factor 19/18 costs only 0.0781 bits. Paying these retries is worthwhile because reducing the Macaulay matrix from the full residual domain to four blocks of ten variables saves much more. This is why the paper analyzes both the algebraic reduction and the root-retry probability, rather than quoting an MQ dimension alone.

A fixed-core-free selected-degree-free frontier. The aggregate spectrum also bounds the density at which the full restricted Jacobian loses rank. At a descended root, 19-Frobenius cyclically permutes the four eigenblocks and carries every nonzero Jacobian minor to a nonzero conjugate minor. The 55 near-square family-count patterns therefore collapse to 16 Frobenius orbits, and one representative per orbit suffices. This removes both a production selected-degree premise and any preferred square-core Jacobian premise.

Table 3: Fixed-core-free, selected-degree-free homotopy frontier. "Per key" charges the robust aggregate-spectrum and acceptance bounds. "Normalized" also charges the inverse of the exact random-XOF structural-preflight lower bound 0.9423222679. Headroom is measured against 143/207/272.

Level	(h, K)	patterns $\to$ orbits	per key	normalized	headroom
I	$(48, 50)$	$55 \to 16$	138.324	138.410	4.590
III	$(68, 70)$	$55 \to 16$	179.758	179.844	27.156
V	$(88, 90)$	$55 \to 16$	220.832	220.918	51.082

The artifact enumerates all 55 patterns, computes their 16 exact cyclic orbits, and sums the exact multihomogeneous Bézout and homotopy ledgers for one representative from each orbit. Every unused equation and every final verification check is charged. The straightforward largest one-core output ceilings are about 1.890 PiB, 2.076 ZiB, and 2416.552 YiB; hence this route is included to remove logical assumptions, not as a resident-memory claim. The separately conditioned eigenblock-core and streamed-XL branches provide lower-output and lower-state alternatives.

Conditional concrete estimates. Table 4 reports the base-2 expected-cost exponents produced by the *pinned* solver models. "Same-quotient generic" applies the pinned generic-MQ estimator to exactly the same symmetry-quotient residual system; "gain" is therefore a like-for-like solver-structure comparison, not a comparison with the submission's different gate methodology. Every repeated solve includes its explicit target or target–slice retry factor.

Table 4: Primary conditional estimates for the public Version 2.3 parameter sets. The tuple is (v, o, ℓ, r). All numerical columns are base-2 expected cost exponents or differences of such exponents in the cited solver model. The exact reductions determine the systems; the rank-spectrum and production solver conditions are listed immediately below.

Level	Parameters	Attack mode	Model	same-quotient generic	Gain
I	$(28, 5, 4, 4)$	structured XL	128.90	136.11	7.21
I	$(48, 16, 2, 2)$	fixed-skew MQ	129.51	129.51	0.00
I	$(28, 4, 4, 5)$	structured XL	128.90	136.11	7.21
III	$(40, 7, 4, 4)$	structured XL	171.76	180.66	8.90
III	$(72, 24, 2, 2)$	fixed-skew MQ	183.51	183.51	0.00
III	$(38, 5, 4, 5)$	structured XL	171.76	180.66	8.90
V	$(50, 9, 4, 4)$	structured XL	214.20	224.75	10.55
V	$(96, 32, 2, 2)$	filtered MQ	235.27	237.92	2.65
V	$(52, 6, 4, 6)$	structured XL	214.20	224.75	10.55

The six $\ell = 4$ rows remain below the numerical values 143, 207, and 272 even after special XL is discarded and the exact quotient systems are fed to the pinned generic-MQ model. That observation is useful robustness, but it is still a model comparison: NIST evaluates categories across all security metrics it regards as relevant, not by one arithmetic exponent alone [21].

Space is not hidden inside those exponents. The arithmetic-minimizing four-block special-XL points have one-vector sizes 65.4 GiB, 118 PiB, and 614 EiB; the last two are not physically plausible attack states. The exact frontier in Section 7.2 supplies lower-state alternatives, and Proposition 39 prices the operator used to realize them. The state-minimizing compact four-block profiles cost $2^{141.70}$, $2^{203.63}$, and $2^{269.35}$ AXN operations with packed eight-array envelopes 17.7 GiB, 2.21 GiB, and 1.01 GiB.

The exact three-plus-one decomposition of Section 7.3 removes the fourth block from the nonlinear solve. Its time-minimizing Level-III and Level-V rows cost $2^{197.02}$ and $2^{250.32}$ with packed states 61.1 GiB and 118.2 GiB. Under the joint stress $\bar{\eta}_L \leq 1/2$ and $\sigma \geq 1/8$, those rows cost $2^{200.94}$ and $2^{254.24}$; a low-state Level-V row costs only $2^{263.40}$ with a 9.55 GiB envelope.

The new Frobenius-channel branch is independent of the selected-degree prediction. The exact normal form $\mathrm{Sym}^2_{\mathbb{F}_{19}}(\mathbb{F}_{19^4}) \cong \mathbb{F}_{19^4} \oplus \mathbb{F}_{19^4} \oplus \mathbb{F}_{19^2}$ turns each base-form group into channels of ordinary degrees 2, 20, and 362. Solving a square mix of the first two channels and filtering all remaining equations gives balanced unit-normalized leading AXN exponents 135.00, 199.06, and 263.03; with $\bar{\eta}_L \leq 1/2$ they are 135.51, 199.57, and 263.55. The robust break-even budgets for the explicit soft-O/implementation multiplier are 7.49, 7.43, and 8.45 bits. This branch is theorem-backed for nonsingular roots once a cheap public leading-Jacobian preflight passes, but it is not a low-memory theorem and its dense straightforward representation can be large.

Taking the best joint-stress row from either XL decomposition gives $2^{141.29}$, $2^{200.49}$, and $2^{263.40}$

at Levels I, III, and V. The corresponding packed states are 1.53 TiB, 55.4 GiB, and 9.55 GiB. Level I enlarges only the random normal-preconditioning field to $\mathbb{F}_{19^{12}}$; the residual equations and charged multiply–accumulates stay over $\mathbb{F}_{19^4}$. These are proved operator-and-accounting statements, not production timings.

Conditions behind the table. For the structured rows, the random-XOF theorem of Theorem 27 gives $\bar{\eta}_L \leq 19^{-1}$ except with probability below $2^{-55.22}$, $2^{-123.19}$, and $2^{-191.16}$ for the compact profiles, provided the reserved-line slice construction is used. This is an information-theoretic theorem in the random-XOF idealization of the official expansion, not an independence theorem for the deterministic bytes produced by a fixed SHAKE seed. Instantiating the XOF by SHAKE is the usual computational modeling step. A fixed key can instead publish the exact projective rank histogram. The convenient pointwise certificates $\operatorname{rank} T_y \geq h+1$ remain stronger sufficient conditions, not load-bearing assumptions.

At the selected special-XL degree, the compact scalar-Wiedemann rows retain the published one-dimensional-kernel prediction. The robust explicit rows charge only $\sigma \geq 1/8$ in the Las Vegas wrapper and allow $\bar{\eta}_L \leq 1/2$. A certified finite border quotient of larger dimension still gives zero-error extraction by Theorem 37, but its additional block/kernel work is not free and must fit the stated work and state envelope. For the three-plus-one rows, σ jointly covers a complete finite quotient for the three-block subsystem, its extraction and enumeration within the stated work–state reserve, and certified linear completion of the fourth block. A quotient or completion that exceeds the reserve, or a consistent rank-deficient completion, causes a restart unless handled by a separately priced fallback.

The channel-homotopy rows do not use this selected-degree assumption. They instead require one exact nonzero leading-Jacobian witness for the chosen public slice and core. Their arithmetic theorem has a soft-O factor, so the paper reports the normalized leading term together with the full break-even budget κ_{hom} and a dense coefficient-volume ceiling; it does not promote that term to a production timing.

The basis-complete and complete-square rows also avoid the selected-degree assumption, and they do not require a fixed-core Jacobian witness. Their root success follows from the same aggregate spectrum used for collision control. Their displayed costs sum every family-count core (or use the complete square system), charge candidate filtering, and leave the soft-O/implementation factor κ_{hom} explicit. Their principal caveat is output volume, not logical root completeness.

The fixed-skew rows assume $\Theta_{\text{pol}} \leq 19^{-1}$, and the filtered Level-V row assumes $\Psi_{\text{pol}} \leq 19^{-1}$; each costs less than 0.075 bits. Minimum polar ranks 49, 73, and 194 are simple sufficient certificates for the three headline rows, not the load-bearing conditions. The stronger thresholds 96, 144, and 240 make every target solvable. The released finite sampling supports these conditions but does not certify the complete spectra. Likewise, the small-profile Macaulay experiments do not certify the selected production degree.

The later AXN and NAND conversions change only the arithmetic unit charged by the estimators. They are retained as sensitivity calculations, not as the paper's primary evidence and not as a complete gate-level attack comparison.

How to read the paper. Section 2 fixes the background and notation. The exact attack is in Sections 3 to 5. Root existence and retry probabilities are separated in Section 6; the solver and gate models are isolated in Section 7, including the independent Frobenius-channel route of Section 7.4; and Section 8 states exactly which assumptions are tested rather than proved. All long proofs and exhaustive cost-search certificates are deferred to the appendices.

The submitted $q = 16$ matrices are not symmetrized and therefore do not satisfy the hypothesis of our quotient theorem. Returning unchanged to that design, however, would restore the known characteristic-two attack surface discussed in Section 2.4. Conversely, a future odd-characteristic revision that removes public-matrix symmetry, changes the left/right actions, or substantially changes the affine geometry is outside the theorem until its new public forms are analyzed. We make no claim that an unpublished future revision inherits the Version 2.3 attack unchanged.

2 Mathematical Background and Relation to Prior Work

This section fixes the notation and background used throughout: underdetermined MQ, polar forms, Macaulay matrices, and multigraded XL.

2.1 MQ systems, roots, and affine slices

Write $\mathbb{F}_q$ for the finite field with q elements. An MQ system over $\mathbb{F}_q$ is a vector-valued polynomial map of total degree at most two,

$$g = (g_1, \ldots, g_K) : \mathbb{F}_q^N \longrightarrow \mathbb{F}_q^K,$$

together with a target $z \in \mathbb{F}_q^K$. We continue to call g quadratic when some coordinates also contain linear or constant terms. A *root* is an x with $g(x) = z$. The set

$$g^{-1}(z) = \{x : g(x) = z\}$$

is the *fiber* above z: it is the set of all candidate assignments that produce the target. After the affine reconstruction described later, those assignments become full signature candidates. All arithmetic is over the stated finite field; every concrete SNOVA calculation in this paper uses $q = 19$, so additions and multiplications are performed modulo 19.

As a toy example, two equations such as

$$x_1x_2 + x_3^2 = z_1, \qquad x_1^2 + x_2x_3 + x_1 = z_2$$

form an MQ system in three variables. Having more variables than equations makes the system *underdetermined*, but does not make it linear or necessarily easy. Modern MQ attacks often spend most of their time on very large linear-algebra matrices built from the quadratic equations.

An *affine space* is a translate $a + V = \{a + v : v \in V\}$ of a linear subspace. An *affine slice* $a + L \subseteq a + V$ restricts the search to a smaller direction space L. If $h = \dim L$, choosing a basis of L turns the slice into an MQ system in h coordinates. Smaller h makes the algebraic solve cheaper, but it also lowers the chance that the slice contains a root; Section 6 quantifies this tradeoff.

Three logically distinct events recur throughout the paper:

(i) the chosen target and slice contain at least one $\mathbb{F}_q$-rational root;

(ii) the slice contains exactly one such root; and

(iii) the selected Macaulay degree exposes enough ideal relations to build a finite quotient algebra and extract its roots.

The first two are properties of the quadratic map. The third is a statement about a specific solver representation. A proof of root existence is not, by itself, a proof that one solver invocation reaches a complete enough degree. Conversely, the quotient algebra need not be one-dimensional: once its multiplication tables are available, standard zero-dimensional solving can enumerate its roots and the public verifier can select a valid one.

2.2 Rank, polar forms, and collisions

For a scalar quadratic polynomial f, define its polar form by

$$\beta_f(x, y) = f(x + y) - f(x) - f(y) + f(0).$$

In odd characteristic, β_f is bilinear. It is the finite-field analogue of the part of a directional derivative that comes from the quadratic terms. Its matrix rank measures how many independent directions the quadratic part couples.

For a vector-valued map $g = (g_1, \ldots, g_K)$, every nonzero coefficient vector $b \in \mathbb{F}_q^K$ selects a scalar combination $b \cdot g$. The minimum polar rank must be taken over *all* such output directions. Multiplying b by a nonzero scalar does not change the rank, so these vectors can be grouped into projective directions: one-dimensional lines through the origin.

Rank matters because finite-field Fourier (character-sum) identities imply that a scalar quadratic with high polar rank is well balanced. Applied to all nonzero combinations $b \cdot g$, this bounds every target fiber rigorously: when the ranks are high enough relative to K, each fiber has size close to the random-map prediction q^{N-K}. A related *cross-polar rank* measures how strongly a displacement inside a slice interacts with the full residual domain; high cross-polar rank suppresses pairs of distinct slice points that map to the same target. In summary:

Polar rank	controls whether full-space target fibers are nonempty and close to their expected size.
Cross-polar rank	controls collisions inside an affine slice and therefore unique-root retry probabilities.
Macaulay quotient	commuting border tables certify that row reduction exposes a finite quotient algebra; its dimension controls the cost of root extraction.

High rank is not invoked as a vague "randomness" heuristic in the structural proofs: Section 6 gives exact first- and second-moment formulas. The production issue is to certify the relevant weighted rank spectrum. A lower bound in every nonzero direction is one stronger, convenient route, but the exact moment formulas can tolerate a small exceptional locus. Public sampling gives evidence; an exact spectrum bound or a proof is still needed.

2.3 XL and Macaulay matrices

XL ("eXtended Linearization") turns a nonlinear system into a larger linear system. Starting from quadratic equations $f_i = 0$, it multiplies each f_i by selected monomials. For example,

$$f = x_1 x_2 + x_3^2 + x_1 + 1 \quad \Longrightarrow \quad x_1 f = x_1^2 x_2 + x_1 x_3^2 + x_1^2 + x_1.$$

Each resulting polynomial becomes a row. Each monomial—for example $x_1^2 x_2$ or $x_1 x_3^2$—becomes a column. The coefficient matrix is a *Macaulay matrix*. Gaussian elimination on this matrix can reveal enough relations among the monomials to recover a root. Treating monomials as columns is only a linear-algebra device: the monomials are not independent variables in the original problem, and the solver must eventually recover coordinates that satisfy all of their multiplicative relations.

The cost comes from the size and rank of this matrix, not from evaluating the original equations once. Generic XL tracks only total degree. *Special XL* exploits a variable-block structure. A monomial then has a vector of degrees, one entry per block, called its *multidegree*. Building only selected multidegrees can make the Macaulay matrix much smaller.

Our $\ell = 4$ reduction produces four blocks after a change to eigenblock coordinates. A quadratic using one block twice has degree pattern $(2, 0, 0, 0)$ up to permutation; a bilinear equation using two blocks has pattern $(1, 1, 0, 0)$. The stable-lift construction in Section 5 exists precisely to preserve these patterns after affine elimination. A *homogenizer* is one extra variable per block that allows the constant and linear terms to be written with the same multidegree; it is set to one when the root is read out.

If x is a root, the vector of all selected monomial values

$$\nu(x) = (1, x_1, \ldots, x_N, x_1^2, x_1x_2, \ldots)$$

lies in the right kernel of the dehomogenized Macaulay matrix. A one-dimensional right kernel is the easiest case: normalizing the constant coordinate reveals the root immediately. It is not the only recoverable case. If row reduction instead expresses every border monomial x_jm in terms of a finite monomial basis $\mathcal{B}$, those relations give the multiplication matrices of the quotient algebra. The FGLM algorithm converts these tables to a lexicographic Gröbner basis and extracts every algebraic root [8]. Thus a commuting-border certificate gives publicly checkable correctness beyond exact Macaulay corank one. The scalar-Wiedemann exponent reported later still prices the published one-dimensional-kernel case; a larger quotient requires separately charged block/nullspace work.

2.4 Relation to prior attacks

Beullens rewrote the SNOVA verifier in ordered bilinear power-pair coordinates and derived affine-column forgery attacks from rank defects in the resulting outer matrix [3]. Our quotient acts one layer earlier: public-matrix symmetry identifies the power-pair polynomials *before* the outer map sees them.

The broad danger of odd-characteristic symmetrization was not unknown. In a 2025 security presentation, the SNOVA team based its odd-q discussion on the $o\binom{\ell}{2}$ alternating-form count, observed that a wedge-style variant might apply, and said that further cryptanalysis was needed [27]. NIST subsequently described the symmetric odd-characteristic construction as a proposed response rather than a stable third-round design [1]. Version 2.3 also includes a public block-symmetry rejection check [25]. We therefore do not claim novelty for the qualitative warning that symmetrization can create an attack surface or for the rough alternating-form count.

Table 5: Boundary with the closest prior analyses. The distinction is by public construction and cryptanalytic deliverable, not merely terminology.

Work	Setting	What is established
SNOVA team, 2025 [27]	Proposed odd-q response	Qualitative warning that a wedge-style variant may work, the $o\binom{\ell}{2}$ alternating-form sizing criterion, and rejection of symmetric signature blocks; no instantiated direct forgery or concrete cost.
Bros et al., 2026 [4]	Submitted construction over $q = 2^k$	Characteristic-two wedge attack breaking the updated Round-2 parameters; not an analysis of the $q = 19$ symmetric redesign.
This work	Version 2.3 over $q = 19$	Exact $\mathrm{Sym}^2(A)$ affine-column factorization, complete public-key forgery reductions, retry accounting, space-aware costs, and constructive bypass of the proposed rejection rule.

The novelty is the missing cryptanalytic instantiation: an exact symmetric-square factorization inside the affine-column attack; complete residual systems for every public $q = 19$ shape; exact root and retry accounting; an A-linear lift preserving the special-XL grading; and constructive fixed-skew and filtered bypasses of Version 2.3's block-symmetry check. The presentation identified a possible attack surface but gave neither this forgery reduction nor concrete costs. This paper supplies both and separately proves that the deployed rejection condition does not remove the reduction.

Cabarcas et al. exploit stability under the public matrix algebra with a multigraded XL algorithm [5]. Furue et al. expose related multihomogeneous systems through matrix transformations [9]. Our attack starts from the complete affine forgery system, uses a public column lift to retain every output equation, and then restricts to an invariant kernel on which the affine offsets preserve the block grading needed by special XL.

The 2026 wedge attack is already a full break of the updated Round-2 parameters, but its algebra is explicitly characteristic two: it works over $q = 2^k$ and exploits the alternating polar form of the block-ring system [4]. Later work studies that wedge map further [26]. Our target is the proposed odd-characteristic response. Its public matrices are symmetric rather than alternating, and our attack factors the public affine-column feature map through $\mathrm{Sym}^2(A)$; it is not an application of the characteristic-two wedge kernel. Other analyses reinterpret scalar-expanded SNOVA as Unbalanced Oil and Vinegar (UOV), or lift it to extension-field UOV, for key recovery [11, 16, 18]. Symmetric and exterior algebras also appear in recent UOV key-recovery attacks [12, 22]. Our attack is a direct forgery attack: it does not recover the hidden oil subspace that constitutes the UOV-style secret key.

For the generic $\ell = 2$ cores, we reproduce Hashimoto's optimization, including its published boundary value $a = 1$ [10, 3]. When this leaves the elementary system $\mathrm{MQ}(q, 1, 1)$, the artifact solves it by exhaustive evaluation and charges that work conservatively. May, Ostuzzi, and Ressler's *Just Guess* algorithm and the generalized variable partitioning of Asanuma et al. are more recent approaches to underdetermined MQ [17, 2]. Evaluating the released classical Just-Guess formulas on our residual dimensions gives larger estimates than the selected Beullens–Hashimoto modes. We have not implemented the newer generalized partition search. The "generic MQ" numbers below are therefore pinned, reproducible baselines, not a claim to have exhausted every generic solver.

Solver engineering is also moving quickly. Sakata and Takagi use Hilbert-series information to reduce the matrices built by an F_4 Gröbner-basis solver and report a new MQ-challenge record [24]. That is an implementation and challenge result, not a plug-in cost formula for the underdetermined residual systems here, so we do not substitute it into Table 4.

2.5 Notation at a glance

Table 6: Recurring notation. Generic background notation such as N is local; N_0 is the specific common-column dimension in the SNOVA reduction.

Symbol	Meaning
(v, o, ℓ, r)	Public parameter tuple: v vinegar blocks, o oil blocks, block height ℓ, and r columns per signature block.
$n = v + o$	Number of matrix-valued signature blocks.
$M = or\ell$	Number of scalar public output equations.
N	Generic domain dimension for an MQ map $g : \mathbb{F}_q^N \to \mathbb{F}_q^K$; its meaning is local to the section in which the map is defined.
$N_0 = n\ell$	Length of one stacked signature column after the common-column relation.
$N_{\mathrm{raw}} = m_1\ell^2r^2$	Number of labeled verifier feature slots before imposing the relation.
$A = \mathbb{F}_q[S]$	Public commutative matrix algebra generated by S; in the Version 2.3 rows, $\dim_{\mathbb{F}_q} A = \ell$.
m_1	Number of public base-form groups in the pinned implementation.
$K = m_1\binom{\ell+1}{2}$	Maximum number of distinct quadratic power-pair features after the symmetry quotient.
Q_R	Matrix mapping the K quotient features to public output directions.
G_R	A public left inverse of Q_R when Q_R has full column rank.
$C_{R,\mathbf{v}}$	Linear constraint matrix obtained by projecting away the quadratic image.
$g_{R,\mathbf{v}}$	Canonical K-coordinate residual map after the exact output split.
$H_{R,\mathbf{v}}$	Stable A-linear kernel used to choose structured solver slices.
L, h	Slice direction space and its base-field dimension.
$\tau = K - h$	Slice deficit. It enters the exact full-rank retry formula; for the selected $\tau = 10, 14$ slices, the retry count is essentially q^τ.
r_*	Minimum full-space polar rank over all nonzero scalar output combinations.
$e_L(b)$	Cross-polar rank of output direction b relative to slice L.
μ, $\bar{\eta}$	Expected roots on a random target–slice pair and an upper bound on its collision mass.

3 The Affine-Column Framework

This section rewrites the public verifier after an attacker-imposed affine relation among signature columns. The result has three visibly separate parts: quotient quadratic features, linear terms created by the offsets, and a constant term. Proposition 2 then eliminates the affine part exactly.

Three vector spaces appear and should not be confused. The raw public feature vector has length N_{raw}; the verifier outputs M scalar equations; and the common column variable u has length N_0. The matrices below simply move between these three spaces.

3.1 Parameters and public forms

The public parameter tuple is (v, o, ℓ, r). There are v vinegar blocks and o oil blocks, for a total of

$$n = v + o$$

matrix-valued signature blocks. The names "vinegar" and "oil" come from SNOVA's UOV ancestry; in this public forgery attack they serve only to specify dimensions, and the attacker never learns or reconstructs the secret partition. Each block has ℓ rows and r columns. Stacking the jth column of all n blocks gives a vector of length $N_0 = n\ell$; the verifier has $M = or\ell$ scalar outputs. The pinned Version 2.3 implementation defines

$$m_1 = \left\lceil \frac{or}{\ell} \right\rceil, \qquad M = or\ell, \qquad N_0 = n\ell. \tag{3}$$

Here m_1 is the number of public base-form groups used to assemble the outputs. The ceiling is used by the reference implementation; a public analysis script uses a floor and therefore disagrees when $\ell \nmid or$. All dimensions in this paper follow the pinned implementation [25].

Let $S \in \mathrm{Mat}_\ell(\mathbb{F}_q)$ be the public symmetric matrix with irreducible characteristic polynomial. Its powers generate the commutative matrix algebra

$$A = \mathbb{F}_q[S], \qquad \Sigma_\xi = I_n \otimes \xi \quad (\xi \in A).$$

For the Version 2.3 parameters, $A \cong \mathbb{F}_{q^\ell}$ and $\dim_{\mathbb{F}_q} A = \ell$. Multiplication by Σ_ξ applies the same $\ell \times \ell$ matrix ξ independently to every signature block.

The scalar-expanded public base matrices are $P_1, \ldots, P_{m_1} \in \mathrm{Mat}_{N_0}(\mathbb{F}_q)$. In the public $q = 19$ construction, each P_i is symmetric. For $\xi, \eta \in A$, define the bilinear form

$$B_i^{\xi,\eta}(x, y) = x^\top \Sigma_\xi P_i \Sigma_\eta y. \tag{4}$$

Setting $x = y$ makes this a quadratic feature. The power basis $1, S, \ldots, S^{\ell-1}$ gives the ordered power-pair coordinates used in the prior affine-column attack.

3.2 Imposing one common column variable

Write the stacked signature columns as

$$U = [u_0 | \cdots | u_{r-1}], \qquad u_j \in \mathbb{F}_q^{N_0}.$$

The attacker chooses multipliers $R_j \in A$ and offsets $v_j \in \mathbb{F}_q^{N_0}$, with $R_0 = 1$. Write $R = (R_0, \ldots, R_{r-1})$ and $\mathbf{v} = (v_0, \ldots, v_{r-1})$, and impose

$$u_j = \Sigma_{R_j} u + v_j, \qquad 0 \le j < r. \tag{5}$$

Thus all r columns are affine functions of one vector u. This is not an assumption about honest signatures; it defines a publicly checkable subset in which the attacker searches for a forgery.

Before the relation is imposed, the feature vector has

$$N_{\mathrm{raw}} = m_1 \ell^2 r^2$$

coordinates: m_1 base forms, ℓ^2 ordered power pairs, and r^2 ordered column pairs. The official verifier flattens each $r \times \ell$ block in (i, j) order, whereas our feature formulas use the block-transposed (j, i) order. Let $t_{\mathrm{off}} \in \mathbb{F}_q^M$ and $\mathcal{E}_{\mathrm{off}} \in \mathbb{F}_q^{M \times N_{\mathrm{raw}}}$ denote the target and public outer map in the official order. If Π_{out} is the corresponding public permutation, define

$$t = \Pi_{\mathrm{out}} t_{\mathrm{off}}, \qquad \mathcal{E} = \Pi_{\mathrm{out}} \mathcal{E}_{\mathrm{off}}.$$

This is only a coordinate reordering; it changes neither ranks nor solutions.

After substituting (5), products of two variable parts give quadratic terms in u, products of one variable part and one offset give linear terms, and products of two offsets give constants. For a fixed relation $R = (R_j)_{j=0}^{r-1}$, let $\mathcal{R}_R \in \mathbb{F}_q^{N_{\mathrm{raw}} \times m_1 \ell^2}$ expand the remaining ordered quadratic features into the full feature vector, and let $\mathcal{F}_{R,\mathbf{v}} \in \mathbb{F}_q^{N_{\mathrm{raw}} \times N_0}$ collect the offset-linear coefficients. The complete substituted verifier is

$$\mathcal{E}(\mathcal{R}_R z_{\mathrm{ord}}(u) + \mathcal{F}_{R,\mathbf{v}} u) + c_{R,\mathbf{v}} = t, \tag{6}$$

where $z_{\mathrm{ord}}(u)$ contains the $m_1 \ell^2$ values $B_i^{S^a, S^b}(u, u)$ and $c_{R,\mathbf{v}}$ is the public constant term. Put

$$E_R = \mathcal{E}\mathcal{R}_R \in \mathbb{F}_q^{M \times m_1 \ell^2}, \qquad \Lambda_{R,\mathbf{v}} = \mathcal{E}\mathcal{F}_{R,\mathbf{v}} \in \mathbb{F}_q^{M \times N_0}. \tag{7}$$

3.3 Separating quadratic and affine output directions

By Theorem 3, the ordered feature vector duplicates every off-diagonal power pair. Let $z_{\mathrm{sym}}(u)$ contain one coordinate for each unordered pair $0 \le a \le b < \ell$, and let D copy each off-diagonal coordinate into its two ordered positions. Then $z_{\mathrm{ord}}(u) = D z_{\mathrm{sym}}(u)$. Define

$$Q_R = E_R D \in \mathbb{F}_q^{M \times K}, \qquad K = m_1 \binom{\ell + 1}{2}. \tag{8}$$

The columns of Q_R span all output directions that the residual quadratic features can reach.

Now choose W_R whose rows form a basis of the left kernel of Q_R and put

$$C_{R,\mathbf{v}} = W_R \Lambda_{R,\mathbf{v}}. \tag{9}$$

Multiplying by W_R kills every quadratic feature. What remains is therefore an ordinary affine system in u. This elementary projection is the key to retaining the *complete* verifier rather than dropping output equations that do not lie in the quadratic image.

Q_R	sends the K distinct quadratic features to the M public output coordinates; its column space is the quadratic image.
W_R	keeps precisely the output directions orthogonal to that image, so $W_R Q_R = 0$.
$C_{R,\mathbf{v}} = W_R \Lambda_{R,\mathbf{v}}$	records the ordinary linear equations that remain in those output directions after the offsets are introduced.

Lemma 1 (Canonical residual coordinates)**.** *Assume* $\operatorname{rank} Q_R = K$*. Choose a public left inverse* $G_R \in \mathbb{F}_q^{K \times M}$ *satisfying* $G_R Q_R = I_K$*. Then*

$$\begin{bmatrix} G_R \\ W_R \end{bmatrix} \in \mathbb{F}_q^{M \times M} \tag{10}$$

is invertible. Writing $b = t - c_{R,\mathbf{v}}$*, the complete substituted verifier is equivalent to the pair*

$$C_{R,\mathbf{v}} u = W_R b, \qquad g_{R,\mathbf{v}}(u) = G_R b, \qquad g_{R,\mathbf{v}}(u) := z_{\mathrm{sym}}(u) + G_R \Lambda_{R,\mathbf{v}} u. \tag{11}$$

In particular, the polar map of $g_{R,\mathbf{v}}$ *is exactly the polar map of* z_{sym}*; offsets and affine elimination do not change it. If* b *is uniform in* $\mathbb{F}_q^M$*, then the two right-hand sides in* (11) *are independent and uniform in* $\mathbb{F}_q^{M-K}$ *and* $\mathbb{F}_q^K$*, respectively.*

Proof. If a vector $v \in \mathbb{F}_q^M$ is killed by both G_R and W_R, then $W_R v = 0$ implies $v = Q_R a$ for some $a \in \mathbb{F}_q^K$. Applying G_R gives $a = G_R Q_R a = 0$, so $v = 0$ and the square matrix in (10) is invertible. Applying its two block rows to $Q_R z_{\mathrm{sym}}(u) + \Lambda_{R,\mathbf{v}} u = b$ gives (11); conversely, invertibility recovers the original equation. Affine maps have zero polar. The final statement follows because an invertible linear change sends a uniform vector to a uniform vector on the product space. $\square$

A two-output example. Suppose that, after the column substitution, two verifier outputs over $\mathbb{F}_{19}$ are

$$y_1 = q(u) + \ell_1(u), \qquad y_2 = 2q(u) + \ell_2(u),$$

where q is quadratic and ℓ_1, ℓ_2 are affine. The combination $2y_1 - y_2 = 2\ell_1(u) - \ell_2(u)$ cancels the quadratic term. Solving this affine equation and substituting it back leaves one quadratic equation; neither original output was dropped. The rows of W_R perform the same cancellation simultaneously for the complete verifier.

Proposition 2 (Exact affine reduction)**.** *Let $\rho = \operatorname{rank} Q_R$, $\lambda = \operatorname{rank} C_{R,\mathbf{v}}$, and $\delta = M - \rho - \lambda$. Then*

$$\operatorname{rank}[Q_R \ \Lambda_{R,\mathbf{v}}] = \rho + \lambda,$$

where $[Q_R \ \Lambda_{R,\mathbf{v}}]$ denotes horizontal matrix concatenation. After row reduction, exactly δ independent affine consistency conditions remain on the target. For every target satisfying those conditions, Gaussian elimination leaves ρ equations of degree at most two in $N_0 - \lambda$ variables, with exactly the same solution set as (6).

In particular, if $\rho = K$ and $\lambda = M - K$, every target passes the affine consistency test*, and the residual system has K equations of degree at most two on an affine translate of* $\ker C_{R,\mathbf{v}}$*. This conclusion does not assert that the residual system has a root.*

Proof. The row space of W_R is $\ker Q_R^{\mathsf{T}}$. Since W_R has full row rank,

$$\ker W_R = \left(\ker Q_R^{\mathsf{T}}\right)^{\perp} = \operatorname{im} Q_R.$$

Thus W_R realizes the quotient map $\mathbb{F}_q^M \to \mathbb{F}_q^M / \operatorname{im} Q_R$. The image of the columns of $\Lambda_{R,\mathbf{v}}$ in this quotient has dimension $\operatorname{rank}(W_R \Lambda_{R,\mathbf{v}}) = \lambda$, which proves $\operatorname{rank}[Q_R \ \Lambda_{R,\mathbf{v}}] = \rho + \lambda$.

Write $b = t - c_{R,\mathbf{v}}$. Projecting the substituted verifier by W_R gives

$$C_{R,\mathbf{v}} u = W_R b.$$

The codomain of $C_{R,\mathbf{v}}$ has dimension $M - \rho$, while its image has dimension λ. Solvability therefore imposes exactly $(M - \rho) - \lambda = \delta$ independent affine conditions on the target. For a target satisfying the affine consistency conditions, parameterize the solution space of these affine equations by $N_0 - \lambda$ free variables and substitute into a complementary set of ρ rows. This leaves ρ equations of degree at most two and uses only invertible row operations and exact affine elimination, so the solution set is unchanged. If $\rho = K$ and $\lambda = M - K$, then $\delta = 0$ and $C_{R,\mathbf{v}}$ is surjective onto its $M - K$-dimensional codomain; hence every target passes the affine consistency test. □

Operational meaning. The attacker computes Q_R, G_R, W_R, $C_{R,\mathbf{v}}$, and the canonical residual map $g_{R,\mathbf{v}}$ from the public key before running an MQ solver. If the ranks are unfavorable, the relation or offsets are changed. When $\operatorname{rank} Q_R = K$ and $\operatorname{rank} C_{R,\mathbf{v}} = M - K$, every hash target determines a nonempty affine space $a(t) + \ker C_{R,\mathbf{v}}$ on which the canonical map $g_{R,\mathbf{v}}$ supplies exactly K equations of degree at most two. This is an exact reformulation that discards no verifier equation; whether the residual equations actually have a root is analyzed in Section 6. Conversely, every root of the residual system maps back through the column relation to a solution of the complete substituted verifier.

4 The Symmetry Quotient

The central reduction is the formal version of (1). Readers may interpret $\mathrm{Sym}^2(A)$ simply as the vector space spanned by *unordered* pairs from A; the tensor notation records why the identification is basis-independent.

For a fixed base form P_i, define

$$q_{i,a,b}(u) = u^\top \Sigma_{S^a} P_i \Sigma_{S^b} u.$$

Transposition gives $q_{i,a,b} = q_{i,b,a}$. In coordinates, the feature vector therefore contains two copies of every off-diagonal polynomial. The theorem below shows that this is not an accident of the power basis: every alternating label $\xi \otimes \eta - \eta \otimes \xi$ vanishes. Write $\mathbb{F}_q[u]_2^{\mathrm{hom}}$ for the vector space of homogeneous quadratic polynomials in the coordinates of u.

Theorem 3 (Symmetric-square factorization)**.** *Let $A = \mathbb{F}_q[S]$ have dimension d, and assume $S^\top = S$ and $P_i^\top = P_i$. For each i, the map*

$$A \otimes_{\mathbb{F}_q} A \longrightarrow \mathbb{F}_q[u]_2^{\mathrm{hom}}, \qquad \xi \otimes \eta \longmapsto u^\top \Sigma_\xi P_i \Sigma_\eta u$$

is invariant under $\xi \otimes \eta \mapsto \eta \otimes \xi$. It therefore factors through

$$\mathrm{Sym}^2_{\mathbb{F}_q}(A) = (A \otimes_{\mathbb{F}_q} A) \big/ \mathrm{span}\{\xi \otimes \eta - \eta \otimes \xi\}.$$

Consequently, the direct sum over the m_1 base forms has rank at most $m_1\binom{d+1}{2}$. After any public linear map to M outputs—in particular, for every relation-dependent matrix Q_R—the rank is at most

$$\min\left\{M, m_1\binom{d+1}{2}\right\}. \tag{12}$$

If the minimal polynomial of S has degree ℓ, then $d = \ell$ and, in the power basis,

$$z_{\mathrm{ord}}(u) = D z_{\mathrm{sym}}(u). \tag{13}$$

Proof. Every element of A is symmetric. Since the displayed expression is a scalar,

$$u^\top \Sigma_\xi P_i \Sigma_\eta u = \left(u^\top \Sigma_\xi P_i \Sigma_\eta u\right)^\top = u^\top \Sigma_\eta P_i \Sigma_\xi u.$$

Thus every alternating label $\xi \otimes \eta - \eta \otimes \xi$ lies in the kernel. Taking the direct sum over i gives the first rank bound, and a linear map to M public outputs gives the additional codomain cap. When the powers of S form a basis of A, the factorization is exactly the coordinate identity (13). □

Why the outer map cannot repair the loss. The public outer map receives only the values of these quadratic features. If two labeled inputs already define the same polynomial, no later linear combination can separate them. The outer map may have full rank on the K-dimensional quotient, but it cannot enlarge that quotient.

For $\ell = 4$, each base form loses six of its sixteen ordered coordinates: $\ell^2 = 16$ becomes $\binom{\ell+1}{2} = 10$. For $\ell = 2$, four ordered coordinates become three. These reductions occur for every relation and every symmetric public key.

The theorem is an upper bound: additional public dependencies could make the quadratic image even smaller. The concrete attacks separately check $\operatorname{rank} Q_R = K$ before solving. Thus the value K

used in the cost tables is not inferred from symmetry alone; it is the exact rank observed in the public preflight for the chosen relation.

The factorization itself uses only transposition symmetry and is valid in every characteristic. Odd characteristic is needed only for the familiar direct-sum interpretation $A \otimes A = \mathrm{Sym}^2(A) \oplus \bigwedge^2(A)$. The submitted $q = 16$ system is outside this theorem because its public base matrices are not symmetrized.

4.1 A canonical Frobenius-channel normal form

The dimension statement $\dim_{\mathbb{F}_q} \mathrm{Sym}^2(A) = \binom{d+1}{2}$ can be sharpened into an explicit coordinate system. This matters algorithmically: the coordinates have different Frobenius degrees, so an attacker need not treat all ten quotient coordinates as unrelated quadratics over $\mathbb{F}_q$.

For $x, y \in A = \mathbb{F}_{q^d}$, write $[x \otimes y]_{\mathrm{sym}}$ for the class of $x \otimes y$ in $\mathrm{Sym}^2_{\mathbb{F}_q}(A)$. For $j > 0$, define the symmetric $\mathbb{F}_q$-bilinear map

$$\phi_j([x \otimes y]_{\mathrm{sym}}) = xy^{q^j} + x^{q^j}y, \qquad \phi_0([x \otimes y]_{\mathrm{sym}}) = xy.$$

When d is even and $j = d/2$, the value of ϕ_j is fixed by $q^{d/2}$-Frobenius and therefore lies in $\mathbb{F}_{q^{d/2}}$.

Theorem 4 (Frobenius-distance normal form)**.** *Let $A = \mathbb{F}_{q^d}$. The direct sum of the maps above is an $\mathbb{F}_q$-linear isomorphism*

$$\mathrm{Sym}^2_{\mathbb{F}_q}(A) \xrightarrow{\sim} \begin{cases} A^{(d+1)/2}, & d \text{ odd}, \\ A^{d/2} \oplus \mathbb{F}_{q^{d/2}}, & d \text{ even}. \end{cases}$$

For even d, the A-coordinates are $\phi_0, \ldots, \phi_{d/2-1}$ and the last coordinate is $\phi_{d/2}$.

Proof. Extend scalars to an algebraic closure and identify $A \otimes_{\mathbb{F}_q} \overline{\mathbb{F}}_q \cong \overline{\mathbb{F}}_q^d$. Index the primitive idempotents by $i \in \mathbb{Z}/d\mathbb{Z}$, so Frobenius cyclically shifts the indices. In these coordinates, ϕ_0 records the d diagonal unordered pairs $\{i, i\}$. For $0 < j < d/2$, the ith embedding of ϕ_j records the unordered pair $\{i, i + j\}$; these form one orbit of size d. When d is even and $j = d/2$, the pair $\{i, i + d/2\}$ is repeated at $i + d/2$, leaving one orbit of size $d/2$, exactly the coordinates of $\mathbb{F}_{q^{d/2}}$. The possible Frobenius distances partition all unordered pairs. The displayed map is therefore a coordinate permutation after scalar extension, hence an isomorphism over $\mathbb{F}_q$. $\square$

For the public $\ell = 4$ parameters, put $A = \mathbb{F}_{q^4}$ and $B = \mathbb{F}_{q^2} \subset A$. The theorem specializes to

$$\mathrm{Sym}^2_{\mathbb{F}_q}(A) \xrightarrow{\sim} A \oplus A \oplus B, \qquad [x \otimes y]_{\mathrm{sym}} \longmapsto (xy,\ xy^q + x^q y,\ xy^{q^2} + x^{q^2} y). \tag{14}$$

The companion certificate constructs the resulting 10×10 matrix in $A = \mathbb{F}_{19}[t]/(t^4 - t - 1)$ and verifies rank ten and determinant eight modulo 19.

Corollary 5 (Polynomial shape of the three channels)**.** *Let $L \cong A^s$ be an A-linear slice. After the invertible dual output change induced by (14), the ten quotient coordinates belonging to each public base form become*

$$F_{i,0} : A^s \to A, \qquad F_{i,1} : A^s \to A, \qquad F_{i,2} : A^s \to B.$$

Their homogeneous parts are, after choosing A-coordinates on L, of the forms

$$F_{i,0}^{(2)}(x) = \sum_{j,k} c_{i,j,k}^{(0)} x_j x_k,$$
$$F_{i,1}^{(q+1)}(x) = \sum_{j,k} c_{i,j,k}^{(1)} x_j x_k^q,$$
$$F_{i,2}^{(q^2+1)}(x) = \sum_{j,k} c_{i,j,k}^{(2)} x_j x_k^{q^2}.$$

Translation to a target-dependent affine coset adds only terms in (x, x^q) to $F_{i,1}$ and in (x, x^{q^2}) to $F_{i,2}$. Consequently their ordinary total degrees over A are at most 2, $q+1$, and q^2+1. Moreover, a uniform quotient target becomes an independent uniform element of $A^{m_1} \oplus A^{m_1} \oplus B^{m_1}$.

Proof. Over a splitting field, Lemma 7 groups the ten unordered eigenblock features by cyclic distance: diagonal, adjacent, and opposite. Descent around the diagonal orbit gives an A-valued ordinary quadratic; descent around the adjacent orbit evaluates one variable block at x and the next at x^q; and the opposite orbit evaluates at (x, x^{q^2}) and has only two conjugates. Translation gives the stated lower-degree terms. The target claim follows because the dual of an invertible $\mathbb{F}_q$-linear coordinate change is invertible. □

5 Structured Stable Lifts

Proposition 2 permits arbitrary offsets v_j. Generic offsets are useful for making the affine equations full rank, but they create linear terms that mix all coordinates. That mixing destroys the block grading used by the selected published $\ell = 4$ special-XL model. We therefore need offsets that do two jobs at once:

(i) span the missing affine output directions, so that the full verifier is retained; and

(ii) confine all offset-linear terms to a small public row space, so that a large invariant kernel remains available for special XL.

The following construction achieves both. Choose one vector $w \in \mathbb{F}_q^{N_0}$ and multipliers $\Gamma_0, \dots, \Gamma_{r-1} \in A$, and set

$$v_j = \Sigma_{\Gamma_j} w. \tag{15}$$

All offsets are therefore generated from the same w by the public algebra A. Define the row space

$$\mathcal{U}_w = \operatorname{span}_{\mathbb{F}_q} \left\{ w^\mathsf{T} \Sigma_\xi P_i \Sigma_\eta : 1 \le i \le m_1,\ \xi, \eta \in A \right\} \subseteq (\mathbb{F}_q^{N_0})^*. \tag{16}$$

This space contains every linear form that can arise from crossing a variable term with one of the structured offsets. Its a priori bound is $m_1 d^2$, not $m_1\binom{d+1}{2}$: one argument is the fixed offset vector w rather than the variable u on both sides, so reversing the two algebra labels need not leave this *linear* form unchanged.

A subspace $H \subseteq \mathbb{F}_q^{N_0}$ is called A-linear if $\Sigma_\xi H \subseteq H$ for every $\xi \in A$. Since $A \cong \mathbb{F}_{q^\ell}$, this is extension-field linearity. The earlier version of this analysis took the A-orbit only of the projected affine constraints and then required a maximum-rank equality to infer that all offset-linear rows were killed. That equality is unnecessary. Take instead the complete right A-orbit of the *entire* offset-linear feature matrix. For the basis $1, S, \dots, S^{d-1}$, form

$$\mathcal{O}^{\mathcal{F}}_{R,\mathbf{v}} = \begin{bmatrix} \mathcal{F}_{R,\mathbf{v}} \\ \mathcal{F}_{R,\mathbf{v}}\Sigma_S \\ \vdots \\ \mathcal{F}_{R,\mathbf{v}}\Sigma_{S^{d-1}} \end{bmatrix}, \qquad \rho_{\mathcal{F}} = \operatorname{rank} \mathcal{O}^{\mathcal{F}}_{R,\mathbf{v}}. \tag{17}$$

Theorem 6 (Unconditional stable-lift kernel)**.** *Retain $S^{\mathsf{T}} = S$ and $P_i^{\mathsf{T}} = P_i$, use structured offsets (15), and suppose $A \cong \mathbb{F}_{q^d}$. Then:*

(i) *$\mathcal{U}_w$ is closed under right multiplication by A and $\dim_{\mathbb{F}_q} \mathcal{U}_w \le m_1 d^2$;*

(ii) *every row of $\mathcal{F}_{R,\mathbf{v}}$ and $\mathcal{O}^{\mathcal{F}}_{R,\mathbf{v}}$ lies in $\mathcal{U}_w$;*

(iii) $\operatorname{row}(\mathcal{O}^{\mathcal{F}}_{R,\mathbf{v}})$ *is an A-subspace, so $d \mid \rho_{\mathcal{F}}$; and*

(iv)

$$H^{\mathcal{F}}_{R,\mathbf{v}} = \ker \mathcal{O}^{\mathcal{F}}_{R,\mathbf{v}} \tag{18}$$

is A-linear, satisfies $\mathcal{F}_{R,\mathbf{v}} H^{\mathcal{F}}_{R,\mathbf{v}} = 0$, and has

$$\dim_A H^{\mathcal{F}}_{R,\mathbf{v}} = n - \frac{\rho_{\mathcal{F}}}{d} \ge n - m_1 d. \tag{19}$$

No orbit-rank equality or genericity assumption is required. A rank smaller than $m_1 d^2$ only enlarges the attacker's stable search space.

Proof. See Section A.1. □

Interpretation. The stable slice now exists for every structured-offset instance. Moving inside $H^{\mathcal{F}}_{R,\mathbf{v}}$ changes the quadratic features but none of the offset-linear features. The offsets can therefore cover the missing affine output directions, while the nonlinear solver retains a clean A-linear block decomposition. The former full-orbit-rank test has become a harmless dimension measurement rather than an attack condition.

The stable-lift logic in four unconditional arrows. Structured offsets put every offset-linear row in $\mathcal{U}_w$; taking the complete right-A orbit of those rows makes their row space A-stable; its kernel is therefore A-linear and automatically annihilates $\mathcal{F}_{R,\mathbf{v}}$; and the bound $\rho_{\mathcal{F}} \le m_1 d^2$ guarantees at least $n - m_1 d$ extension-field search dimensions. Any unexpected dependency helps rather than hurts.

5.1 From an A-linear kernel to four variable blocks

Special XL needs variables divided into blocks with controlled multidegrees. The algebra A supplies those blocks after extending scalars to a field Ω in which the minimal polynomial of S splits. Over Ω, multiplication by S diagonalizes into ℓ eigenspaces. The primitive idempotent e_a is simply the projection onto the ath eigenspace. This scalar extension is a solver coordinate change, not a relaxation of the signature problem: the recovered point must still descend to $\mathbb{F}_q$ and pass the original verifier.

Let $\mathbf{e}_a \in \mathbb{Z}^\ell$ denote the ath standard unit vector. A quadratic using only block a has multidegree $2\mathbf{e}_a$; a bilinear term using blocks a and b has multidegree $\mathbf{e}_a + \mathbf{e}_b$.

Lemma 7 (Eigenblock coordinates of the quotient)**.** *Let $\Omega/\mathbb{F}_q$ be a splitting field of the minimal polynomial of S, and let $e_1, \dots, e_\ell$ be the primitive idempotents of $A \otimes_{\mathbb{F}_q} \Omega$. If L is an A-linear space of A-dimension s, then*

$$L \otimes_{\mathbb{F}_q} \Omega = L_1 \oplus \cdots \oplus L_\ell, \qquad L_a = \Sigma_{e_a}(L \otimes_{\mathbb{F}_q} \Omega), \qquad \dim_\Omega L_a = s.$$

For each i, the eigenblock features

$$x^{\mathsf{T}}\Sigma_{e_a}P_i\Sigma_{e_b}x, \qquad 1 \le a \le b \le \ell,$$

are an invertible linear recombination of the symmetric power-pair features. They have multidegree $2\mathbf{e}_a$ when $a = b$ and $\mathbf{e}_a + \mathbf{e}_b$ when $a < b$.

Proof. Finite fields are perfect, so the irreducible minimal polynomial of S is separable. Over Ω, the power basis and the primitive-idempotent basis of $A \otimes \Omega \cong \Omega^{\ell}$ are related by an invertible Vandermonde matrix. Passing to the symmetric square gives an invertible change between the unordered power-pair features and the unordered eigenblock features. The idempotent e_a projects $L \otimes \Omega$ onto L_a. Hence a diagonal feature uses only variables in L_a, while an off-diagonal feature is bilinear in the variables from L_a and L_b, giving the stated multidegrees. □

Corollary 8 (Stable affine slices)**.** *Assume the public checks*

$$\operatorname{rank} Q_R = K, \qquad \operatorname{rank} C_{R,\mathbf{v}} = M - K \tag{20}$$

pass, put $H_{R,\mathbf{v}} = H^{\mathcal{F}}_{R,\mathbf{v}}$, and let $L \subseteq H_{R,\mathbf{v}}$ be an A-linear subspace of A-dimension s (equivalently, base-field dimension $h = \ell s$). By Theorem 6, such an L is always available whenever $s \le n - m_1\ell$. For every target, the affine equations in Proposition 2 have a solution space $a(t) + \ker C_{R,\mathbf{v}}$ for a publicly computed base point $a(t)$. For every $y \in \ker C_{R,\mathbf{v}}$, the affine slice $a(t) + y + L$ is contained in that space, and the full offset-linear feature $\mathcal{F}_{R,\mathbf{v}}u$ is constant on the slice.

After extension to a splitting field, use the eigenblock coordinates of Lemma 7. Translation by $a(t) + y$ creates only linear and constant terms in the same one or two blocks as the corresponding quadratic term. Introducing one homogenizing variable per block—an auxiliary variable that is set to one after solving—therefore turns each complete affine equation into a homogeneous equation of multidegree $2\mathbf{e}_a$ or $\mathbf{e}_a + \mathbf{e}_b$. If the restricted quadratic feature map has total rank K, then the disjoint monomial supports of these unordered multidegrees force rank m_1 in every component: there are $\binom{\ell+1}{2}$ components, each has rank at most m_1, and their ranks sum to $K = m_1\binom{\ell+1}{2}$.

Proof. See Section A.2. □

What the preflight certifies. The two ranks in (20) say that the quotient has its full expected rank and that the offsets supply every remaining affine output direction. The stable kernel and the identity $\mathcal{F}_{R,\mathbf{v}}H_{R,\mathbf{v}} = 0$ are now theorem consequences, not rank hypotheses. Computing $\rho_{\mathcal{F}}$ merely reports the actual kernel dimension, which can only exceed the conservative lower bound. The restricted quadratic rank then certifies that no multidegree component was lost on the chosen slice. All remaining tests are public and occur before an expensive solver run.

5.2 Decoupling slice selection from fresh public coefficients

The public $q = 19$ expansion has more structure than an arbitrary symmetric matrix. In the coordinate order used by the reference implementation, write the public variable space as an A-linear direct sum

$$\mathcal{V}_{\text{pub}} = W \oplus V' \oplus O_{\text{pub}}, \qquad \dim_A W = 1, \quad \dim_A V' = v - 1, \quad \dim_A O_{\text{pub}} = o. \tag{21}$$

Here W is one reserved public vinegar line, V' is the sum of the remaining public vinegar lines, and O_{pub} is the public oil-coordinate summand. These are public coordinate labels; the attack does not recover the secret oil space.

Choose a nonzero $w \in W$ for (15). Define the *cross data* to consist of the public outer-map data together with the restrictions

$$P_{11,i}|_{W\times(W\oplus V')}, \qquad P_{12,i}|_{W\times O_{\mathrm{pub}}}, \qquad 1 \le i \le m_1, \tag{22}$$

and exclude both $P_{11,i}|_{V'\times V'}$ and the compressed P_{22} block. The relation R, the multipliers Γ_j, and a canonical bounded list of A-linear candidate slices are chosen as deterministic functions of this cross data.

Lemma 9 (Coordinate separation in the pinned public expansion)**.** *In the pinned $q = 19$ reference implementation, every independent scalar entry of an upper-triangular diagonal block of P_{11} is read from a distinct public-expansion byte and only then copied to its transposed position. Off-diagonal P_{11} blocks are likewise filled entry by entry and their transposes are copied afterward. The bytes used for P_{12}, P_{21}, and the public mixing data occupy disjoint later positions. Finally,* `convert_bytes_to_GF` *reduces each byte coordinatewise modulo* 19*. Consequently, the raw byte coordinates underlying $P_{11,i}|_{V'\times V'}$ are pairwise distinct and disjoint from every byte coordinate in the cross data of* (22)*. Under the random-XOF idealization they are therefore mutually independent, independent of the cross data, and each has maximum atom* $14/256$*.*

Proof. This is a direct audit of the loops in `expand_public`: the expansion cursor is incremented once for each independent scalar before the symmetry copy is performed, and the cursor is never reused. The conversion routine then applies reduction modulo 19 separately to each occupied byte. The last sentence is the random-XOF interpretation of these disjoint input coordinates, not a claim of literal statistical independence for a fixed SHAKE seed [25]. □

Lemma 10 (Biased-product affine-subspace bound)**.** *Let $X = (X_1, \dots, X_N)$ have independent coordinates in $\mathbb{F}_q$, each with maximum atom at most ρ. If $T : \mathbb{F}_q^N \to \mathbb{F}_q^r$ has rank r, then for every $b \in \mathbb{F}_q^r$,*

$$\Pr[T(X) = b] \le \rho^r. \tag{23}$$

Equivalently, every affine subspace of codimension r has probability at most ρ^r under this product distribution.

Proof. Choose r pivot coordinates for T. After conditioning on every nonpivot coordinate, row reduction forces the pivot coordinates successively to one specified value each. Independence and the maximum-atom bound give a factor at most ρ per pivot. □

The following observation removes the former determinant-preflight loss for the affine source. It is the alternating analogue of Lemma 26.

Lemma 11 (Alternating affine-source injection)**.** *Assume q is odd, choose*

$$R_j = S^j, \qquad \Gamma_0 = 1, \qquad \Gamma_1 = 0, \tag{24}$$

and let $w \ne 0$ span the reserved A-line W. In the raw rows with column pair $(j, k) = (1, 0)$, quotienting the quadratic columns $\mathcal{R}_R D$ leaves one copy of $\bigwedge^2_{\mathbb{F}_q}(A)$ per public base form. If $0 \ne c = \sum_\nu a_\nu \wedge b_\nu \in \bigwedge^2(A)$, the coefficient map of the corresponding offset-linear form on V' has transpose

$$\Psi_{c,w}(x) = \sum_\nu \big((b_\nu w) \odot (a_\nu x) - (a_\nu w) \odot (b_\nu x)\big). \tag{25}$$

The map $x \mapsto \Psi_{c,w}(x)$ is injective on V'. Consequently, in the random-XOF idealization,

$$\Pr\left[\operatorname{rank}(\pi_{\mathrm{alt}}\mathcal{F}_{R,\mathbf{v}}|_{V'}) < m_1\binom{d}{2}\right] \le \frac{q^{m_1\binom{d}{2}}-1}{q-1}\,\rho^{d(v-1)}, \tag{26}$$

where π_{alt} denotes the displayed alternating quotient. On the complementary event,

$$\operatorname{rank}[\mathcal{R}_R D\ \mathcal{F}_{R,\mathbf{v}}|_{V'}] \ge K + m_1\binom{d}{2}. \tag{27}$$

Proof. As the power labels vary, the $(1,0)$ raw rows identify with $A\otimes A$; the extra factor $R_1 = S$ is invertible. Their quadratic columns are ordinary symmetrization, whose image and kernel in odd characteristic are respectively $\mathrm{Sym}^2(A)$ and $\bigwedge^2(A)$. The structured offset contributes $B_i(bw, ax)$ on a pure tensor $a\otimes b$, which gives (25) on an alternating tensor.

For $x \ne 0$ in V', the A-lines W and Ax are disjoint. Project the symmetric tensor in (25) to the cross summand $W \otimes Ax$ and identify both lines with A. The result is $-c$ after invertible multiplication in the two factors, and is therefore nonzero. Thus the transpose coefficient map has rank $\dim_{\mathbb{F}_q} V' = d(v-1)$.

Now take a nonzero dual tuple $(c_1,\ldots,c_{m_1}) \in (\bigwedge^2 A)^{m_1}$ and choose i with $c_i \ne 0$. Condition on every public-form block except $P_i|_{W\times V'}$. The event that this tuple lies in the left kernel of the alternating affine map imposes a rank-$d(v-1)$ affine system on those fresh symbols, so Lemma 10 bounds it by $\rho^{d(v-1)}$. Union-bound over projective nonzero dual tuples. Trivial intersection between the symmetric and alternating summands then gives (27). □

Theorem 12 (Random-XOF density of the complete structured preflight)**.** *Continue with (24), put $J = W \oplus O_{\mathrm{pub}}$, and define*

$$\delta_{\mathrm{src}} = \frac{q^{m_1\binom{d}{2}}-1}{q-1}\rho^{d(v-1)}, \tag{28}$$

$$\delta_{\mathrm{proj}} = \rho^{m_1\binom{d+1}{2}} + \left(\frac{|A|^{1+o}-1}{|A|-1} - 1\right)\rho^{m_1 d^2}, \tag{29}$$

$$p_M(\rho) = \prod_{c=1}^{M}(1-\rho^c). \tag{30}$$

Under the random-XOF idealization, with probability at least

$$p_{\mathrm{pre}} = p_M(\rho)(1-\delta_{\mathrm{src}})(1-\delta_{\mathrm{proj}}), \tag{31}$$

all of the following hold simultaneously:

(i) $\operatorname{rank} Q_R = K$ *and* $\operatorname{rank} C_{R,\mathbf{v}} = M - K$*;*

(ii) $\operatorname{row}(\mathcal{F}_{R,\mathbf{v}}) = \mathcal{U}_w$ *and* $H^{\mathcal{F}}_{R,\mathbf{v}} \cap J = \{0\}$*; and*

(iii) projection of the entire stable kernel to V' is injective, so every subspace of it automatically passes (32).

For the six official $\ell = d = 4$ shapes, the exact bounds are:

shape	$m_1\binom{4}{2}$	*proved/needed*	$\log_2 \delta_{\mathrm{src}}$	$\log_2 \delta_{\mathrm{proj}}$	p_{pre}
I-a		$30/30$	-329.54	-209.63	$> 0.9423222679 - 2^{-209}$
I-b		$30/30$	-329.54	-209.63	$> 0.9423222679 - 2^{-209}$
III-a		$42/42$	-479.81	-293.49	$> 0.9423222679 - 2^{-293}$
III-b		$42/30$	-446.27	-293.49	$> 0.9423222679 - 2^{-293}$
V-a		$54/54$	-596.54	-377.34	$> 0.9423222679 - 2^{-377}$
V-b		$54/54$	-630.08	-377.34	$> 0.9423222679 - 2^{-377}$

All decimals are rounded downward at the displayed precision; the artifact stores the exact rational values.

Proof. Every row of $\mathcal{F}_{R,\mathbf{v}}$ lies in $\mathcal{U}_w$. Conversely, the $(1,0)$ raw column pair in the proof of Lemma 11, as its two power labels vary, exposes every generator $w^\mathsf{T}\Sigma_\xi P_i \Sigma_\eta$ of $\mathcal{U}_w$. Thus the two row spaces agree; since $\mathcal{U}_w$ is right-A stable, the complete orbit matrix has the same row space and $H^{\mathcal{F}}_{R,\mathbf{v}} = \mathcal{U}_w^\perp$.

If $0 \neq x \in W$, then $Ax = W$. The condition $x \in \mathcal{U}_w^\perp$ forces every symmetric restriction $P_i|_{W\times W}$ to vanish, imposing $m_1\binom{d+1}{2}$ independent scalar conditions. If $x \in J \setminus W$, then Ax is an A-line disjoint from W; the $W \times O_{\mathrm{pub}}$ symbols map surjectively to $\mathrm{Bil}(W, Ax)$, so the same condition imposes $m_1 d^2$ independent scalar conditions. The event depends only on the A-line Ax. There is one line W and $(|A|^{1+o} - 1)/(|A| - 1) - 1$ other lines in J. Apply Lemma 10 and union-bound to obtain (29). The source event of Lemma 11 uses only $W \times V'$ symbols, whereas this projection event uses only the symmetric $W \times W$ symbols and $W \times O_{\mathrm{pub}}$ symbols. They are independent by Lemma 9.

On the source event, choose $M - K$ affine columns whose alternating images are independent and adjoin all K quadratic columns. This gives an M-dimensional source subspace with the quadratic columns as a distinguished subset. Conditional on that public-form data, each independent outer-map row has a surjective image in $\mathbb{F}_q^M$. If the previous rows span a subspace of codimension c, Lemma 10 bounds the chance that the next row remains in it by ρ^c. Multiplying the complementary probabilities gives $p_M(\rho)$. Invertibility on the chosen source columns implies both $\operatorname{rank} Q_R = K$ and $\operatorname{rank}[Q_R\ \Lambda_{R,\mathbf{v}}] = M$; then Proposition 2 gives $\operatorname{rank} C_{R,\mathbf{v}} = M - K$. The outer-map bytes are disjoint from both public-form regions, so the three success probabilities multiply. Finally, $H^{\mathcal{F}}_{R,\mathbf{v}} \cap J = \{0\}$ is exactly injectivity of its projection to the complementary summand V'. $\square$

Lemma 13 (Reserved-line decoupling)**.** *With $w \in W$, the matrices Q_R, $C_{R,\mathbf{v}}$, $\mathcal{O}^{\mathcal{F}}_{R,\mathbf{v}}$, the stable kernel $H_{R,\mathbf{v}}$, and every candidate slice selected canonically from that kernel depend only on the cross data in (22). In particular, they are independent of the internal public coefficients $P_{11,i}|_{V'\times V'}$ in the random-XOF idealization.*

A candidate used in the seeded analysis additionally passes the exact public projection preflight

$$L \cap (W \oplus O_{\mathrm{pub}}) = \{0\}, \qquad \textit{equivalently} \qquad \operatorname{rank}(\pi_{V'}|_L) = \dim_{\mathbb{F}_q} L. \tag{32}$$

The full-key restricted-quadratic-rank and verifier preflights may be applied after the bounded candidate list is fixed.

Proof. The quotient matrix Q_R is determined by the public outer map and the relation. Every row created by a structured offset is a linear combination of forms $(\Sigma_\xi w)^\mathsf{T} P_i \Sigma_\eta$. Its left argument remains inside W, so only the W-row restrictions in (22) occur: P_{22} and the internal $V' \times V'$ block never appear. The same remains true after taking the complete right A-orbit in (17). Kernels, canonical row-reduced bases, and candidate subspaces derived from these matrices are therefore functions of the cross data alone. The last checks are ordinary public rank tests. Selecting the first full-key candidate that passes them from a list of size J costs only a union factor J in the seeded failure bound below. $\square$

The purpose of this construction is not to hide information from the attacker—all coefficients are public after key generation. It is to make a valid probability statement: the slice is fixed using one disjoint portion of the public seed expansion, while the internal P_{11} coefficients that force its polar rank remain fresh. Theorem 27 turns this data separation into an explicit aggregate-spectrum theorem.

5.3 Passing the verifier's block-symmetry check

The public verifier does more than evaluate the quadratic map. It also rejects signatures containing too many symmetric blocks. A root of the reduced MQ system is useful only if it passes this final check. We therefore build rejection compatibility into the reduction rather than treating it as an unmodeled success probability.

Version 2.3 rejects a square odd-characteristic signature if one of its blocks is symmetric when $\ell > 2$, and rejects an $\ell = 2$ signature if more than $\lfloor n/4 \rfloor$ blocks are symmetric [25]. Rectangular blocks are not subject to this rule.

The structured $\ell = 4$ rows. Only the square rows with $r = \ell = 4$ are subject to the verifier's strict "no symmetric block" rule; the rectangular $r = 5, 6$ rows have no such check. For a relation $R = (R_0, R_1, R_2, R_3)$, define the local skew map

$$\mathsf{S}_R : \mathbb{F}_q^4 \longrightarrow \bigwedge^2 \mathbb{F}_q^4, \qquad \mathsf{S}_R(x)_{a,c} = (R_c x)_a - (R_a x)_c \quad (0 \le a < c < 4). \tag{33}$$

For any offsets, one signature block is symmetric exactly when $\mathsf{S}_R(x)$ equals a fixed offset-dependent vector. Hence the public preflight rank $\mathsf{S}_R = 4$ makes the symmetry equation have either zero or one solution in that block. Because the n block variables are disjoint, the accepted subset of the full common-column space has density

$$a_{4,n} \ge (1 - q^{-4})^n. \tag{34}$$

No independence from the MQ equations is assumed; the accepted-singleton theorem below uses only this global density.

Lemma 14 (An exact injective-skew relation for the official $q = 19$ matrix)**.** *For the official $\ell = 4$ matrix*

$$S = \begin{pmatrix} 1 & 2 & 3 & 0 \\ 2 & 3 & 0 & 1 \\ 3 & 0 & 1 & 2 \\ 0 & 1 & 2 & 15 \end{pmatrix} \quad \textit{over } \mathbb{F}_{19},$$

the powers relation $R_j = S^j$, $0 \le j < 4$, has

$$[\mathsf{S}_R] = \begin{pmatrix} 1 & 1 & 3 & 0 \\ 14 & 8 & 5 & 8 \\ 10 & 3 & 7 & 6 \\ 5 & 14 & 7 & 16 \\ 3 & 18 & 9 & 0 \\ 18 & 12 & 7 & 1 \end{pmatrix}, \qquad \operatorname{rank}_{\mathbb{F}_{19}} \mathsf{S}_R = 4.$$

Thus the injective-skew preflight is nonvacuous. The attack retains a relation only when this rank and the independent Q_R, affine, orbit, and restricted quadratic-rank preflights all pass.

Proof. The six rows are the coefficients of $(S^c x)_a - (S^a x)_c$ for $0 \le a < c < 4$. Row reduction modulo 19 gives four pivots. The companion artifact recomputes S, its powers, the displayed matrix, and its rank from integer data. $\square$

For the three square rows, $n = 33, 47, 59$. Combining (34) with Theorem 20 changes the compact retry denominator from $1 - 19^{-1}$ to $a_{4,n} - 19^{-1}$, and the robust denominator from $1/2$ to $a_{4,n} - 1/2$. The largest resulting surcharge is only 0.001307 bits. It is included in the artifact and is smaller than the rounding precision of the displayed cost tables. This is a probabilistic bypass with an exact density bound: unlike the fixed-skew $\ell = 2$ construction below, it does not claim that every point of the search space is accepted.

A deterministic $\ell = 2$ construction. The first two parameter levels use offsets that force a fixed nonzero skew in every block.

Proposition 15 (Fixed-skew $\ell = 2$ lift)**.** *For the Version 2.3 $\ell = 2$ matrix*

$$S = \begin{pmatrix} 1 & 2 \\ 2 & 15 \end{pmatrix}, \qquad R_0 = I, \qquad R_1 = 9I + 10S = \begin{pmatrix} 0 & 1 \\ 1 & 7 \end{pmatrix},$$

choose offsets satisfying

$$(v_1)_{b,0} - (v_0)_{b,1} = 1 \tag{35}$$

for every block b. Then every block of every signature satisfying the column relation has

$$U_b[0,1] - U_b[1,0] = 1. \tag{36}$$

Consequently every solution passes the $\ell = 2$ symmetric-block rejection check. If the public preflight also gives $\operatorname{rank} Q_R = K$ and $\operatorname{rank} C_{R,\mathbf{v}} = M - K$, every target passes the affine consistency test and leaves K equations of degree at most two in $N_0 - M + K$ variables. Every root of that residual system is a valid signature.

Proof. The first row of R_1 is the second row of $R_0 = I$. Hence the variable part of $U_b[0,1] - U_b[1,0]$ vanishes. The remaining offset difference is one by (35), proving (36). No block is symmetric. The final statement is the full-lift case of Proposition 2. $\square$

Corollary 16 (Symmetric-block rejection is structurally insufficient)**.** *For the public $\ell = 2$ Version 2.3 verifier, any countermeasure whose only additional condition is that a signature contain sufficiently few symmetric blocks leaves the fixed-skew affine family of Proposition 15 untouched: every block in every candidate has the prescribed nonzero skew 1. Such a rule can reject honest-looking symmetric blocks, but it cannot exclude this attack family without testing some further property.*

Proof. Equation (36) holds identically on the entire affine family, before any MQ equation is solved. Hence none of its blocks is symmetric, independently of the target or root. $\square$

Thus the fixed-skew construction has no rejection probability: every MQ root is acceptable. The Level-V headline attack uses a faster alternative in which targets are filtered before solving.

The zero-offset $\ell = 2$ filter. Set $R_0 = I$, $R_1 = 0$, and use zero offsets. Each block then has the form

$$\begin{pmatrix} x & 0 \\ y & 0 \end{pmatrix},$$

so it is symmetric exactly when $y = 0$. These rejection events are nonzero hyperplanes on disjoint block coordinates. This relation is chosen carefully: a relation that forced the two off-diagonal entries to agree would make every block symmetric and would be useless.

Let N be the residual-domain dimension ($N = N_0$ for this zero-offset lift) and let $g : \mathbb{F}_q^N \to \mathbb{F}_q^K$ be the residual map of degree at most two. For each nonzero $b \in \mathbb{F}_q^K$, take the polar rank of the scalar quadratic $b \cdot g$, and let r_* be the minimum of these ranks. Thus r_* measures the worst output direction; a large r_* forces every target fiber to have nearly its expected size.

Put

$$t_{\mathrm{rej}} = \lfloor n/4 \rfloor + 1, \qquad B = \binom{n}{t_{\mathrm{rej}}}, \qquad \alpha = 1 - q^{-K},$$

and let $\mathcal{X}_{\mathrm{acc}}$ be the assignments accepted by the block-symmetry filter. At least t_{rej} symmetric blocks are needed for rejection. Because the n hyperplanes use disjoint coordinates, the exact density of accepted assignments is

$$a_n := \frac{|\mathcal{X}_{\mathrm{acc}}|}{q^N} = q^{-n} \sum_{k=0}^{\lfloor n/4 \rfloor} \binom{n}{k} (q-1)^{n-k}. \tag{37}$$

Proposition 17 (Accepted targets for the $\ell = 2$ filter)**.** *Every target has an accepted root if*

$$1 - Bq^{-t_{\mathrm{rej}}} > \alpha(1+B)q^{K - r_*/2}. \tag{38}$$

For a uniform target $z \in \mathbb{F}_q^K$,

$$\Pr\left[g^{-1}(z) \cap \mathcal{X}_{\mathrm{acc}} \neq \varnothing\right] \geq \frac{a_n}{1 + \alpha q^{K - r_*/2}}. \tag{39}$$

Moreover, if $1 - \alpha q^{K - r_*/2} > 0$, every target z satisfies

$$\frac{|g^{-1}(z) \cap \mathcal{X}_{\mathrm{acc}}|}{|g^{-1}(z)|} \geq 1 - \frac{Bq^{-t_{\mathrm{rej}}} + \alpha B q^{K - r_*/2}}{1 - \alpha q^{K - r_*/2}}, \tag{40}$$

whenever the right-hand side is positive.

Proof. See Section A.3. □

The three conclusions answer different questions. Readers interested only in the concrete attack may retain the following takeaway. The global condition $r_* \geq 240$ gives an accepted root for every Level-V target. The much weaker condition $r_* \geq 194$ makes the expected number of uniform targets to an accepted root less than $2^{0.075}$ by Corollary 34; this is the condition charged in the headline. More generally, (38) is a sufficient worst-case condition, (39) controls expected retries, and (40) says that, target by target, almost every root is accepted when the rank is high enough.

6 Root Existence, Uniqueness, and Retry Bounds

After the exact affine reduction, the structured attack repeatedly chooses a hash target and an affine coset of the selected slice direction space. We call one such pair a *target–slice trial*. The attack asks whether the resulting MQ system has a recoverable root. Three events must not be conflated:

(i) the slice contains at least one base-field root;

(ii) it contains exactly one base-field root; and

(iii) the selected Macaulay degree exposes a finite quotient algebra from which all algebraic roots can be extracted.

This section proves statements about the first two events. The third is the separate special-XL solving condition in Section 7.2; exact corank one is only its simplest special case.

Let V be the full residual direction space, let $L \subseteq V$ be the chosen slice direction space, and write $h = \dim L$. For a map of degree at most two $g : V \to \mathbb{F}_q^K$, a random-map heuristic predicts

$$\mu = q^{h-K}$$

roots in a random affine coset of L above a random target. When $h = K - \tau$ with $\tau \geq 0$, this mean is $q^{-\tau}$, suggesting about q^{τ} trials. The obstruction detected by a second-moment analysis is excess collisions: distinct points in the same slice mapping to the same target. Polar derivatives count those collisions exactly, so the argument below is a theorem rather than a random-system heuristic. Full cross-polar rank is a clean special case, but it is far stronger than the attack needs. The load-bearing headline condition is the aggregate bound $\bar{\eta}_L \leq q^{-1}$. The pointwise condition $\operatorname{rank} T_y \geq h + 1$ for every nonzero slice direction is a convenient sufficient certificate; it keeps the unique-root retry count within a factor $q/(q-1)$ of q^{K-h}. The theorem first states the exact formulas and then quantifies both sparse defects and weaker uniform rank cushions.

Write the vector-valued polar map as

$$\beta(x, y) = g(x + y) - g(x) - g(y) + g(0).$$

For $b \in \mathbb{F}_q^K \setminus \{0\}$, define the cross-polar map

$$\Phi_{b,L} : V \longrightarrow L^*, \qquad \Phi_{b,L}(x)(y) = b \cdot \beta(x, y), \tag{41}$$

and let $e_L(b) = \operatorname{rank} \Phi_{b,L}$. Put

$$e_* = \min_{b \neq 0} e_L(b), \qquad \Theta_L(g) = \sum_{b \neq 0} q^{-e_L(b)}. \tag{42}$$

For $y \in L \setminus \{0\}$, also define the directional derivative

$$T_y : V \longrightarrow \mathbb{F}_q^K, \qquad T_y(x) = \beta(x, y), \tag{43}$$

write $r(y) = \operatorname{rank} T_y$, and let $\kappa(y)$ be one when $-(g(y) - g(0)) \in \operatorname{im} T_y$ and zero otherwise. The indicator $\kappa(y)$ records whether two points separated by y can collide at all; the rank $r(y)$ then determines how many such pairs exist.

Theorem 18 (Root collisions and the rank-spectrum bound)**.** *Choose an affine coset C uniformly from V/L and choose $z \in \mathbb{F}_q^K$ uniformly. Let*

$$Z = |\{x \in C : g(x) = z\}|, \qquad \mu = q^{h-K},$$

and define

$$\eta_0 = \sum_{y \in L \setminus \{0\}} \kappa(y) q^{-r(y)}, \qquad \bar{\eta} = \sum_{y \in L \setminus \{0\}} q^{-r(y)}. \tag{44}$$

Then

$$\mathbb{E}Z = \mu, \tag{45}$$

$$\mathbb{E}[Z(Z-1)] = \mu\eta_0, \tag{46}$$

$$\eta_0 \le \bar{\eta} = \mu(1 + \Theta_L(g)) - 1. \tag{47}$$

Consequently,

$$\mathrm{Var}(Z) \le \mu^2 \Theta_L(g), \tag{48}$$

$$\Pr[Z > 0] \ge \frac{\mu}{1 + \eta_0} \ge \frac{1}{1 + \Theta_L(g)}. \tag{49}$$

If $h = K - \tau$ and $e_* \ge h - \Delta$, then

$$\Theta_L(g) < q^{\tau+\Delta}, \qquad \mathbb{E}[\text{trials to a root-containing target–coset pair}] < 1 + q^{\tau+\Delta}. \tag{50}$$

This last bound controls the search for a root-containing target–coset pair, not uniqueness of the root. Here Δ is the maximum loss from full cross-polar rank h.

Proof. See Section A.4. □

How to read the theorem. The first moment $\mathbb{E}Z = \mu$ is exactly the random-map value. The second factorial moment $\mathbb{E}[Z(Z-1)]$ counts ordered pairs of distinct roots and therefore measures collisions. The compatible collision mass η_0 is exact; $\bar{\eta}$ drops the compatibility test and is a convenient upper bound. The sum $\Theta_L(g)$ describes the entire cross-polar rank spectrum, not merely its minimum. A few isolated rank defects may have little effect, whereas low rank in many output directions can substantially increase retries.

$\mu = q^{h-K}$	expected number of roots in a random target–slice pair
η_0	exact expected number of compatible collisions per planted root
$\bar{\eta}$	rank-only upper bound on that collision mass
directional rank $r(y) \ge h+1$	unique-root trial count at most $q^{K-h}/(1-q^{-1})$

Corollary 19 (Unique-root trials and the planted distribution)**.** *In the setting of Theorem 18,*

$$\Pr[Z = 1] \ge \mu(1 - \eta_0) \ge \mu(1 - \bar{\eta}), \tag{51}$$

$$\Pr[Z > 0] \ge \mu\left(1 - \frac{\bar{\eta}}{2}\right), \qquad (\bar{\eta} < 2), \tag{52}$$

$$\Pr[Z \ge 2 \mid Z > 0] \le \frac{\bar{\eta}}{2 - \bar{\eta}}, \qquad (\bar{\eta} < 2). \tag{53}$$

In particular, if $\bar{\eta} < 1$, the expected number of independent trials until $Z = 1$ is at most

$$\frac{1}{\mu(1 - \bar{\eta})}. \tag{54}$$

Let U *be the uniform target–coset pair conditioned on* $Z > 0$*, and let* P *be the planted distribution obtained by choosing a uniform coset, a uniform point in it, and the point's image as target. If* $\bar{\eta} < 1$*, then*

$$\Pr_{\mathsf{P}}[Z \geq 2] \leq \eta_0 \leq \bar{\eta}, \qquad d_{\mathrm{TV}}(\mathsf{U}, \mathsf{P}) \leq \bar{\eta}. \tag{55}$$

Proof. See Section A.5. □

Theorem 20 (Accepted singleton before affine elimination)**.** *Let* U *be a finite-dimensional* $\mathbb{F}_q$*-space, let* $\mathsf{C} : U \to \mathbb{F}_q^c$ *be surjective, let* $L \subseteq \ker \mathsf{C}$ *have dimension* h*, and let* $g : U \to \mathbb{F}_q^K$ *have degree at most two. Choose* $d \in \mathbb{F}_q^c$ *and* $z \in \mathbb{F}_q^K$ *independently and uniformly, then choose uniformly an affine coset* $\mathcal{A}$ *of* L *contained in* $\mathsf{C}^{-1}(d)$*. For an arbitrary accepted set* $\mathcal{X} \subseteq U$ *of density* $a = |\mathcal{X}|/|U|$*, put*

$$Z = |\{x \in \mathcal{A} : g(x) = z\}|, \qquad Z_{\mathcal{X}} = |\{x \in \mathcal{A} \cap \mathcal{X} : g(x) = z\}|, \qquad \mu = q^{h-K}.$$

Define $T_y(x) = g(x+y) - g(x) - g(y) + g(0)$ *and* $\bar{\eta}_L = \sum_{0 \neq y \in L} q^{-\operatorname{rank} T_y}$*, as above. Then*

$$\mathbb{E} Z_{\mathcal{X}} = a\mu, \tag{56}$$

$$\mathbb{E}[Z(Z-1)] = \mu\eta_0 \leq \mu\bar{\eta}_L, \tag{57}$$

$$\Pr[Z = Z_{\mathcal{X}} = 1] \geq \mu(a - \eta_0) \geq \mu(a - \bar{\eta}_L). \tag{58}$$

Consequently, if $a > \bar{\eta}_L$*, the expected number of independent trials to an accepted root that is the unique base-field root of its trial is at most*

$$\frac{q^{K-h}}{a - \bar{\eta}_L}. \tag{59}$$

The bound requires no independence between acceptance and the polynomial map.

Proof. Because $L \subseteq \ker \mathsf{C}$ and C is surjective, sampling d and then a uniform L-coset in its fiber is exactly uniform over all cosets in U/L. Every accepted point is therefore selected with probability $q^{h-\dim U}$, and its image matches uniform z with probability q^{-K}, which proves (56). The ordered-pair calculation is identical to Theorem 18 and gives (57). Pointwise,

$$\mathbf{1}_{\{Z = Z_{\mathcal{X}} = 1\}} \geq Z_{\mathcal{X}} - Z(Z-1) :$$

for $Z = 0$ both sides vanish, for $Z = 1$ the right side is one exactly when the root is accepted, and for $Z \geq 2$ the right side is nonpositive. Taking expectations proves (58) and then (59). □

Theorem 21 (Accepted nonsingular roots from the same aggregate spectrum)**.** *Assume the setting of Theorem 20. Write the residual map as* $g = q_2 + q_1 + q_0$*, where* q_i *is homogeneous of degree* i*, and define the restricted formal Jacobian*

$$D_L g(x) : L \longrightarrow \mathbb{F}_q^K, \qquad D_L g(x)[y] = \beta(x, y) + q_1(y).$$

Let $\mathcal{X} \subseteq U$ *be the signatures accepted by the public verifier, with density* α*, and let* Y *be the number of points in the sampled fiber–coset trial that lie in* $\mathcal{X}$*, map to the sampled residual target, and satisfy* $\operatorname{rank} D_L g(x) = h$*. Then*

$$\Pr[Y > 0] \geq \mu \, \frac{(\alpha - \bar{\eta}_L/(q-1))^2}{1 + \bar{\eta}_L}, \qquad \mu = q^{h-K}, \tag{60}$$

whenever $\alpha > \bar{\eta}_L/(q-1)$*. Hence the expected number of independent trials to an accepted root at which the full restricted residual Jacobian has rank* h *is at most the reciprocal of the right-hand side.*

The theorem does not require that the root be unique, and it assumes no independence among acceptance, Jacobian rank, the unused residual equations, or the polynomial map.

Proof. For a nonzero direction $y \in L$, the condition $D_L g(x)[y] = 0$ is an affine system in x whose linear part is $T_y : x \mapsto \beta(x, y)$. It therefore has at most $q^{\dim U - \operatorname{rank} T_y}$ solutions. Union-bound over projective directions $[y] \in \mathbb{P}(L)(\mathbb{F}_q)$ to obtain

$$\frac{|\{x \in U : \operatorname{rank} D_L g(x) < h\}|}{|U|} \leq \sum_{[y] \in \mathbb{P}(L)(\mathbb{F}_q)} q^{-\operatorname{rank} T_y} = \frac{\bar{\eta}_L}{q-1}. \tag{61}$$

Averaging over the uniform fiber–coset and residual target exactly as in Theorem 20 gives

$$\mathbb{E}Y \geq \mu \left(\alpha - \frac{\bar{\eta}_L}{q-1} \right).$$

Moreover $Y(Y-1) \leq Z(Z-1)$, where Z counts all residual roots in the trial, so Theorem 18 gives $\mathbb{E}[Y(Y-1)] \leq \mu \bar{\eta}_L$. Since $Y \leq Z$ and $\mathbb{E}Z = \mu$,

$$\mathbb{E}[Y^2] = \mathbb{E}Y + \mathbb{E}[Y(Y-1)] \leq \mu(1 + \bar{\eta}_L).$$

Cauchy–Schwarz in the form $\Pr[Y > 0] \geq (\mathbb{E}Y)^2 / \mathbb{E}[Y^2]$ proves (60). ☐

Total variation distance is the largest discrepancy that the two distributions can assign to any event. Thus, when $\bar{\eta}$ is tiny, an experiment that plants a known root samples essentially the same target–coset distribution as a uniformly chosen trial conditioned on being solvable. This justifies using planted instances to probe solver behavior; it does not remove the separate finite-size-to-production extrapolation.

Corollary 22 (A one-rank cushion suffices)**.** *Let*

$$r_{\min}(L) = \min_{y \in L \setminus \{0\}} \operatorname{rank} T_y.$$

If $r_{\min}(L) \geq h + c$ for an integer $c \geq 1$, then

$$\bar{\eta} \leq (q^h - 1) q^{-(h+c)} < q^{-c}. \tag{62}$$

Consequently, the expected number of independent target–slice trials until a unique base-field root is at most

$$\frac{q^{K-h}}{1 - q^{-c}}. \tag{63}$$

For the headline choice $q = 19$ and $c = 1$, the extra exponent over q^{K-h} is at most

$$\log_2 \frac{19}{18} = 0.0780026\ldots \ \textit{bits}.$$

Proof. There are $q^h - 1$ nonzero vectors in L. Substitute $r(y) \geq h + c$ into the definition of $\bar{\eta}$, and then apply (54). ☐

Corollary 23 (Full cross-polar rank)**.** *The following are equivalent:*

(i) $e_L(b) = h$ *for every* $b \neq 0$*;*

(ii) T_y *is surjective for every* $y \in L \setminus \{0\}$*.*

Under these conditions,

$$\eta_0 = \bar{\eta} = \mu - q^{-K}.$$

If $h = K - \tau$ with $\tau \geq 0$, the expected number of trials until a unique base-field root is at most

$$\mathsf{Retry}_{K,\tau} := \frac{q^\tau}{1 - q^{-\tau} + q^{-K}}. \tag{64}$$

Proof. If some T_y is not surjective, a nonzero b annihilates its image, so the transpose of $\Phi_{b,L}$ has the nonzero kernel vector y and $e_L(b) < h$. The converse is the same argument in reverse. Under either equivalent condition, $r(y) = K$ and $\kappa(y) = 1$ for every nonzero y, giving $\eta_0 = \bar{\eta} = (q^h - 1)q^{-K} = \mu - q^{-K}$. Substitute this value into (54). $\square$

Full cross-polar rank gives the slightly sharper factor $\mathsf{Retry}_{K,\tau}$, but the headline does not rely on it. In the running Level-I example, $h = 40$, $K = 50$, and $\tau = 10$. The weaker global condition $r_{\min}(L) \geq 41$ gives at most $19^{10} \cdot 19/18 = 2^{42.56}$ target–slice trials. The former full-rank assumption required rank 50 in every nonzero direction; the new condition requires only rank 41.

Equation (47) distinguishes sparse rank defects from a pervasive loss. If at most T nonzero coefficient vectors b are defective and each has defect $h - e_L(b) \leq \Delta$, then

$$\bar{\eta} \leq \mu - q^{-K} + Tq^{-K}(q^{\Delta} - 1). \tag{65}$$

If defects are counted projectively, T in this formula must include all $q - 1$ nonzero scalar multiples of each projective direction. Under full cross-polar rank and $q = 19$, (53) is below $2^{-43.47}$ for $\tau = 10$ and below $2^{-60.47}$ for $\tau = 14$; the corresponding total-variation bounds are below $2^{-42.47}$ and $2^{-59.47}$.

Proposition 24 (Exact averaging over random A-linear slices)**.** *Let $A \cong \mathbb{F}_{q^\ell}$, let $H \subseteq V$ be an A-linear space of A-dimension D, and choose L uniformly from the A-linear subspaces of H having A-dimension s. Define $r(y) = \operatorname{rank} T_y$ using the full residual domain V. Then*

$$\mathbb{E}_L[\bar{\eta}_L] = \frac{q^{\ell s} - 1}{q^{\ell D} - 1} \sum_{y \in H \setminus \{0\}} q^{-r(y)}. \tag{66}$$

If a fresh L is sampled for each trial and the right-hand side is $\eta_{\mathrm{av}} < 1$, then the average probability of a unique base-field root is at least $q^{\ell s - K}(1 - \eta_{\mathrm{av}})$.

Proof. Every fixed nonzero $y \in H$ belongs to the same fraction of the s-dimensional A-subspaces, namely $(q^{\ell s} - 1)/(q^{\ell D} - 1)$. Average the definition of $\bar{\eta}_L$ term by term. The unique-root statement follows by averaging (51) over L. $\square$

This identity gives a second exact certification route. A small exceptional rank-defect locus in the stable kernel can be harmless even when it prevents a worst-direction theorem for one fixed slice. The proposition concerns retry probability only; a randomly selected slice must still pass the public restricted-rank and rejection preflights before its special-XL cost model is used.

Corollary 25 (Most random slices are good)**.** *In the setting of Proposition 24, for every $\varepsilon > 0$,*

$$\Pr_L[\bar{\eta}_L > \varepsilon] \leq \frac{\eta_{\mathrm{av}}}{\varepsilon}. \tag{67}$$

In particular, if $\eta_{\mathrm{av}} \leq \delta\varepsilon$, at least a $1 - \delta$ fraction of the A-linear slices satisfy $\bar{\eta}_L \leq \varepsilon$.

Proof. This is Markov's inequality applied to the nonnegative random variable $\bar{\eta}_L$ and the exact expectation in (66). $\square$

6.1 A random-XOF theorem for the structured rank spectrum

In the pinned $q = 19$ public expansion, each generated field symbol is a distinct XOF output byte reduced modulo 19 [25]. In the random-XOF model these bytes are independent and uniform; hence every symbol has maximum point probability

$$\rho = \max_{a\in\mathbb{F}_{19}} \Pr[X \bmod 19 = a] = \frac{14}{256}. \tag{68}$$

The slight modulo bias is retained rather than replaced by a uniform-field heuristic. The theorem below is information-theoretic under this idealization; it does not assert literal independence among bytes generated from one fixed SHAKE seed.

Write $a \odot b$ for the class of $a \otimes b$ in a symmetric square.

Lemma 26 (Cross-tensor injection)**.** *Let U be an A-vector space, let $0 \neq y \in U$, and let $0 \neq c \in \mathrm{Sym}^2_{\mathbb{F}_q}(A)$. If $c = \sum_\nu a_\nu \odot b_\nu$, define the linear map*

$$\Xi_{c,y}(x) = \sum_\nu \big((a_\nu y) \odot (b_\nu x) + (b_\nu y) \odot (a_\nu x)\big) \in \mathrm{Sym}^2_{\mathbb{F}_q}(U). \tag{69}$$

When q is odd,

$$\ker \Xi_{c,y} \subseteq Ay. \tag{70}$$

Proof. If $x \notin Ay$, the A-lines Ay and Ax are disjoint. Project $\mathrm{Sym}^2(U)$ onto the cross summand between these two lines and identify that summand with $Ay \otimes Ax$. Under the identifications $Ay \cong A \cong Ax$, the projection of (69) is the ordinary symmetrization of c. Symmetrization is injective on $\mathrm{Sym}^2(A)$ in odd characteristic: diagonal tensors acquire the nonzero factor 2, and the off-diagonal symmetric tensors remain distinct. Since $c \neq 0$, the projection is nonzero. Thus $\Xi_{c,y}(x) \neq 0$ whenever $x \notin Ay$. □

Theorem 27 (Decoupled seeded-spectrum bound)**.** *Assume the random-XOF model above and the reserved-line construction of Section 5.2. Suppose* $\operatorname{rank} Q_R = K$*,* $\operatorname{rank} C_{R,\mathbf{v}} = M - K$*, and the $\mathbb{F}_q$-linear slice $L \subseteq \ker C_{R,\mathbf{v}}$ has base-field dimension h and passes (32). Put*

$$D_0 = V' \cap \ker C_{R,\mathbf{v}}, \qquad t_0 = \dim_{\mathbb{F}_q} D_0 - \ell, \qquad t = \ell(v-2) - (M-K). \tag{71}$$

Then $t_0 \geq t$, and, over the fresh internal coefficients $P_{11,i}|_{V'\times V'}$,

$$\Pr[b \in \ker T_y^{\mathsf{T}}] \leq \rho^t \qquad (0 \neq y \in L,\ 0 \neq b \in \mathbb{F}_q^K), \tag{72}$$

$$\mathbb{E}[\bar\eta_L] \leq (q^h - 1)(q^{-K} + \rho^t), \tag{73}$$

$$\Pr[\bar\eta_L > \varepsilon] \leq \frac{(q^h-1)(q^{-K}+\rho^t)}{\varepsilon} \qquad (\varepsilon > 0). \tag{74}$$

More sharply, after subtracting the unavoidable full-rank baseline,

$$\Pr\Big[\bar\eta_L > (q^h-1)q^{-K} + \varepsilon\Big] \leq \frac{(q^h-1)(1-q^{-K})\rho^t}{\varepsilon} \qquad (\varepsilon > 0). \tag{75}$$

In particular,

$$\Pr[\bar\eta_L > q^{-1}] \leq q(q^h-1)(q^{-K}+\rho^t). \tag{76}$$

If a list of at most J candidates is fixed from the cross data and the first candidate passing all later full-key preflights is selected, the probability that the selected candidate violates either tail is at most J times the corresponding right-hand side. The slice need not be A-linear; A-linearity is required later only by the multigraded XL and eigenblock solvers.

Proof. The dimension bound follows from

$$\dim D_0 \geq \dim V' - (M-K) = \ell(v-1) - (M-K).$$

Fix $y \neq 0$ and $b \neq 0$. By Lemma 1, the canonical residual map is z_{sym} plus an affine term. Therefore the nonzero residual output coefficient b is itself a nonzero tuple $(c_1, \dots, c_{m_1}) \in \bigoplus_i \mathrm{Sym}^2(A)$ in the fixed quotient-feature basis. Choose i with $c_i \neq 0$. The projection $\bar{y} = \pi_{V'}(y)$ is nonzero by (32).

For $x \in D_0$, the coefficient vector of the fresh symmetric matrix $P_{11,i}|_{V' \times V'}$ in the scalar polar equation $b \cdot T_y(x) = 0$ is precisely the tensor $\Xi_{c_i,\bar{y}}(x)$. Contributions involving the W component of y or the public-oil component use only the already fixed cross data; P_{22} does not occur because x is supported on V'. By Lemma 26, the kernel of this coefficient map on D_0 has dimension at most ℓ. Its rank is therefore at least $t_0 \geq t$.

Select t test vectors in D_0 whose coefficient vectors are independent. The event $b \in \ker T_y^{\mathsf{T}}$ implies the resulting rank-t affine system in the independent upper-triangular $P_{11,i}|_{V' \times V'}$ symbols. After row reduction, condition on all nonpivot symbols. Each pivot symbol is forced to one value and has probability at most ρ; independence gives probability at most ρ^t. This proves (72).

Finally,

$$q^{-\operatorname{rank} T_y} = q^{-K} |\ker T_y^{\mathsf{T}}| = q^{-K} \left(1 + \sum_{b \neq 0} \mathbf{1}_{\{b \in \ker T_y^{\mathsf{T}}\}} \right).$$

Taking expectation and summing the indicators gives both

$$\mathbb{E}[q^{-\operatorname{rank} T_y}] \leq q^{-K} + \rho^t \quad \text{and} \quad \mathbb{E}[q^{-\operatorname{rank} T_y} - q^{-K}] \leq (1 - q^{-K})\rho^t.$$

Sum over the $q^h - 1$ nonzero $y \in L$ to obtain (73). Markov's inequality applied first to $\bar{\eta}_L$ and then to its nonnegative excess above $(q^h - 1)q^{-K}$ gives (74) and (75). For a bounded candidate list, a bad selected candidate implies that at least one member of the fixed list is bad, so the union bound applies even when the later exact preflights use the full key. □

Corollary 28 (Numerical bad-spectrum bounds)**.** *For the six structured Version 2.3 rows, the lower bound t and the resulting probabilities are as follows. A slash in the t column gives the values for the two shapes at that level. "Compact" refers to Table 11; "robust" refers to Table 12.*

Level	t	*compact* (s,τ)	$\log_2 \Pr[\bar{\eta} > 1/19]$	*robust* (s,τ)	$\log_2 \Pr[\bar{\eta} > 1/2]$
I	74/74	$(9, 14)$	< -55.22	$(10, 10)$	< -41.48
III	110/114	$(10, 30)$	< -123.19	$(11, 26)$	< -109.45
V	138/146	$(11, 46)$	< -191.16	$(12, 42)$	< -177.41

For the compact Level-I row, even a cross-data-generated list of 2^{20} candidates leaves the union bound below $2^{-35.22}$. These probabilities are under that idealized key distribution. They do not replace a deterministic certificate when a claim is made about one already fixed public seed.

Proof. Substitute $q = 19$, $\rho = 14/256$, and the public dimensions into (74). The exact high-precision evaluations are included in the companion artifact. □

Corollary 29 (Full-dimensional fresh ordinary slices)**.** *For an $\ell = 4$ row, let $D_0 = V' \cap \ker C_{R,\mathbf{v}}$. The rank of $C_{R,\mathbf{v}}$ gives the deterministic lower bound*

$$\dim_{\mathbb{F}_{19}} D_0 \geq 4(v-1) - (M-K). \tag{77}$$

For all six public rows this lower bound is at least K. Hence the attacker can select, from cross data alone, a K-dimensional ordinary $\mathbb{F}_{19}$-linear slice $L \subseteq D_0$. Put

$$B_K = 1 - 19^{-K} + \frac{1}{19}.$$

In the random-XOF idealization, $\bar{\eta}_L \le B_K$ except with the probabilities in the final column below. The bound uses (75) with $\varepsilon = 1/19$.

Level	*parameters*	K	$\dim D_0 \ge$	$t \ge$	$\log_2 \Pr[\bar{\eta}_L > B_K]$
I	$(28,5,4,4)$	*50*	*78*	*74*	< -93.61
I	$(28,4,4,5)$	*50*	*78*	*74*	< -93.61
III	$(40,7,4,4)$	*70*	*114*	*110*	< -159.59
III	$(38,5,4,5)$	*70*	*118*	*114*	< -176.36
V	$(50,9,4,4)$	*90*	*142*	*138*	< -192.02
V	$(52,6,4,6)$	*90*	*150*	*146*	< -225.56

For the worst accepted density at each level, Theorem 21 therefore gives an accepted full-Jacobian root in one target–coset trial with probability at least 0.43163*,* 0.43153*, and* 0.43145*, respectively. The corresponding expected trial counts are below* 2.318*. These ordinary slices are used by the complete-square homotopy route; they need not preserve the A-linear grading required by special XL.*

Proof. The intersection inequality $\dim(V' \cap \ker C) \ge \dim V' - \operatorname{rank} C$ gives (77). The table substitutes the six public dimensions, uses $t = \dim D_0 - 4$, and applies (75). Because $h = K$, the deterministic baseline is $(19^K - 1)19^{-K} = 1 - 19^{-K}$. The final root probabilities follow from (60) with $\bar{\eta}_L = B_K$ and the exact square-row acceptance lower bounds of Section 5.3. □

Proposition 30 (Exact projective rank-histogram certificate)**.** *Choose an $\mathbb{F}_q$-basis $y_0, \ldots, y_{h-1}$ of a fixed slice L and write*

$$T(Y) = \sum_{i=0}^{h-1} Y_i T_{y_i} \in \mathbb{F}_q[Y_0, \ldots, Y_{h-1}]^{K \times N}.$$

For $0 \le j < h$, use the disjoint normalized projective chart $Y_0 = \cdots = Y_{j-1} = 0$, $Y_j = 1$, put $R_j = \mathbb{F}_q[Y_{j+1}, \ldots, Y_{h-1}]$, and let $T^{(j)}$ be the resulting matrix over R_j. For $t \ge 0$, define

$$I_{j,t} = \left\langle Y_i^q - Y_i \ (i > j), \quad \text{all } (t+1) \times (t+1) \text{ minors of } T^{(j)} \right\rangle \subseteq R_j. \tag{78}$$

Set $n_{j,\le t} = \dim_{\mathbb{F}_q}(R_j / I_{j,t})$, put $n_{j,\le -1} = 0$, and define

$$N_t = \sum_{j=0}^{h-1} (n_{j,\le t} - n_{j,\le t-1}). \tag{79}$$

Then N_t is exactly the number of projective directions $[y] \in \mathbb{P}(L)(\mathbb{F}_q)$ with $\operatorname{rank} T_y = t$, and

$$\bar{\eta}_L = (q-1) \sum_t N_t q^{-t}. \tag{80}$$

Consequently, the quotient dimensions in (79) give a finite, independently checkable certificate of the exact aggregate collision mass. In particular, $\operatorname{rank} T_y \ge r+1$ for every $y \ne 0$ if and only if every $I_{j,r}$ is the unit ideal.

The same construction applied to the polar matrix pencil indexed by $b \in \mathbb{F}_q^K$ gives exact projective counts N_t^{pol} and hence

$$\Theta_{\mathrm{pol}} = (q-1)\sum_t N_t^{\mathrm{pol}} q^{-t}, \qquad \Psi_{\mathrm{pol}} = (q-1)\sum_t N_t^{\mathrm{pol}} q^{-t/2}. \tag{81}$$

A certificate may consist of reduced Gröbner bases and their standard monomial sets, or of polynomial reduction transcripts proving the unit-ideal claims and quotient dimensions; verification needs no probabilistic sampling.

Proof. The chosen charts contain exactly one representative of each projective $\mathbb{F}_q$-direction. The field equations identify

$$R_j/\langle Y_i^q - Y_i : i > j\rangle \cong \prod_{a \in \mathbb{F}_q^{h-j-1}} \mathbb{F}_q$$

by evaluation. Every ideal in this product of fields is radical, and its quotient dimension is the number of coordinates on which all of its generators vanish. The minors in (78) vanish exactly when $\operatorname{rank} T^{(j)}(a) \le t$. Thus $n_{j,\le t}$ counts the rank-at-most-t points in chart j, and differencing gives (79). Every projective direction has $q-1$ nonzero scalar representatives and rank is invariant under base-field scaling, which proves (80). The polar formulas are identical. □

Proposition 31 (A fully priced exhaustive spectrum computation)**.** *Let $P_h = (q^h-1)/(q-1)$ be the number of projective directions in an h-dimensional slice, and suppose $K \le N$. The exact histogram in Proposition 30 can also be computed directly, without Gröbner bases, in at most*

$$4P_hK^2N \tag{82}$$

base-field operations and $O(hKN)$ stored base-field elements. The algorithm enumerates the disjoint normalized charts, updates T_y along a q-ary Gray code, row-reduces each $K \times N$ matrix, and accumulates its rank.

For the model-state profiles in Table 10, charging the paper's $b_1 = 22.29$ bit operations per base-field operation gives the following worst-case exponents, rounded upward to the displayed precision:

Level	(h, K, N)	*exact spectrum*	*structured solve*	*combined*
I	$(32, 50, 102)$	156.21	140.13	156.21
III	$(36, 70, 146)$	174.69	205.47	205.47
V	$(44, 90, 182)$	209.71	262.53	262.53

Here "combined" is the logarithm of the sum of the certification and solve work. Thus, for the displayed Level-III and Level-V low-state attacks, an exact favorable value of $\bar{\eta}_L$ is not an unpriced production step: the complete enumeration is dominated by the modeled solve. At Level I, direct enumeration is too expensive for a below-143 claim, so the structured certificate of Proposition 30 or another proof remains necessary. The computation can of course return an unfavorable spectrum; the proposition prices certification, not the probability that a randomly selected slice passes it.

Proof. Precompute the h coefficient matrices T_{y_i}. A standard q-ary Gray code changes one coordinate at a time, so forming the next matrix costs at most KN additions. Gaussian elimination on a $K \times N$ matrix uses fewer than K^2N multiply-add pairs and KN normalizations. Counting each addition, multiplication, or division separately and allowing slack for chart changes gives (82). Substitution of the displayed parameters gives the table; the machine-readable artifact performs the integer and logarithmic calculations. □

Corollary 32 (Full-space target availability)**.** *Let $g : \mathbb{F}_q^N \to \mathbb{F}_q^K$ have degree at most two. Let r_* be the minimum polar rank of a nonzero output combination, and put $\alpha = 1 - q^{-K}$. For a uniform target,*

$$\Pr[g^{-1}(z) \neq \varnothing] \geq \frac{1}{1 + \alpha q^{K-r_*}}. \tag{83}$$

Hence the expected number of target trials is at most $1 + \alpha q^{K-r_*}$. If $r_* \geq 2K$, every target has a root.

Proof. See Section A.6. □

Corollary 33 (Aggregate full-space rank spectra)**.** *For $b \in \mathbb{F}_q^K \setminus \{0\}$, let $\rho(b)$ be the polar rank of the scalar quadratic $b \cdot g$, and put*

$$\Theta_{\mathrm{pol}}(g) = \sum_{b \neq 0} q^{-\rho(b)}, \qquad \Psi_{\mathrm{pol}}(g) = \sum_{b \neq 0} q^{-\rho(b)/2}. \tag{84}$$

For a uniform target z,

$$\Pr[g^{-1}(z) \neq \varnothing] \geq \frac{1}{1 + \Theta_{\mathrm{pol}}(g)}. \tag{85}$$

More generally, if $\mathcal{X} \subseteq \mathbb{F}_q^N$ has density a, then

$$\Pr[g^{-1}(z) \cap \mathcal{X} \neq \varnothing] \geq \frac{a}{1 + \Psi_{\mathrm{pol}}(g)}. \tag{86}$$

Thus the expected target counts are at most $1 + \Theta_{\mathrm{pol}}(g)$ and $(1 + \Psi_{\mathrm{pol}}(g))/a$, respectively. These bounds depend on the complete rank spectra, not their minima; a small exceptional low-rank locus can be charged exactly rather than controlling the whole pencil.

Proof. Take $L = V$ in Theorem 18. Then $e_V(b) = \rho(b)$, so (49) is exactly (85). For the second statement, the quadratic character-sum bound gives, for every target,

$$|g^{-1}(z)| \leq q^{N-K} \left(1 + \sum_{b \neq 0} q^{-\rho(b)/2}\right) = q^{N-K}(1 + \Psi_{\mathrm{pol}}(g)).$$

Double-counting the aq^N points of $\mathcal{X}$ by their images shows that at least $aq^K/(1 + \Psi_{\mathrm{pol}}(g))$ targets have a preimage in $\mathcal{X}$, proving (86). □

Corollary 34 (Small expected-target cushions)**.** *For every integer $c \geq 1$,*

$$\Theta_{\mathrm{pol}}(g) \leq q^{-c} \quad \Longrightarrow \quad \mathbb{E}[\text{uniform targets to a nonempty fiber}] \leq 1 + q^{-c}, \tag{87}$$

$$\Psi_{\mathrm{pol}}(g) \leq q^{-c} \quad \Longrightarrow \quad \mathbb{E}[\text{uniform targets to an accepted root}] \leq \frac{1 + q^{-c}}{a_n}. \tag{88}$$

The convenient minimum-rank conditions

$$r_* \geq K + c \quad \Longrightarrow \quad \Theta_{\mathrm{pol}}(g) < q^{-c}, \tag{89}$$

$$r_* \geq 2K + 2c \quad \Longrightarrow \quad \Psi_{\mathrm{pol}}(g) < q^{-c} \tag{90}$$

are sufficient, but not necessary. In particular, at $q = 19$ and $c = 1$, either aggregate condition adds less than $\log_2(20/19) < 0.075$ bits. For the Level-V filter, $n = 128$, $K = 96$, and

$$1 - a_{128} \leq \binom{128}{33} 19^{-33} < 2^{-38.4},$$

so $\Psi_{\mathrm{pol}}(g) \le 19^{-1}$*—in particular,* $r_* \ge 194$*—makes the total expected-target overhead less than* 0.075 *bits as well.*

Proof. Apply Corollary 33. Moreover,

$$\Theta_{\mathrm{pol}}(g) \le (q^K - 1)q^{-r_*} < q^{K-r_*}, \qquad \Psi_{\mathrm{pol}}(g) \le (q^K - 1)q^{-r_*/2} < q^{K-r_*/2},$$

which proves the minimum-rank implications. The final estimate is the union bound for at least 33 zero hyperplane coordinates among 128 independent blocks. □

For a structured $\ell = 4$ lift, an invertible output transformation separates the $M - K$ affine coordinates from the remaining K coordinates of a uniform target. Conditioning on the affine coordinates fixes a base point; the other coordinates remain uniform, and translation does not change the polar maps. We take $V = \ker C_{R,\mathbf{v}}$ and let L be the selected stable slice, of dimension $h = \ell s = K - \tau$. The root ledger uses the aggregate bound $\bar{\eta}_L \le q^{-1}$, so (54) bounds the unique-root retry factor by $q^\tau/(1 - q^{-1})$. The minimum-rank condition $r_{\min}(L) \ge h + 1$ is sufficient, and full cross-polar rank would improve the exponent by less than 0.079 bits. The bound can now be justified in three distinct ways: the random-XOF idealized-expansion theorem Theorem 27, a fixed-key projective histogram, or the random A-slice average of Proposition 24. Reaching the degree at which the solver exposes a finite quotient remains a separate condition.

For the fixed-skew $\ell = 2$ lift, take $L = V$ to be the complete residual space. By Proposition 15, every root is accepted. The headline uses $\Theta_{\mathrm{pol}} \le 19^{-1}$ from Corollary 33, which adds less than 0.075 bits of expected target search. The minimum-rank values $r_* \ge K + 1$, namely 49, 73, and 97 at Levels I, III, and V, are sufficient but not necessary. The stronger condition $r_* \ge 2K$ makes every target solvable. Under the AXN conversion, the nominal arithmetic gaps remain positive for global ranks at least 46, 68, and 89. These budget and all-target minimum ranks concern the converted operation model, not a complete solver circuit; the sampled evidence in Section 8 is not a certificate. The Level-V filter attack instead uses the accepted-root version of Corollary 34.

Theorem 35 (End-to-end Las Vegas wrapper)**.** *Fix a public key and an attacker-chosen relation that pass the exact affine and rejection preflights. Suppose a residual solver has the following certified behavior: when it reports success, it returns a finite candidate set containing all base-field roots of that residual system; every returned point is then checked by the original public verifier. Let* $\sigma \in (0, 1]$ *be a lower bound on the solver's conditional probability of producing such a completeness certificate, conditioned on the root event stated below.*

(a) For a structured slice L *of dimension* h*, let* a *be a lower bound on the density of assignments accepted by the original verifier in the full common-column space. If* $\bar{\eta}_L < a$*, the expected number of target–slice solver trials before a certified valid signature is at most*

$$\frac{q^{K-h}}{(a - \bar{\eta}_L)\sigma}. \tag{91}$$

Here the conditioning event for σ *is that the trial has exactly one base-field root and that root is accepted. Take* $a = 1$ *for the rectangular rows and* $a = (1 - q^{-4})^n$ *for the square* $\ell = 4$ *rows.*

(b) For a full-space fixed-skew lift, if the aggregate polar spectrum is Θ_{pol}*, the expected number of target–solver trials is at most*

$$\frac{1 + \Theta_{\mathrm{pol}}}{\sigma}. \tag{92}$$

Here the conditioning event is that the target fiber is nonempty.

(c) *For a filtered lift whose accepted assignments have density a, if the aggregate square-root spectrum is Ψ_{pol}, the expected number of target–solver trials is at most*

$$\frac{1+\Psi_{\mathrm{pol}}}{a\sigma}. \tag{93}$$

Here the conditioning event is that the target has an accepted root.

In every case the algorithm is Las Vegas: it may restart, but it never returns an invalid signature.

Proof. For the structured case, the invertible target split of Lemma 1 makes the affine right-hand side and residual target independent and uniform. Theorem 20 then gives an accepted unique-root probability at least $q^{h-K}(a - \bar{\eta}_L)$. Multiplying by the conditional certificate probability σ and inverting gives (91). The other two root-event probabilities are at least $1/(1+\Theta_{\mathrm{pol}})$ and $a/(1+\Psi_{\mathrm{pol}})$ by Corollary 33; the same calculation proves Equations (92) and (93). Completeness of a reported candidate set prevents a missed root from being mistaken for a proof of infeasibility, and the original verifier removes all algebraic or reconstruction artifacts. □

Where the numerical model enters. The compact special-XL rows set $\sigma = 1$ at the selected degree. The robust explicit rows set only $\sigma = 1/8$ and charge the resulting three bits. A transcript satisfying Theorem 37 certifies success on an individual trial. For any other measured or proved value, the correction is transparent: add $\log_2(1/\sigma)$ bits. The seeded root-spectrum theorem and the production solver-certificate rate are therefore separate inputs and cannot be silently conflated.

Certified attack procedure. The forger (i) runs the public rank and rejection preflights; (ii) samples an exactly uniform official target as described next; (iii) forms the exact residual system and, for the structured rows, a target–slice trial; (iv) runs the selected solver and accepts a solver claim only with a completeness certificate; and (v) reconstructs the full signature and invokes the unmodified verifier. A failed preflight, empty fiber, absent certificate, or failed verifier check causes a restart, never an unchecked output.

6.2 Sampling exactly uniform targets with the official conversion

The probability theorems above assume a uniform target in $\mathbb{F}_q^K$. Reducing binary hash chunks modulo 19 introduces a small bias, so uniformity cannot simply be assumed. The attacker removes it without changing the verifier: it repeatedly chooses a message–salt input, runs the official hash-to-field conversion, and keeps the input only when each underlying binary chunk lies in a range that makes the resulting base-19 digits exactly uniform.

Concretely, the official $q = 19$ conversion reads at most fifteen base-19 digits from each 64-bit chunk. If a uniform b-bit chunk X is to supply a digits, the attacker accepts that message–salt candidate only if

$$X < \left\lfloor \frac{2^b}{q^a} \right\rfloor q^a. \tag{94}$$

Conditioned on this event, $X \bmod q^a$ is uniform, so the extracted digits are jointly uniform. Applying (94) independently to every chunk makes the retained official target exactly uniform. The invertible output-coordinate change used in Section 3.2 preserves uniformity. Conditioning on the values of the $M - K$ affine-output coordinates—or on their passing values in a filtered lift—then leaves the remaining K residual target coordinates exactly uniform, which is the distribution assumed in the root theorems. This is ordinary rejection sampling performed by the forger over message–salt candidates; the verifier continues to apply the unmodified official conversion. Across all Version 2.3 lengths the acceptance probability is at least 0.152462, or fewer than 6.6 candidates on average.

The Level-V $\ell = 2$ filter needs about $19^{32} > 2^{128}$ raw targets before the $M - K = 32$ consistency conditions pass, so the 128-bit salt alone does not supply enough distinct inputs. An existential forger may vary a chosen-message prefix as well as the salt. Distinct prefix–salt pairs are distinct random-oracle inputs and therefore give independent targets in the model. The artifact's deliberately loose target-generation envelope is below 2^{32} gates per target retained by the unbiased hash-chunk sampler (the AXN charge is no larger than the NAND envelope used there). Even after multiplication by the 19^{32} filtering factor, this term is more than 70 exponent bits below the cost-driving MQ solve.

7 Concrete Costs

The algebraic reduction determines the residual MQ systems exactly. Turning those systems into numerical security estimates requires a separate cost model. We report that model in layers so that a reader can see which numbers are structural, which come from an MQ solver estimate, and which are merely a change of arithmetic units.

Table 7: How to interpret the cost accounting.

Layer	What it means
Residual system	Exact numbers of equations, variables, blocks, and public rank conditions after the reduction.
Solver model	Published generic-MQ or special-XL formula for the dominant finite-field work at those dimensions.
AXN conversion	Secondary sensitivity calculation replacing the solver formula's coarse field-operation factor by an explicit two-input AND/XOR/XNOR circuit for one signed multiply-accumulate.
Not included	A complete implementation of sparse matrix construction, pivoting and control, memory addressing and traffic, communication, or an optimized end-to-end attack netlist.

A cost exponent c means modeled work proportional to 2^c in the stated unit. A difference of ten exponent bits is a factor of about 2^{10}, not ten percent. Because the solver formulas count dominant arithmetic rather than a complete machine, any difference from a NIST reference is only a *nominal arithmetic gap*; it is not an implementation margin or a category claim.

7.1 Cost units and circuit-unit sensitivity

The cited attacks first count finite-field operations and then apply coarse bit-cost factors [3, 5]. NIST lists 2^{143}, 2^{207}, and 2^{272} as nominal classical-gate references, while also requiring category comparisons across every metric it considers relevant to practical security [21]. We retain each attack's field-operation count and replace only its dominant field factor by explicit circuits. The resulting figures sharpen one metric; they do not by themselves establish a category break. Our primary AXN basis assigns unit cost to two-input AND, XOR, and XNOR gates, with free constants, wires, fan-out, and structural sharing. This is the gate family used by the concrete AES-128 circuit on NIST's circuit list [19]. A two-input-NAND conversion provides a separate basis-sensitivity check.

Table 8: Constructive AES calibration circuits. Full charges the entire key space for the first-pair filter; expected search is one bit lower, and the lazy confirmation overhead is negligible under the stated ideal-cipher model. These circuits are not lower bounds on optimal AES implementations and do not replace NIST's nominal references.

Level	pairs	NIST nominal	AXN gates/filter	AXN full	NAND gates/filter	NAND full
I	2	143	30,081	142.88	106,616	144.70
III	3	207	34,227	207.06	121,735	208.89
V	3	272	41,543	271.34	147,772	273.17

For generic MQ, the published factor is

$$b_1 = (\log_2 19)^2 + \log_2 19 = 22.29. \tag{95}$$

Using canonical five-bit representatives and exact Barrett reduction, the validated circuits for one signed multiply-accumulate, $c + ab$ or $c - ab$, over $\mathbb{F}_{19}$ use 402 and 441 AXN gates, respectively; their NAND translations use 781 and 835 gates. To avoid assuming a particular sign schedule in elimination, we charge

$$g_1^{\mathrm{AXN}} = 441, \qquad g_1^{\mathrm{NAND}} = 835. \tag{96}$$

Both signs are checked exhaustively on all 19^3 valid input triples by the generator and by an independent scalar evaluator.

For special XL, we work in the isomorphic sparse basis $\mathbb{F}_{19}[t]/(t^4 - t - 1)$. In this basis, the element $13 + 13t + 15t^2 + 2t^3$ is a root of the official defining polynomial; evaluating its powers supplies the explicit basis change to $\mathbb{F}_{19}[S]$. Two-level Karatsuba uses nine base-field multiplications, and $t^4 = t + 1$ makes reduction additive. The validated circuits for $c \pm ab$ over $\mathbb{F}_{19^4}$ use 6,077 and 6,081 AXN gates, or 11,086 and 11,102 NAND gates. We therefore charge

$$g_4^{\mathrm{AXN}} = 6{,}081, \qquad g_4^{\mathrm{NAND}} = 11{,}102. \tag{97}$$

The artifact checks the complete scalar basis tensor, 4,096 random extension products, the basis isomorphism, and both translated netlists. Charging a signed multiply-accumulate for every multiplication in the special-XL formula is conservative.

Same-basis AES calibration. The release builds variable-key AES-128/192/256 one-pair key-filter circuits with the NIST 113-gate S-box, the complete key schedule, encryption, and a final ciphertext comparator. To identify the actual key rather than merely a key that matches one pair, surviving candidates are checked lazily on enough independent plaintext–ciphertext pairs to leave at most 2^{-128} expected wrong survivors in the ideal-cipher model: a total of two pairs for AES-128 and a total of three for AES-192 and AES-256. The additional average work is negligible because only survivors reach those checks. Every first-pair circuit is validated on the FIPS test vector and eight additional keys against an independent AES implementation.

The NIST-listed 28,600-gate AES-128 circuit has full-key-space exponent $128 + \log_2(28{,}600) = 142.80$, numerically consistent with the nominal value 143. Across our nine rows, the AXN arithmetic estimates are 9.06–53.78 bits below the constructive full-space AES circuits and 9.18–54.44 bits below the nominal NIST references. The NAND cross-check gives attack ranges $2^{133.12}$–$2^{134.74}$, $2^{175.99}$–$2^{188.74}$, and $2^{218.42}$–$2^{240.50}$.

Scope of the conversion. The AXN and NAND figures price the field-arithmetic schedule counted by the published estimators; they do not include solver control, memory traffic, or a complete sparse-linear-algebra implementation. Public preprocessing and exact-uniform target generation are bounded separately, so the dominant unresolved inputs are the global production rank spectra and the production-scale solving models.

7.2 Structured special XL for $\ell = 4$

The exact reduction supplies an A-linear slice of A-dimension s, hence base-field dimension $4s$ when $\ell = 4$. After extension to the splitting field, this becomes four blocks of s variables. For each unordered pair of blocks there are m_1 independent equations of degree two: diagonal equations use one block twice, and off-diagonal equations use two blocks once each. Special XL records a separate degree for each block and selects a multidegree at which the associated Macaulay matrix is predicted to have the needed rank. A reader need not manipulate the series below: here it is only a compact device for choosing the matrix degree and counting the resulting monomial columns.

After adding one homogenizing variable to each block, the Hilbert series used by the published special-XL model is

$$H_{m_1,s}(\mathbf{t}) = \frac{\prod_{a=1}^{4}(1-t_a^2)^{m_1}\prod_{1\le a<b\le 4}(1-t_at_b)^{m_1}}{\prod_{a=1}^{4}(1-t_a)^{s+1}}. \tag{98}$$

For $\mathbf{d} = (d_1, \ldots, d_4)$, put

$$M_s(\mathbf{d}) = \prod_{a=1}^{4}\binom{s+d_a}{d_a}. \tag{99}$$

The search chooses a multidegree whose signed Hilbert coefficient is negative. This sign change is a rank-prediction trigger in the cited special-XL model: a random Macaulay matrix is predicted to have full column rank, while a system with a planted solution is predicted to have a one-dimensional right kernel [5]. We use the same degree and row-reduction cost. Theorem 37 shows that one-dimensionality is not necessary for correctness: any certified commuting-table finite border quotient can be solved and checked. It does not make a larger kernel free. Unless extra block/nullspace work is separately charged, the displayed scalar-Wiedemann exponent retains the published $D = 1$ prediction. The published upper bound for one target–slice trial is $3s^2\big(M_s(\mathbf{d})\big)^2$ extension-field multiplications. This is a Wiedemann-style sparse-linear-algebra count, not a license to materialize a dense $M_s(\mathbf{d}) \times M_s(\mathbf{d})$ matrix [3, 5, 6]. The next proposition makes the corresponding state claim precise.

Cabarcas et al. handle an underdetermined system by specializing k coordinates, where $4 \mid k$, and multiplying the work by q^k. Our attack uses a different retry loop: it fixes a $4s$-dimensional slice and samples the target and affine coset. The headline condition $\bar{\eta}_L \le q^{-1}$ and (54) give the conservative retry factor

$$\mathsf{Retry}_\tau^+ = \frac{q^\tau}{1-q^{-1}}, \qquad \tau = K - 4s.$$

The modeled work is therefore

$$\mathsf{Retry}_\tau^+\, 3s^2 M_s(\mathbf{d})^2 \tag{100}$$

extension-field multiplications. Relative to the old full-rank factor, this adds at most $\log_2(19/18) = 0.0780026\ldots$ bits. The minimum-rank condition $r_{\min}(L) \ge 4s+1$ from Corollary 22 is one sufficient certificate. The more general spectral formulas (44) and (66) can give a smaller factor and tolerate an exceptional low-rank locus.

The computation takes place over $\mathbb{F}_{19^4}$. The published convention assigns each extension-field multiplication

$$b_4 = 2\left(\log_2 19^4\right)^2 + \log_2 19^4 = 594.43 \tag{101}$$

bit operations. The AXN arithmetic conversion replaces b_4 by the conservative signed multiply-accumulate bound g_4^{AXN} from (97).

Proposition 36 (Streaming the special-XL Macaulay operator)**.** *Let $A_{\mathbf{d}}$ be the special-XL Macaulay matrix at multidegree $\mathbf{d}$, with $M = M_s(\mathbf{d})$ columns. With the convention that a binomial coefficient with negative lower argument is zero, its row count is*

$$\begin{aligned} R_s(\mathbf{d}) = m_1 \sum_{a=1}^{4} \binom{s+d_a-2}{d_a-2} \prod_{c\neq a} \binom{s+d_c}{d_c} \\ + m_1 \sum_{1\le a<b\le 4} \binom{s+d_a-1}{d_a-1}\binom{s+d_b-1}{d_b-1} \prod_{c\notin\{a,b\}} \binom{s+d_c}{d_c}. \end{aligned} \tag{102}$$

Every row can be generated from the public quadratic coefficients and its multiplier monomial using $O(s^2)$ scratch space; no matrix entry table is needed.

Put $R = R_s(\mathbf{d})$. Let $E/\mathbb{F}_{19^4}$ be a finite extension, choose the entries of a random diagonal $\Delta \in E^{R\times R}$ independently and uniformly, and put

$$B = A_{\mathbf{d}}^{\mathsf{T}} \Delta A_{\mathbf{d}} \in E^{M\times M}. \tag{103}$$

A product Bx can be evaluated by one pass over the rows of $A_{\mathbf{d}}$, accumulating $\Delta_{jj}\langle a_j, x\rangle a_j^{\mathsf{T}}$. It uses $O(M)$ field elements of resident vector state and $O(R_s(\mathbf{d})(s+1)^2)$ field operations. If $r = \operatorname{rank} A_{\mathbf{d}}$, then

$$\Pr_{\Delta}[\ker B \neq \ker A_{\mathbf{d}}] \le \frac{r}{|E|} \le \frac{M}{|E|}. \tag{104}$$

Consequently scalar Wiedemann needs $O(M)$ field-vector state to seek a one-dimensional right kernel, and block size b needs $O(bM)$ state to seek a b-dimensional kernel, under the standard generic-projection assumptions of black-box Wiedemann. The matrix is never materialized.

For the scalar $b = 1$ realization used below, put $\Omega = \mathbb{F}_{19^4}$ and $E = \mathbb{F}_{19^8}$. If $t = (s+1)^2$, one product by B costs fewer than $5Rt$ signed Ω multiply–accumulates: an E element times an Ω scalar uses two Ω multiplications, and an E product uses three by Karatsuba. Producing the scalar Wiedemann sequence and reconstructing one null vector uses fewer than $3M$ products by B. Thus

$$16R(s+1)^2 M \tag{105}$$

signed Ω multiply–accumulates is a rounded upper bound per solver trial. A straightforward seeded implementation ledger reserves eight length-M arrays over E for the iterate, accumulator, scalar sequence, reconstruction, and polynomial-work buffers.

For the robust Level-I profile only, take instead $E' = \mathbb{F}_{19^{12}}$, a cubic extension of Ω. A sparse dot product and row accumulation use at most $6t$ Ω multiplications, and naive cubic multiplication uses nine more. Since $6t + 9 < 9t$ for $t = (s+1)^2 \ge 4$, the same three-M product ledger gives the rounded bound

$$32R(s+1)^2 M \tag{106}$$

signed Ω multiply–accumulates per trial. Storing the three Ω components in three bytes each uses nine bytes per E' coordinate; this is more conservative than bit-packing an E' element.

Proof. The two sums in (102) count multiplier monomials for, respectively, the m_1 diagonal equations of multidegree $2e_a$ and the m_1 off-diagonal equations of multidegree e_a+e_b. A homogenized quadratic row has at most $(s+1)^2$ terms, which proves the streaming row and transpose products.

For the probability bound, factor $A_{\mathbf{d}} = UV$, where $U \in E^{R\times r}$ has full column rank and $V \in E^{r\times M}$ has full row rank. If $U^\mathsf{T}\Delta U$ is invertible, then

$$\ker(A_{\mathbf{d}}^\mathsf{T}\Delta A_{\mathbf{d}}) = \ker V = \ker A_{\mathbf{d}}.$$

By Cauchy–Binet, $\det(U^\mathsf{T}\Delta U)$ is a nonzero multilinear polynomial of total degree r in the diagonal entries: a nonzero $r \times r$ minor of U contributes its squared determinant as a nonzero coefficient. The Schwartz–Zippel bound gives (104). Standard scalar and block Wiedemann recover null vectors from black-box products with linear vector storage; block variants can recover a basis under their usual random-projection conditions [7, 6, 13].

For the explicit scalar ledger, write an E element in an Ω-basis. Each sparse dot product and each row accumulation uses at most $2t$ Ω products, while multiplying the dot product by the diagonal preconditioner uses three. Since $4t+3 < 5t$ for $t \geq 4$, one normal-operator product costs less than $5Rt$. The usual scalar sequence requires fewer than $2M$ products and null-vector reconstruction fewer than M more; rounding 15 to 16 gives (105). For a cubic extension, the corresponding row cost is at most $6t+9 < 9t$; multiplying by the same three-M product count and rounding 27 to 32 gives (106). The eight-array figures are implementation reserves rather than asymptotic claims. $\square$

Theorem 37 (Recovery from a certified finite Macaulay quotient)**.** *Fix a target and affine slice, and let $I \subseteq \Omega[x_1,\ldots,x_h]$ be the ideal of its dehomogenized residual equations. Let $\mathcal{B}$ be a finite divisor-closed set of monomials containing 1, with $|\mathcal{B}| = D$, and let $\partial\mathcal{B}$ be its border. Suppose the selected Macaulay row space provides, for every $\beta \in \partial\mathcal{B}$, a certified relation*

$$g_\beta = \beta - \sum_{b\in\mathcal{B}} c_{\beta,b} b \in I. \tag{107}$$

Define $D \times D$ formal multiplication matrices $M_1,\ldots,M_h$ from these rewrite rules and from the tautological products that remain in $\mathcal{B}$. If

$$M_iM_j = M_jM_i \qquad (1 \leq i,j \leq h), \tag{108}$$

then $G = \{g_\beta : \beta \in \partial\mathcal{B}\}$ is a $\mathcal{B}$-border basis of the zero-dimensional ideal $J = (G)$, $\mathcal{B}$ is an Ω-basis of $\Omega[x]/J$, and $J \subseteq I$. Consequently $V(I) \subseteq V(J)$ and the attacker can enumerate every base-field root of the original slice system from the multiplication tables. A dense FGLM conversion costs $O(hD^3)$ operations in Ω; finite-field factorization and root extraction add randomized polynomial time in D, h, and $\log|\Omega|$. Every extra point is removed by the original equations and public verifier. The one-dimensional-kernel method is the special case $D = 1$.

Proof. The relations in (107) form a border prebasis for the order ideal $\mathcal{B}$. The commuting-multiplication-matrix characterization of border bases states that such a prebasis is a border basis exactly when its formal multiplication matrices commute [14]. For completeness, the mechanism is as follows. Commutativity makes

$$\pi : \Omega[x_1,\ldots,x_h] \longrightarrow \mathrm{End}_\Omega(\Omega^{\mathcal{B}}), \qquad x_i \longmapsto M_i,$$

an algebra homomorphism. Divisor closure gives $\pi(b)e_1 = e_b$ for every $b \in \mathcal{B}$. By construction, $\pi(g_\beta)e_1 = 0$ for every border relation. Since the matrices commute, for every $b \in \mathcal{B}$,

$$\pi(g_\beta)e_b = \pi(g_\beta)\pi(b)e_1 = \pi(b)\pi(g_\beta)e_1 = 0.$$

Thus each border relation acts as zero. The standard border-division theorem then gives unique normal forms in $\operatorname{span}_\Omega \mathcal{B}$, so $\Omega[x]/J$ has basis $\mathcal{B}$ and the M_i are its multiplication matrices. The supplied Macaulay row combinations certify $g_\beta \in I$, hence $J \subseteq I$ and therefore $V(I) \subseteq V(J)$.

FGLM converts the multiplication tables to a lexicographic Gröbner basis in $O(hD^3)$ field operations [8]. Standard zero-dimensional solving over finite fields then enumerates the geometric points of J; inverse eigenblock and affine transformations reconstruct the complete signature. Checking membership in the base field, the original slice equations, and the public verifier makes the procedure Las Vegas. $\square$

How much quotient dimension is already priced. The per-trial row-reduction term in (100) is $3s^2 M_s(\mathbf{d})^2$. FGLM has arithmetic complexity $O(hD^3)$, but the hidden universal constant should not silently be set to one. We therefore report two break-even scales. The constant-free scale is

$$D_{\text{raw}} = \left\lfloor \left(\frac{3s^2 M_s(\mathbf{d})^2}{h} \right)^{1/3} \right\rfloor. \tag{109}$$

For a deliberately extravagant implementation reserve, we also charge $2^{30} hD^3$ extension-field operations and put

$$D_{30} = \left\lfloor \left(\frac{3s^2 M_s(\mathbf{d})^2}{2^{30} h} \right)^{1/3} \right\rfloor. \tag{110}$$

Any extraction implementation costing at most $2^{30} hD^3$ is arithmetically subsumed by the modeled row reduction whenever $D \leq D_{30}$. These bounds answer only whether *postprocessing already-supplied multiplication tables* changes the leading arithmetic exponent. They do not price obtaining a D-dimensional nullspace, constructing all border relations, or the extra block Wiedemann work when $D > 1$. They are also *not* storage-feasibility claims: dense multiplication tables require $\Theta(hD^2)$ field elements, and every real run must impose an explicit memory cap (or certify a sparse/streaming representation). Thus $D > 1$ is a correctness fallback, not a free extension of the displayed scalar cost. The arithmetic break-even values are:

Table 9: Arithmetic-only break-even dimensions for FGLM postprocessing once multiplication tables are available. D_{raw} ignores the fixed implementation constant; D_{30} reserves a factor 2^{30}. The table does not price block/nullspace construction or the $\Theta(hD^2)$ storage of dense multiplication tables.

m_1	profile	h	D_{raw}	D_{30}	$\log_2 D_{30}$
5	time minimum	40	16,021,697	15,646	13.93
5	model-state	32	110,695	108	6.75
5	AXN-state	36	1,151,871	1,124	10.13
7	time minimum	60	280,271,069,473	273,702,216	28.03
7	state minimum	40	370,128	361	8.50
9	time minimum	76	92,610,866,101,092	90,440,298,926	36.40
9	state minimum	44	224,424	219	7.77

Las Vegas wrapper. For each target–slice trial, row-reduce the Macaulay matrix. If a one-dimensional kernel appears, use the direct normalization method. More generally, continue when the row space supplies the border relations in Theorem 37, construct the quotient multiplication matrices, and enumerate its roots. Discard every point that fails base-field descent or the original verifier. Failure to obtain a complete finite quotient at that degree causes a retry or a controlled increase of the multidegree; it can never cause acceptance of an invalid signature.

Precisely what the displayed special-XL costs assume. The exponents in Table 10 use (i) the aggregate condition $\bar{\eta}_L \leq 19^{-1}$, which gives the factor Retry_τ^+, and (ii) the published Hilbert-series prediction that a successful selected-degree trial has a one-dimensional right kernel. That is the case priced by scalar Wiedemann. The finite-quotient theorem proves zero-error correctness for $D > 1$, and Table 9 shows that subsequent FGLM work can be small, but neither statement prices the additional block/nullspace work needed to obtain all D border relations. Accordingly a production $D > 1$ transcript must charge that work and its success probability separately. What remains unproved is the selected-degree $D = 1$ prediction (or a fully priced larger-D replacement).

Before stating the frontier, note that a block degree $d_a = 1$ does not remove block a or any of its affine variables. Its monomial columns contain the homogenizer and every block variable. It only means that no multiple of a diagonal equation of multidegree $2\mathbf{e}_a$ fits in that target component; (102) assigns zero rows to that unavailable sector. Off-diagonal equations still constrain the block, and the same signed Hilbert coefficient and border-certificate tests apply. Degree one is therefore the natural minimum for a recovery matrix containing every variable.

Table 10: Certified points on the special-XL time–state frontier for the six $\ell = 4$ rows. "One vector" uses three bytes per $\mathbb{F}_{19^4}$ coordinate. The model-state rows globally minimize M among negative-coefficient profiles with $d_i \geq 1$ and published-model exponent at most 143/207/272. The AXN-state rows are the corresponding minima under the secondary AXN bounds; at Level V one row satisfies both.

m_1	profile	(s, τ)	$\mathbf{d}$	$\log_2 M$	R/M	Model	AXN	one vector
5	time minimum	$(10, 10)$	$(3, 3, 3, 4)$	34.45	2.69	128.90	132.25	65.4 GiB
5	model-state	$(8, 18)$	$(1, 2, 3, 3)$	23.39	1.99	140.13	143.49	31.5 MiB
5	AXN-state	$(9, 14)$	$(2, 2, 3, 4)$	28.83	2.36	134.34	137.70	1.33 GiB
7	time minimum	$(15, 10)$	$(4, 4, 6, 6)$	55.30	4.06	171.76	175.12	118 PiB
7	model-state	$(9, 34)$	$(1, 1, 2, 4)$	21.91	1.81	205.47	208.82	11.3 MiB
7	AXN-state	$(10, 30)$	$(1, 2, 3, 3)$	25.82	1.96	196.61	199.97	170 MiB
9	time minimum	$(19, 14)$	$(5, 6, 6, 6)$	67.68	4.58	214.20	217.56	614 EiB
9	model/AXN-state	$(11, 46)$	$(1, 2, 2, 3)$	24.66	1.73	262.53	265.89	76.0 MiB

Proposition 38 (Certified low-state frontier)**.** *Restrict only to multidegrees with $d_i \geq 1$, so every variable block is represented. Over all slice dimensions available to both $\ell = 4$ parameter sets at each level and all multidegrees with negative coefficient in* (98)*, the three model-state rows of Table 10 globally minimize $M_s(\mathbf{d})$ subject to published-model exponents at most* 143, 207, *and* 272*. The three AXN-state rows globally minimize $M_s(\mathbf{d})$ subject to AXN arithmetic exponents at most the same references.*

Proof. The exact finite search and monotone tail certificate are given in Section C and in the machine-readable frontier artifact. ☐

The original exact search checks every admissible s and all 42,356 sorted multidegrees whose modeled work could match or beat the time incumbent. It recovers the three time minima in Table 10, uniquely up to block permutation. A second exact search proves Proposition 38. The model-state rows lower the simple sufficient cross-polar ranks from 41/61/77 to 33/37/45; the Level-I and Level-III AXN-state rows require ranks 37 and 41. For the three model-state rows, random diagonal preconditioning over $E = \mathbb{F}_{19^8}$ fails with probability at most 0.00065, 0.00024, and 0.00157 by (104). The Level-I and Level-III AXN-state bounds are 0.0281 and 0.00350 and can be reduced geometrically by repetition. The existing full-size rank campaign tested the time-minimizing slices, not these smaller ones. The reserved-line construction therefore selects each low-state A-subspace from cross data and applies Theorem 27; a fixed key may additionally publish the exact histogram.

Thus a below-reference point need not carry an astronomical state vector. The table's three-byte figures describe vectors over $\mathbb{F}_{19^4}$ and isolate the column-count frontier. Over the rigorous preconditioning field $E = \mathbb{F}_{19^8}$, the three AXN-state rows use packed one-vector states 2.22 GiB, 283.2 MiB, and 126.8 MiB. The next result goes beyond an asymptotic $O(M)$ statement: it prices the exact streamed row count and a complete scalar-Wiedemann product ledger. It still assumes that the selected production degree has the published one-dimensional-kernel behavior.

Table 11: Explicit streamed normal-operator accounting for the globally minimal state-aware profiles below the numerical references. "Stream" charges (105), AXN arithmetic over $\mathbb{F}_{19^4}$, $\bar{\eta}_L \le 19^{-1}$, expected $\mathbb{F}_{19^8}$ preconditioner restarts, and a factor $1 + 2^{-10}$ for extraction. "Spectrum" is complete direct projective enumeration; "combined" is the sum of the two work factors.

Level	(s,τ)	$\mathbf{d}$	M	R/M	Stream	Spectrum	Combined
I	$(9,14)$	$(2,2,3,4)$	475,832,500	2.358	141.70	173.20	173.20
III	$(10,30)$	$(1,2,3,3)$	59,383,896	1.956	203.63	191.68	203.63
V	$(11,46)$	$(1,2,2,3)$	26,574,912	1.725	269.35	209.71	269.35

Proposition 39 (Explicit streamed state frontier)**.** *Restrict to $d_i \ge 1$, $M < 19^8$, and all slice dimensions available to both $\ell = 4$ shapes at the indicated level. Among every negative-coefficient profile in (98) whose explicit streamed AXN work from (105) is at most 2^{143}, 2^{207}, or 2^{272}, respectively, the three rows of Table 11 globally minimize M. Their streamed margins are 1.30, 3.37, and 2.65 bits. One packed length-M vector over $\mathbb{F}_{19^8}$ occupies 2.22 GiB, 283 MiB, and 127 MiB; the eight-array implementation reserve occupies 17.7 GiB, 2.21 GiB, and 1.01 GiB.*

Proof. For each admissible s, both $M_s(\mathbf{d})$ and $R_s(\mathbf{d})$ are coordinatewise nondecreasing in $\mathbf{d}$. The all-one profile therefore gives a monotone lower bound on streamed work. Every profile with M below the displayed incumbent lies in a finite sorted degree box determined by $\binom{s+D}{D}(s+1)^3 < M_{\text{incumbent}}$. Exhaustive exact Hilbert coefficient evaluation over those boxes finds no feasible smaller M; the machine-readable certificate records every box and the monotone tail closure. $\square$

This calculation is deliberately separate from the published $3s^2M^2$ estimate: it prices the actual streamed row count, the extension used for normal preconditioning, and the full scalar-Wiedemann product ledger. At Levels III and V, complete exact projective-spectrum computation is cheaper than the compact solve and can therefore be run as public precomputation without changing the leading exponent. At Level I direct enumeration costs $2^{173.20}$, but the reserved-line theorem supplies the aggregate bound over the random-XOF idealization. A compressed determinantal certificate is needed only for a deterministic fixed-seed claim below the numerical reference.

An implementation-level joint stress frontier. The compact profiles above optimize state under the headline values $\bar{\eta}_L \le 1/19$ and $\sigma = 1$. We now weaken both inputs at once and price the actual streamed operator rather than the published $3s^2M^2$ estimator. Every row below allows

$$\bar{\eta}_L \le \frac{1}{2}, \qquad \sigma \ge \frac{1}{8}, \tag{111}$$

charges the exact row count, homogenizing coordinates, preconditioner restarts, and the extraction reserve. Levels III and V use (105). Level I uses the cubic-preconditioner bound (106); only the random normal preconditioner moves to $\mathbb{F}_{19^{12}}$.

Table 12: Robust explicit streamed frontier. All work columns are base-2 AXN exponents. "State" is the conservative eight-array envelope. Exact Hilbert enumeration and monotone tail closure prove minimum state among every negative-coefficient profile with $d_i \geq 1$ fitting the indicated explicit bound and preconditioner field.

Level	(s,τ)	$\mathbf{d}$	preconditioner	M	robust	margin	state
I	$(10,10)$	$(3,3,3,4)$	$\mathbb{F}_{19^{12}}$	23,417,049,656	141.29	1.71	1.53 TiB
III	$(11,26)$	$(1,2,3,5)$	$\mathbb{F}_{19^8}$	1,488,195,072	200.49	6.51	55.4 GiB
V	$(12,42)$	$(2,2,2,3)$	$\mathbb{F}_{19^8}$	342,874,805	264.03	7.97	12.8 GiB

Proposition 40 (Robust explicit state frontier)**.** *Restrict to $d_i \geq 1$ and to all slice dimensions available to both structured shapes at each level. For the Level-I field $\mathbb{F}_{19^{12}}$ and the Level-III/V field $\mathbb{F}_{19^8}$, the rows of Table 12 globally minimize $M_s(\mathbf{d})$ among every negative-Hilbert profile satisfying (111) whose explicit streamed work is at most 2^{143}, 2^{207}, or 2^{272}. Their one-vector states are 196.3 GiB, 6.93 GiB, and 1.60 GiB; the displayed state column reserves eight such arrays.*

Proof. Both $M_s(\mathbf{d})$ and $R_s(\mathbf{d})$ are coordinatewise nondecreasing. For fixed largest degree D, the profile $(1,1,1,D)$ therefore lower-bounds streamed work. The all-one profile excludes the smaller slice dimensions, and $M < M_{\text{incumbent}}$ together with the relevant work bound leaves a finite sorted degree box for every remaining s. Exhaustive integer Hilbert coefficient evaluation finds no feasible smaller M. The artifact records every box, tested profile count, and monotone tail closure. □

This joint stress test is stronger than the earlier estimator-only sensitivity calculation. It says that the numerical conclusion does not require a perfect rank spectrum or a solver certificate on every root-containing trial. The seeded theorem is stronger still on the first input: for these robust slices, $\Pr[\bar{\eta}_L > 1/2]$ is below $2^{-41.48}$, $2^{-109.45}$, and $2^{-177.41}$ in the random-XOF idealization.

There is also meaningful degree slack at Level V. The baseline robust run plus all four one-coordinate degree-increment neighbors costs $2^{270.95}$ in total, still below 2^{272}. This is a first-shell sensitivity statement, not a theorem that one of those neighbors must provide a certificate. A higher-state Level-V profile $(s,\tau,\mathbf{d}) = (13,38,(1,2,3,5))$ costs only $2^{257.02}$ under the same joint stress and uses a 262.8 GiB eight-array envelope, leaving 14.98 bits of margin.

For comparison, once the seeded $\bar{\eta}_L \leq 1/19$ event holds, the compact profiles remain below their references provided only that the production certificate rates exceed 0.405, 0.0968, and 0.159 at Levels I, III, and V. These are break-even values, not measured production rates. The 112 planted four-block Macaulay experiments reported in Section 8 all produce the predicted one-dimensional kernel, but the production transcript remains the central missing artifact.

After charging the one-rank retry cushion, the time-minimizing structured AXN rows retain 10.75, 31.88, and 54.44 bits of nominal arithmetic gap at Levels I, III, and V. Because the modeled row-reduction work is quadratic in the Macaulay column count, those budgets tolerate column-count growth factors of about 41.7, 6.2×10^4, and 1.6×10^8, respectively. The Level-III and Level-V time rows remain below their nominal references if every block degree rises by one; the Level-I rows tolerate any two single-block degree increments. These are sensitivity statements inside the same arithmetic model, not production timings.

7.3 An exact three-plus-one decomposition

The four-block system has ten unordered equation families: four diagonal families and six cross families, each containing m_1 equations. The final block need not participate in the nonlinear Macaulay solve. Write $\mathcal{E}_{ab} = 0$ for the m_1 dehomogenized equations supported on blocks a and b, where $1 \le a \le b \le 4$. Affine and constant terms are included; when one block is fixed, a cross family is affine-linear in the other block.

Theorem 41 (Exact three-plus-one completion)**.** *Let $\mathcal{I}_{123}$ be the ideal generated by the six families $\mathcal{E}_{ab}$ with $1 \le a \le b \le 3$. Suppose a certified first-stage solver returns a finite set $\mathcal{S} \subseteq \Omega^{3s}$ containing every point of $V(\mathcal{I}_{123})$. For each $p = (x_1, x_2, x_3) \in \mathcal{S}$, substitute p into $\mathcal{E}_{14}, \mathcal{E}_{24}, \mathcal{E}_{34}$ and write the resulting $3m_1$ affine-linear equations as*

$$A_p x_4 = b_p, \qquad A_p \in \Omega^{3m_1 \times s}. \tag{112}$$

Apply the following certified rule:

(i) discard p if (112) is inconsistent;

(ii) if $\operatorname{rank} A_p = s$, solve for its unique x_4 and retain the completion exactly when $\mathcal{E}_{44} = 0$;

(iii) if (112) is consistent but $\operatorname{rank} A_p < s$, declare the trial incomplete unless a separately priced solver enumerates every completion.

Whenever no incomplete case occurs, the retained set contains every root of the full four-block system. Base-field descent, affine reconstruction, and the original verifier remove every extraneous point. The resulting algorithm is Las Vegas.

Proof. Let (x_1, x_2, x_3, x_4) be a full root. Its first three blocks lie in $V(\mathcal{I}_{123})$ and therefore occur in $\mathcal{S}$. At that partial root, the true fourth block satisfies (112). If the trial is certified complete, the system cannot be a consistent rank-deficient case; it is therefore full-column-rank and its unique solution is the true x_4. The diagonal check retains it. Conversely, every retained completion satisfies all ten equation families. The final public-verifier check gives zero-error output. ◻

The first-stage Hilbert series is the three-block specialization

$$H^{(3)}_{m_1,s}(t_1, t_2, t_3) = \frac{\prod_{a=1}^{3}(1 - t_a^2)^{m_1} \prod_{1 \le a < b \le 3}(1 - t_a t_b)^{m_1}}{\prod_{a=1}^{3}(1 - t_a)^{s+1}}. \tag{113}$$

For $\mathbf{d} = (d_1, d_2, d_3)$, define

$$M_s^{(3)}(\mathbf{d}) = \prod_{a=1}^{3} \binom{s + d_a}{d_a}, \tag{114}$$

$$\begin{aligned} R_s^{(3)}(\mathbf{d}) = {} & m_1 \sum_{a=1}^{3} \binom{s + d_a - 2}{d_a - 2} \prod_{c \ne a} \binom{s + d_c}{d_c} \\ & + m_1 \sum_{1 \le a < b \le 3} \binom{s + d_a - 1}{d_a - 1} \binom{s + d_b - 1}{d_b - 1} \prod_{c \notin \{a,b\}} \binom{s + d_c}{d_c}. \end{aligned} \tag{115}$$

The same streamed normal-operator proof as Proposition 36 gives the per-trial bound

$$16 R_s^{(3)}(\mathbf{d})(s+1)^2 M_s^{(3)}(\mathbf{d}) \tag{116}$$

signed $\mathbb{F}_{19^4}$ multiply–accumulates with eight packed $\mathbb{F}_{19^8}$ arrays. The retry exponent remains $\tau = K - 4s$: the decomposition changes how a root-containing trial is solved, not the probability that the full slice contains an accepted singleton. In the robust rows, $\sigma \ge 1/8$ now denotes the joint conditional probability that the first-stage degree supplies a complete finite quotient and that every potentially extending partial root passes the certified linear-completion rule of Theorem 41.

The scalar ledger still uses the published one-dimensional-kernel prediction for the first-stage planted system. If a larger quotient is returned, every additional nullspace, border-basis, enumeration, and completion operation—and its resident state—must fit the displayed $1+2^{-10}$ reserve. Otherwise the trial is declared incomplete or the extra work is priced separately. Thus σ is a certificate-within-budget probability, not merely the probability that the first-stage ideal is finite.

Table 13: Exact three-plus-one streamed profiles. "Compact" uses $\bar{\eta}_L \le 1/19$ and $\sigma = 1$; "robust" uses $\bar{\eta}_L \le 1/2$ and $\sigma \ge 1/8$. Both columns charge the exact square-row acceptance density, expected normal-preconditioner restarts, and a $1+2^{-10}$ extraction reserve. State is an eight-array packed $\mathbb{F}_{19^8}$ envelope.

Level	profile	(s,τ)	$\mathbf{d}$	$M^{(3)}$	$R^{(3)}/M^{(3)}$	compact	robust	state
I	time boundary	$(9,14)$	$(4,4,5)$	1,023,472,450	2.891	144.25	148.17	38.1 GiB
III	time minimum	$(11,26)$	$(3,3,6)$	1,639,770,496	2.614	197.02	200.94	61.1 GiB
V	time minimum	$(13,38)$	$(3,4,4)$	3,172,064,000	2.312	250.32	254.24	118.2 GiB
V	robust low-state	$(12,42)$	$(2,3,5)$	256,214,140	2.183	259.48	263.40	9.55 GiB

The state column uses five-byte packed representatives of $\mathbb{F}_{19^8}$. A simpler implementation storing one 64-bit word per coordinate multiplies those figures by 8/5; in particular, the Level-V low-state row occupies 15.27 GiB rather than 9.55 GiB and remains a commodity-memory profile.

Proposition 42 (Certified three-plus-one frontier)**.** *Restrict to $d_i \ge 1$, $M_s^{(3)}(\mathbf{d}) < 19^8$, and every slice dimension available to both structured shapes at a level. The first three rows of Table 13 globally minimize compact and robust streamed work within this class, uniquely up to block permutation. The Level-I minimum is above 2^{143}. At Level V, the last row globally minimizes $M_s^{(3)}(\mathbf{d})$ among robust rows below 2^{272}.*

Proof. For $d_i \ge 1$, the three cross-family contributions imply

$$\frac{R_s^{(3)}(\mathbf{d})}{M_s^{(3)}(\mathbf{d})} \ge \frac{3m_1}{(s+1)^2}.$$

Hence (116) is at least $48m_1(M_s^{(3)}(\mathbf{d}))^2$. The incumbent work bound and $M^{(3)} < 19^8$ give a finite monomial-count cap for each s. For sorted degrees $d_1 \le d_2 \le d_3$,

$$M_s^{(3)}(\mathbf{d}) \ge (s+1)^2 \binom{s+d_3}{d_3},$$

so every remaining degree tail closes monotonically. Exact integer expansion of (113) over the resulting finite boxes gives the displayed optima. The machine-readable certificate records every box, coefficient, row count, and tail closure. □

The decomposition does not improve Level I, where the retry exponent is too large. At Level III it supplies a substantially faster finite-state point, although the four-block robust frontier has slightly lower state. At Level V it is decisive: the time row leaves 17.76 robust bits of margin, and the low-state row improves the four-block robust point from $2^{264.03}$ and 12.8 GiB to $2^{263.40}$ and 9.55 GiB. The selected-degree caveat has not disappeared from this XL branch; it has been narrowed to a three-block finite quotient plus a publicly checkable linear-completion condition, all obtained within the stated work–state reserve.

7.4 A selected-degree-free Frobenius-channel route

The preceding XL routes exploit the four eigenblocks after extending scalars. The Frobenius normal form of Section 4.1 gives a different attack over $A = \mathbb{F}_{19^4}$ itself. It is complementary rather than uniformly better: it replaces the production solving-degree assumption by a symbolic homotopy theorem, but exposes a soft-O implementation factor and a much larger straightforward data volume.

Write the transformed quotient equations on an A-linear slice as

$$F_0(x) = z_0 \in A^{m_1}, \qquad F_1(x) = z_1 \in A^{m_1}, \qquad F_2(x) = z_2 \in \mathbb{F}_{19^2}^{m_1}.$$

By Corollary 5, F_0 has ordinary degree at most two, F_1 has degree at most 20, and F_2 has degree at most 362. Choose a whole coordinates of F_0 and b whole coordinates of F_1, where

$$a + b = s, \qquad 0 \le a, b \le m_1. \tag{117}$$

These give a square system of s equations in s variables over A. Solve that core completely, then check every unused coordinate of all three channels, base-field descent, affine reconstruction, the rejection rule, and the original verifier. No equation is discarded: unused equations are filters.

A public separability certificate. Let $J_{a,b}(x)$ be the Jacobian of the square core, and let $D^{\mathrm{top}}_{a,b}(x)$ be the highest-degree homogeneous part of $\det J_{a,b}(x)$. A degree-two row contributes degree one to the Jacobian; a degree-twenty row has highest part $x_j x_k^{19}$, whose formal derivative in characteristic 19 has degree 19. Hence

$$\deg \det J_{a,b} \le a + 19b. \tag{118}$$

Translation to the target-dependent affine coset changes only lower-degree Jacobian terms. Therefore one exact public witness $x_0 \in A^s$ with $D^{\mathrm{top}}_{a,b}(x_0) \neq 0$ certifies, once per chosen slice and core, that the Jacobian determinant is not the zero polynomial for every target trial. Such a witness is a cheap preflight, not a solver heuristic.

Theorem 43 (Accepted nonsingular channel roots)**.** *Assume the setting of Theorem 20, with an A-linear slice of A-dimension s, and suppose the square channel core satisfies the public leading-Jacobian certificate above. Let α be the density of signatures accepted by the public rejection rule, put*

$$\delta_{a,b} = \frac{a + 19b}{|A|} = \frac{a + 19b}{19^4}, \qquad \mu = 19^{4s-K},$$

and suppose $\bar{\eta}_L \le \eta$. Then

$$\Pr[\textit{the trial has an accepted full root nonsingular for the core}] \ge \mu \frac{(\alpha - \delta_{a,b})^2}{1 + \eta}. \tag{119}$$

In particular, if $\alpha > \delta_{a,b}$, the expected number of target–slice trials to such a root is at most

$$19^{K-4s} \frac{1 + \eta}{(\alpha - \delta_{a,b})^2}. \tag{120}$$

The root need not be unique, and no independence between singularity, acceptance, and the unused filter equations is assumed.

Proof. The nonzero Jacobian determinant has at most $(a + 19b)|A|^{s-1}$ zeros in A^s by the finite-field Schwartz–Zippel bound. For a uniform full quotient target, each fixed slice point satisfies all K

scalar residual equations with probability 19^{-K}. If Y counts accepted full roots that are nonsingular for the core, then

$$\mathbb{E}Y \geq \mu(\alpha - \delta_{a,b}).$$

Furthermore $Y(Y-1) \leq Z(Z-1)$, so Theorem 18 gives $\mathbb{E}[Y(Y-1)] \leq \mu\eta$ and hence $\mathbb{E}[Y^2] \leq \mu(1+\eta)$. Cauchy–Schwarz yields $\Pr[Y > 0] \geq (\mathbb{E}Y)^2/\mathbb{E}[Y^2]$, which is (119). □

The core can now be solved by a theorem-backed algorithm. We state the precise finite-field form needed here because the published proposition phrases its field hypothesis as a characteristic bound.

Lemma 44 (Finite-cardinality multihomogeneous symbolic homotopy)**.** *Let k be a perfect finite field. Partition $N = \sum_{j=1}^{m} n_j$ variables into blocks $X_1, \dots, X_m$ of sizes $n_1, \dots, n_m$, and let $f_1, \dots, f_N$ be nonconstant polynomials represented by a straight-line program of length L. Suppose f_i has multidegree at most $\mathbf{d}_i = (d_{i,1}, \dots, d_{i,m})$. Define*

$$C_{\mathbf{n}}(\mathbf{d}) = [\vartheta_1^{n_1} \cdots \vartheta_m^{n_m}] \prod_{i=1}^{N} \left(\sum_{j=1}^{m} d_{i,j} \vartheta_j \right), \tag{121}$$

$$C_{\mathbf{n}'}(\mathbf{d}') = \operatorname{coeffsum} \left(\prod_{i=1}^{N} \left(\vartheta_0 + \sum_{j=1}^{m} d_{i,j} \vartheta_j \right) \bmod \langle \vartheta_0^2, \vartheta_1^{n_1+1}, \dots, \vartheta_m^{n_m+1} \rangle \right), \tag{122}$$

and put $B = C_{\mathbf{n}}(\mathbf{d})$ and

$$E = \max \left\{ \max_{1 \leq j \leq m} \sum_{i=1}^{N} d_{i,j}, 8B^2 \right\}. \tag{123}$$

If $|k| \geq E$, a randomized symbolic-homotopy algorithm returns a zero-dimensional parametrization containing every nonsingular root with probability at least $7/8$; on every run it returns a subset of the nonsingular roots or reports failure. Its arithmetic cost is

$$\widetilde{O}\left(B C_{\mathbf{n}'}(\mathbf{d}') \left(L + \sum_{i,j} d_{i,j} + N^2 \right) N \right) \tag{124}$$

operations in k.

Proof. Safey El Din and Schost prove the same algorithm and cost when the characteristic is zero or at least $\max\{\max_j \sum_i d_{i,j}, 8(N-1)B^2\}$ [23, Proposition 5]. We replace the two uses of that large-characteristic hypothesis by field-cardinality arguments.

For block j, their start system uses $D_j = \sum_i d_{i,j}$ affine moment-curve forms. Choose arbitrary distinct $\alpha_{j,0}, \dots, \alpha_{j,D_j-1} \in k$ and set

$$\kappa_{j,u}(X_j) = X_{j,1} + \alpha_{j,u} X_{j,2} + \cdots + \alpha_{j,u}^{n_j-1} X_{j,n_j} + \alpha_{j,u}^{n_j}.$$

Any n_j of these forms have an invertible Vandermonde coefficient matrix, whereas any $n_j + 1$ give an inconsistent affine system because the augmented Vandermonde matrix is invertible. The proof of their start-system lemma therefore applies verbatim: the product system has exactly B roots, all simple, and they are enumerated within the same soft-linear cost. The first term in (123) supplies the distinct nodes.

It remains to choose the separating linear form. The cited proof identifies at most B^2 nonzero vectors $v \in \mathcal{A}^N$, where $\mathcal{A}$ is a domain containing k, on none of which the chosen linear form may vanish. Choose its coefficient vector λ uniformly from k^N. For fixed $v \neq 0$, the k-linear map $\lambda \mapsto \lambda \cdot v$ is nonzero, so its kernel contains at most $|k|^{N-1}$ coefficient vectors. Hence

$$\Pr_{\lambda \leftarrow k^N}[\lambda \cdot v = 0] \leq |k|^{-1}, \qquad \Pr[\lambda \text{ is not well separating}] \leq B^2/|k| \leq 1/8.$$

This replaces the one-parameter family in their separating-form lemma and removes the factor $N-1$ from the field-size threshold. Their auxiliary algorithm accepts an arbitrary well-separating linear form, so no later step changes.

The homotopy curve, Newton lifting, specialization, squarefree cleaning, and zero-dimensional parametrization use only field operations and inversion at nonsingular points. Since finite fields are perfect, the remainder of the correctness and arithmetic-cost proof is unchanged. This is a finite-cardinality adaptation proved here; the cited proposition itself is stated under its original characteristic hypothesis. □

For the channel core, $N = s$, the degrees are a copies of two and b copies of twenty, and therefore

$$B = 2^a 20^b, \qquad B^+ = B\left(1 + \frac{a}{2} + \frac{b}{20}\right). \tag{125}$$

A conservative straight-line program first computes the $s(s+1)/2$ unordered products for the degree-two rows. For the twisted rows it computes every x_j^{19} with at most six multiplications and every ordered product $x_j x_k^{19}$. Charging one coefficient multiplication and one addition per possible term gives

$$L_{a,b,s} = \mathbf{1}_{a>0}\left[\binom{s+1}{2} + 2a\left(\binom{s+1}{2} + s + 1\right)\right] + \mathbf{1}_{b>0}\Big[6s + s^2 + 2b(s^2 + 2s + 1)\Big]. \tag{126}$$

Choose the smallest r for which $(19^4)^r \geq E$ and run the homotopy over an r-degree extension $\widetilde{A}/A$. Multiplication in that extension can be reduced to $c_r = 2r - 1$ pointwise multiplications in A by evaluation and interpolation at $2r - 1$ distinct A-points. This counts only the leading pointwise products. We therefore expose, rather than hide, a multiplier κ_{hom} containing every soft-O factor, inversion, evaluation and interpolation operation, filter and factorization cost, memory traffic, and implementation constant.

Define the unit-normalized leading AXN work

$$W^{\text{lead}}_{a,b,s}(\eta) = \frac{8}{7}\, 19^{K-4s} \frac{1+\eta}{(\alpha - \delta_{a,b})^2}\, c_r g_4^{\text{AXN}}\, BB^+(L_{a,b,s} + 2a + 20b + s^2)s. \tag{127}$$

The factor 8/7 repeats the homotopy until its good random choice occurs. The actual implementation ledger is $\kappa_{\text{hom}} W^{\text{lead}}_{a,b,s}$; (127) is not asserted to upper-bound the suppressed factor.

Table 14: Balanced Frobenius-channel homotopy profiles for all six $\ell = 4$ Version 2.3 rows. Compact uses $\eta = 1/19$; robust uses $\eta = 1/2$. The square-row lower bound $\alpha = (1 - 19^{-4})^n$ is charged for both shapes at each level. "+7" is the sensitivity value obtained by setting $\kappa_{\text{hom}} = 2^7$, not a claim that 2^7 bounds the suppressed factor. The last column is a straightforward dense coefficient-volume ceiling, not a proved resident-state lower bound.

Level	$(s; a, b; \tau)$	B	B^+	r	compact	robust	robust +7	dense ceiling
I	$(9; 5, 4; 14)$	5,120,000	18,944,000	3	135.00	135.51	142.51	6.03 PiB
III	$(10; 7, 3; 30)$	1,024,000	4,761,600	3	199.06	199.57	206.57	341.5 TiB
V	$(11; 9, 2; 46)$	204,800	1,146,880	3	263.03	263.55	270.55	17.94 TiB

Proposition 45 (Certified channel-core ledger)**.** *For the three rows of Table 14, the Jacobian singular-density bounds are respectively*

$$\frac{81}{19^4}, \qquad \frac{64}{19^4}, \qquad \frac{47}{19^4}.$$

The compact unit-normalized leading exponents are 134.99722*,* 199.06332*, and* 263.03426*; the robust exponents are* 135.50818*,* 199.57429*, and* 263.54523*. Thus the robust break-even values for* $\log_2 \kappa_{\mathrm{hom}}$ *are* 7.49182*,* 7.42571*, and* 8.45477 *bits. Among every whole-channel square core satisfying* (117)*, the displayed rows minimize the straightforward dense coefficient-volume ceiling subject to retaining seven robust sensitivity bits below the numerical reference. All statements are reproduced by exact integer enumeration in the companion certificate.*

Proof. Substitute (125) and (126) into (127), use the exact accepted densities from Section 5.3, and enumerate $1 \le s \le 2m_1$ and $\max(0, s - m_1) \le a \le \min(m_1, s)$ with $b = s - a$. For fixed s, replacing a degree-twenty equation by a degree-two equation decreases B, B^+, the degree sum, and the SLP bound; the exhaustive certificate nevertheless checks every mix. It also records the extension-cardinality threshold, packed output size, dense coefficient-volume ceiling, and every break-even factor. □

The final zero-dimensional parametrizations in the three balanced rows contain at most $(s+1)B$ extension-field elements, namely about 342 MiB, 75.2 MiB, and 16.4 MiB in packed form. A straightforward dense parametric-homotopy representation can be much larger, as the final column shows; Lemma 44 is a time theorem, not a low-memory theorem. For comparison, minimizing that dense ceiling subject only to a compact normalized exponent below the reference gives $2^{142.68}$ with 13.7 TiB at Level I, $2^{206.79}$ with 786 GiB at Level III, and $2^{270.06}$ with 29.8 GiB at Level V. Those boundary rows leave essentially no room for suppressed factors.

This branch changes the logical status of the paper. A reviewer may reject the selected-degree extrapolation used by streamed XL, or may instead regard the symbolic-homotopy constants and data movement as too large for a concrete claim. Those objections are no longer the same objection: the homotopy branch has theorem-backed root completeness and no selected degree, while the XL branch has the explicit low-state operator frontier. The unconditional quotient, exact filtering reduction, retry theorem, and final-verifier check are common to both.

7.5 Low-output cores from the original quadratic families

The Frobenius-channel construction raises some equations to degree 20 in order to work directly over A. A faster selected-degree-free route keeps the original degree-two eigenblock equations. It solves only two or three of the four scalar-extension blocks nonlinearly, uses Frobenius descent to reconstruct the omitted blocks, and filters every unused equation.

Fix $b \in \{2, 3\}$ consecutive eigenblocks, each with s variables. Select l_a equations from diagonal family (a, a) and e_{ab} equations from cross family (a, b), with

$$0 \le l_a, e_{ab} \le m_1, \qquad \sum_a l_a + \sum_{a<b} e_{ab} = bs. \tag{128}$$

The resulting square core has exact multihomogeneous Bézout number

$$B = [t_1^s \cdots t_b^s] \prod_{a=1}^{b} (2t_a)^{l_a} \prod_{1 \le a < c \le b} (t_a + t_c)^{e_{ac}}. \tag{129}$$

Its companion quantity B^+ is (122). Both are exact integers, not semi-regularity predictions.

A base-field point becomes, after scalar extension, a Frobenius orbit $(x, x^{19}, x^{19^2}, x^{19^3})$. Thus every genuine residual root restricts to a root of the selected b-block core. Conversely, after the homotopy enumerates the core roots, the attacker checks the Frobenius relations among the selected blocks, reconstructs every omitted block from one selected block, and checks all unused residual equations, rejection, affine reconstruction, and the original verifier. No linear-completion or selected-Macaulay-degree assumption is involved.

A public leading-Jacobian certificate. Let γ_a be the degree of the core-Jacobian determinant in block a; explicitly,

$$\gamma_a = 2l_a + \sum_{c \neq a} e_{\min\{a,c\},\max\{a,c\}} - s. \tag{130}$$

On the descent locus, its leading part pulls back to a polynomial over A of degree at most

$$\Delta = \sum_{a=0}^{b-1} \gamma_a 19^a. \tag{131}$$

The attacker evaluates this leading determinant at one fixed public descent point. A nonzero value certifies that the determinant polynomial is nonzero for every target translate. The companion artifact supplies exact allowable symmetric-form assignments at which this determinant is nonzero for each recommended profile; hence the preflight is a proper algebraic condition, not an assumed possibility.

Theorem 46 (Accepted nonsingular eigenblock-core roots)**.** *Let a selected core satisfy the public leading-Jacobian certificate, let α be the density accepted by the public rejection rule, and suppose $\bar{\eta}_L \leq \eta$. Put $h = 4s$, $\mu = 19^{h-K}$, and $\delta = \Delta/19^4$. Then a random target–coset trial contains an accepted full residual root that is nonsingular for the selected core with probability at least*

$$\max\left\{\mu(\alpha - \eta - \delta), \mu\frac{(\alpha-\delta)^2}{1+\eta}\right\}, \tag{132}$$

where a nonpositive first term is ignored. Finite-cardinality symbolic homotopy followed by the exact filters above is therefore a Las Vegas forger. It uses neither a production solving-degree prediction nor a one-dimensional Macaulay-kernel assumption.

Proof. The fixed leading-Jacobian certificate and (131) bound the density of descent points singular for the core by δ. Let Y count accepted full roots that are nonsingular for the core and let Z count all full roots. The first moment is at least $\mu(\alpha - \delta)$. Since $Y(Y-1) \leq Z(Z-1)$, the aggregate collision theorem gives $\mathbb{E}[Y(Y-1)] \leq \mu\eta$. The singleton inequality $\Pr[Y = 1] \geq \mathbb{E}Y - \mathbb{E}[Y(Y-1)]$ gives the first term of (132); Cauchy–Schwarz with $\mathbb{E}[Y^2] \leq \mu(1+\eta)$ gives the second. Conditional on such a root, Lemma 44 returns it with probability at least $7/8$. Every later operation is an exact public filter. $\square$

The random-XOF model also makes the leading-Jacobian preflight quantitative. Group the fresh $V' \times V'$ coefficients by public base form. If the nonzero determinant polynomial has degree at most d_i in form block i, recursive application of Lemma 10 gives

$$\Pr[\det J_{\text{core}} \neq 0] \geq \beta_{\text{core}} = \prod_{i=1}^{m_1}(1 - d_i\rho). \tag{133}$$

The exact witness fixes the nonzero coefficient needed by that recursion. The core event and the seeded-spectrum event use the same internal coefficient region, so they are combined by the union bound $\beta_{\mathrm{core}} - \varepsilon_{\mathrm{spec}}$, not by an independence claim. They are independent of the complete structural preflight in Theorem 12. Consequently the certified random-key density is

$$p_{\mathrm{eig}} \geq p_{\mathrm{pre}}(\beta_{\mathrm{core}} - \varepsilon_{\mathrm{spec}}). \tag{134}$$

Table 15: Recommended original-family eigenblock cores. Compact uses $\eta = 1/19$ and robust uses $\eta = 1/2$. "Normalized" pays the inverse exact lower bound (134), so it is a work-per-success ledger over a random generated key in the random-XOF idealization. "+8" multiplies the leading homotopy term by 2^8 as a sensitivity point. Output is a packed final-parametrization ceiling, not a peak-memory theorem.

Level	core	$(l_a; e_{ab})$	B	r	robust/key	p_{eig}	normalized	normalized margin	output
I	3×10	$(5,5,5;5,5,5)$	73,793,536	4	134.077	0.129015	137.031	5.969	19.793 GiB
III	3×11	$(7,7,0;7,7,5)$	6,881,280	3	196.246	0.116451	199.348	7.652	1.570 GiB
V	2×13	$(9,8;9)$	16,515,072	3	247.432	0.200117	249.753	22.247	3.015 GiB

For the three rows, the exact per-form degree vectors in (133) are

$$(6,6,6,6,6), \qquad (5,5,5,5,5,4,4), \qquad (3,3,3,3,3,3,3,3,2),$$

giving $\beta_{\mathrm{core}} = 0.1369122816$, 0.1235790793, and 0.2123661755. The robust spectrum-failure probabilities are below $2^{-41.48}$, $2^{-109.45}$, and $2^{-160.42}$, respectively. Before paying key density, the robust break-even budgets for $\log_2 \kappa_{\mathrm{hom}}$ are 8.923, 10.754, and 24.568 bits; setting $\kappa_{\mathrm{hom}} = 2^8$ gives 142.077, 204.246, and 255.432 per certified key. The normalized +8 values are 145.031, 207.348, and 257.753, so the first two levels still need an implementation measurement to turn the normalized sensitivity row into a below-reference claim.

The released search is exhaustive within the stated original-family class: it checks every admissible one-, two-, and three-block square core and records the complete time/output Pareto frontier. It examines 15,663 count vectors at Level I, 87,643 at Level III, and 333,841 at Level V, of which 11,119, 61,442, and 232,706 have nonzero Bézout number. Thus the displayed profiles are not hand-selected examples. Smaller-output points include 872.3 MiB at Level I, 42.1 MiB at Level III, and 477.8 MiB or 47.3 MiB at Level V, with the exact arithmetic tradeoffs in the artifact.

7.6 A Frobenius-orbit-complete homotopy frontier

We now remove the preferred-core premise entirely. Let the four eigenblocks be indexed cyclically by $\mathbb{Z}/4\mathbb{Z}$. For a stable A-linear slice of A-dimension s, write $h = 4s$. If the full restricted residual Jacobian has rank h at a descended root, some h-row minor is nonzero. When $h = K - 2$, each family-count pattern is obtained by deleting two rows from the ten full multidegree families, giving

$$\binom{10+1}{2} = 55$$

patterns.

Lemma 47 (Frobenius transport of minors)**.** *At a point satisfying Frobenius descent,* 19*-Frobenius cyclically rotates the four eigenblocks. It sends the determinant of every square restricted Jacobian minor to the determinant of the cyclically rotated minor, raised to the nineteenth power. In particular, one minor is nonzero if and only if every minor in its cyclic orbit is nonzero.*

Proof. The residual equations and the eigenblock coordinates are defined over $\mathbb{F}_{19}$ before scalar extension. Frobenius therefore commutes with restriction and differentiation, while cyclically permuting the eigenblocks. Applying Frobenius to a determinant raises its value to the nineteenth power, which preserves zero and nonzero values. □

Burnside enumeration gives exactly 16 orbits of the 55 patterns under this cyclic action. The attacker solves one representative from each orbit by the finite-cardinality symbolic-homotopy algorithm, and filters every candidate through all omitted residual equations, all descent equations, rejection, affine reconstruction, and the original verifier. By Lemma 47, every descended root at which the full restricted Jacobian has rank h is nonsingular for at least one solved representative.

Proposition 48 (Orbit-complete fixed-core-free frontier)**.** *For Levels I, III, and V, take respectively* $(h, K) = (48, 50), (68, 70), (88, 90)$*. Under the robust aggregate-spectrum and exact acceptance bounds, the sum over the* 16 *orbit representatives has per-certified-key AXN exponents*

$$138.32403, \qquad 179.75835, \qquad 220.83240.$$

The complete structural preflight of Theorem 12 succeeds in the random-XOF idealization with probability greater than 0.9423222679 *for every official* $\ell = 4$ *shape. Charging its inverse gives the random-key-normalized exponents*

$$138.40973, \qquad 179.84405, \qquad 220.91810.$$

The corresponding margins below 143/207/272 *are* 4.59027*,* 27.15595*, and* 51.08190 *bits.*

Proof. The artifact exhaustively enumerates the 55 patterns, quotients them by the cyclic Frobenius action, and obtains 16 orbits. For every representative it computes the exact multihomogeneous Bézout number, the companion homotopy quantity, extension degree, straight-line-program bound, candidate-filter cost, and robust retry factor. Summing these ledgers gives the per-key exponents. The final values add $-\log_2(0.9423222679) < 0.08571$ bits and the negligible seeded-spectrum tail. □

This theorem-backed branch has no selected Macaulay degree, no exact-corank assumption, and no preferred core-Jacobian condition. Its straightforward largest individual parametrizations remain extremely large: approximately 1.890 PiB, 2.076 ZiB, and 2416.552 YiB. These are dense sequential output ceilings rather than peak-memory lower bounds, but they make clear that the purpose of this branch is logical completeness. The original-equation eigenblock cores of Section 7.5 provide much smaller output ceilings, while the streamed-XL branch provides the low-state frontier.

7.7 The $\ell = 2$ attacks

For Levels I and III, use the fixed-skew lift of Proposition 15. The public preflight gives quotient rank K, affine rank $M - K$, and residual systems of dimensions 48/112 and 72/168. CFXL is a hybrid strategy that specializes some variables and then applies XL-style linear algebra; Hashimoto refines this optimization for underdetermined MQ. Minimizing these variants in Beullens's printed Appendix-A convention gives per-solve exponents 129.44 and 183.44 modeled bit operations [3, 10]. Replacing b_1 by g_1^{AXN} gives per-solve arithmetic exponents 133.75 and 187.74. Every recovered root is accepted. We charge $\Theta_{\mathrm{pol}} \leq 19^{-1}$ from Corollary 33; the minimum ranks $r_* \geq 49$ and $r_* \geq 73$ are sufficient. The resulting headline exponents are 129.51/133.82 and 183.51/187.82 in the Model/AXN columns. The same fixed-skew construction at Level V has per-solve exponents 237.84/242.15 and headline exponents 237.92/242.22 under the same aggregate condition; $r_* \geq 97$ is sufficient. The stronger thresholds 96, 144, and 192 remove target retries entirely.

Table 16: $\ell = 2$ systems, aggregate headline conditions, and sufficient minimum-rank certificates. Model and AXN charge $\Theta_{\text{pol}} \le 19^{-1}$ for fixed skew and $\Psi_{\text{pol}} \le 19^{-1}$ for the filter. "Budget" is the smallest minimum rank preserving positive nominal AXN nominal arithmetic gap; "suff." is a minimum rank implying the aggregate headline condition; "all target" guarantees an accepted root for every target.

Parameters	mode	eqs./vars.	model	AXN	budget	suff.	all target
$(48, 16, 2, 2)$	fixed-skew	48/112	129.51	133.82	46	49	96
$(72, 24, 2, 2)$	fixed-skew	72/168	183.51	187.82	68	73	144
$(96, 32, 2, 2)$	filter	96/256	235.27	239.57	177	194	240
$(96, 32, 2, 2)$	fixed-skew	96/224	237.92	242.22	89	97	192

The optimizer includes Hashimoto's published boundary value $a = 1$. For $\mathrm{MQ}(19, 1, 1)$, the semi-regular Hilbert-series rule used by CFXL never produces the stopping degree at which its linearization would be costed. The artifact therefore solves that elementary system by 19 direct evaluations and deliberately overcharges each evaluation. This lowers six generic quotient baselines and the Level-V fixed-skew fallback; it does not change the selected Level-I or Level-III fixed-skew costs.

The Level-V headline instead uses $R_0 = I, R_1 = 0$ and zero offsets. The quotient has rank K, target filtering enforces the $M - K$ public consistency conditions, and the residual system has 96 equations in all 256 variables. Target search occurs before equation solving, so its cost adds to—rather than multiplying—the cost of a solve. The generic-MQ per-solve AXN arithmetic exponent is 239.50, while exact-uniform consistency filtering is smaller by more than 70 exponent bits. Under the headline aggregate condition $\Psi_{\text{pol}} \le 19^{-1}$, (88) adds less than 0.075 bits, giving Model/AXN exponents 235.27/239.57. The stronger condition $r_* \ge 240$ gives an accepted root for every target. The fixed-skew fallback is slower but makes every recovered root acceptable by construction.

For the filter row, the headline uses $\Psi_{\text{pol}} \le 19^{-1}$; the displayed minimum rank 194 is sufficient by (90), and the all-target threshold includes rejection through (38). For fixed skew, the headline uses $\Theta_{\text{pol}} \le 19^{-1}$; the displayed $r_* \ge K + 1$ is sufficient, while $r_* \ge 2K$ makes every target solvable. The quotient matrix depends only on the fixed public outer map and the relation, not on a generated key. Exhaustive enumeration of all $19^2 = 361$ relations $R_1 = aI + bS$ proves quotient rank K for every relation on all three parameter shapes. Exhaustive optimization of the natural two-block special-XL model gives bit-operation exponents 132.31, 184.27, and 236.36, all worse than the selected generic-MQ modes.

8 Implementation and Evidence

The artifact implements the public key generator, verifier-side feature maps, symmetry quotient, affine reduction, structured lift, rank tests, and cost searches over $\mathbb{F}_{19}$. It does *not* report a production-size end-to-end forgery, a production-size Macaulay solve, or a wall-clock attack timing. Its full-size experiments concern public preprocessing and rank; its solver experiments use smaller tractable profiles. The checks therefore have different logical force. The table below is the intended reading guide for every experimental statement in this section.

Table 17: Evidence hierarchy. "Exact" here means exact for the concrete object checked, not an exhaustive theorem about all production directions.

Check	What is computed	What it does—and does not—establish
Public preflight	Exact ranks and rejection conditions for the chosen key, relation, offsets, and slice	The exact reduction is valid for that concrete attack instance.
Seeded spectrum theorem	Exact min-entropy calculation in the random-XOF idealization	Bounds the aggregate spectrum for the reserved-line construction; not a deterministic claim about one already fixed SHAKE seed.
Rank campaign	Exact ranks for many selected nonzero output or direction vectors	Finite fixed-key evidence and regression coverage; no longer the logical basis of the structured aggregate-spectrum claim.
Macaulay experiment	Exact kernels at smaller tractable dimensions	Evidence for the production solver model; not a production timing or rank proof.
Four-block and 3+1 frontiers	Exact Hilbert coefficients, row and column counts, normal-operator ledgers, linear-completion rules, and monotone tail certificates	Globally certified optima inside the stated solver classes; not a production solve or wall-clock trace.
Frobenius-channel certificate	Exact rank-ten coordinate map, every square core mix, B'ezout and SLP quantities, extension thresholds, Jacobian losses, and break-even factors	Proves the channel algebra and numerical ledger; the soft-O implementation factor and resident memory remain explicit.
Basis-complete/square homotopy	Exact family-count enumeration, multihomogeneous Bézout quantities, extension thresholds, full-square ledger, filters, and spectrum-tail calculations	Removes the fixed-core and selected-degree premises in the stated operation-count model; κ_{hom} and large parametrization volume remain explicit.
Circuit validation	Exhaustive or independent checking of the released field primitive	Exact cost and functionality of that primitive, but not of a complete solver.

A "projective direction" below means a nonzero coefficient vector modulo multiplication by a nonzero field scalar; all scalar multiples define the same polar rank. Sampling many distinct directions remains useful for regression and fixed-key diagnostics. The structured headline now has a proof over the random-XOF idealization of Theorem 27; an exact spectrum computation remains the deterministic route for one fixed public key. A global minimum-rank proof is a still stronger sufficient certificate.

8.1 Public preflight and verifier regression

For a structured $\ell = 4$ instance, the preflight checks

$$\operatorname{rank} Q_R = K, \qquad \operatorname{rank} C_{R,\mathbf{v}} = M - K, \qquad \operatorname{rank} \mathcal{O}_{R,\mathbf{v}} = m_1 \ell^2,$$

together with $\mathcal{F}_{R,\mathbf{v}} H_{R,\mathbf{v}} = 0$, the predicted A-kernel dimension, A-invariance, the restricted quadratic rank, and rejection compatibility. All 121 regenerated full-size instances tested in the artifact pass,

Table 18: Exactly computed cross-polar ranks for the tested structured $\ell = 4$ slice–direction pairs.

Parameters	slices	$h = 4s$	rank tests	sampled minimum
$(28, 5, 4, 4)$	3	40	26,296	40
$(28, 4, 4, 5)$	2	40	24,198	40
$(40, 7, 4, 4)$	2	60	45,658	60
$(38, 5, 4, 5)$	2	60	2,188	60
$(50, 9, 4, 4)$	2	76	2,228	76
$(52, 6, 4, 6)$	2	76	2,228	76
Total	13	–	102,796	maximal

including the official Level-I known-answer key. The first deterministic offset candidate passes for 114 instances; the second passes for the remaining seven.

For fixed skew, the preflight checks the relation and rejection identity and the quotient, affine, and augmented ranks. All 60 full-size keys tested in the artifact pass. The artifact also reproduces the official Level-I public key byte for byte, verifies its known-answer signature against the official hash target, and checks the fixed output-order permutation described in Section 3.2. Eighteen additional regressions cover all nine Version 2.3 shapes. An independent Gauss–Jordan implementation agrees with the rank kernel on 2,036 matrices, including 36 production-sized matrices.

8.2 Rank evidence

Seeded aggregate spectrum. The machine-readable certificate evaluates (73) and (74) to 180 decimal digits for both structured shapes at each level. For the compact profiles it obtains $\log_2$ bad-1/19 bounds -55.22, -123.19, and -191.16; for the robust profiles it obtains bad-1/2 bounds -41.48, -109.45, and -177.41. The calculation retains the actual maximum symbol atom 14/256, rather than pretending the byte-to-field conversion is uniform. It is a theorem-driven calculation from public dimensions, not a sampled-rank experiment.

Structured slices. Every selected slice is tested on all coefficient-basis directions (one nonzero output coefficient at a time) and 1,024 deterministic dense directions (many nonzero coefficients). Selected slices also receive complete normalized weight-two tests (two nonzero coefficients, modulo overall scaling) or additional dense tests. All 102,796 tested cross-polar maps have maximal rank $h = 4s$; the per-shape counts are in Table 18. The dual campaign evaluates 21,992 directional maps T_y on the same 13 slices, and every one is surjective of rank K. All 124,788 coefficient and direction vectors used in these two campaigns are projectively distinct within their respective slice records.

The directional campaign directly tests the simple sufficient certificate for the aggregate headline condition. The sufficient ranks are only $h + 1 = 41, 61, 77$; every tested T_y instead has rank $K = 50, 70, 90$. Thus the sampled directional margins are 9, 9, and 13 ranks, respectively. These finite tests certify neither the global minimum nor the weighted aggregate, but they probe a condition substantially weaker than surjectivity.

Fixed-skew systems. A broad campaign tests every coefficient-basis direction together with combinations having two, three, or many nonzero output coefficients. A second campaign targets the 21 exceptional projective directions that make the local operator on $\mathrm{Mat}_2(\mathbb{F}_{19})$ rank two rather than four. It tests all 381 local projective directions in one block, every exceptional direction in every remaining block, and deterministic two-block exceptional combinations. Together the campaigns

Table 19: Full-size polar-rank evidence for the fixed-skew $\ell = 2$ systems. "Budget" preserves positive AXN nominal arithmetic gap; "suff." is the simple minimum-rank certificate for the aggregate headline condition in Table 16; "all target" guarantees a root for every target.

Parameters	preflights	tests	sampled min.	budget	suff.	all target
$(48, 16, 2, 2)$	20	1,104	110	46	49	96
$(72, 24, 2, 2)$	20	1,296	167	68	73	144
$(96, 32, 2, 2)$	20	1,488	223	89	97	192
Total	60	3,888	–	–	–	–

perform 3,888 exact rank evaluations; their sampled minima and the budget, sufficient, and all-target thresholds appear in Table 19.

Zero-offset filter. The broad and exceptional-direction campaigns perform 29,424 exact rank evaluations on two keys per shape. Their sampled minima are 127, 190, and 254 in dimensions 128, 192, and 256. For the Level-V filter, the sampled minimum 254 exceeds the simple sufficient minimum-rank value 194 by 60 and the all-target threshold 240 by 14. If the global minimum is at least 254, then (40) gives an accepted fraction of at least 0.999999999 in every target fiber. The experiment supports that hypothesis but does not certify the global minimum.

8.3 Solver and probability checks

The special-XL search examines every cost-competitive multidegree and recovers the three time optima in Table 10. Separate exact searches verify every candidate that can match the compact and robust state incumbents and close all larger degrees by coordinatewise monomial-count monotonicity, proving Propositions 38 to 40. They record exact Hilbert coefficients, M, R, extension-field preconditioner bounds, operator constants, and complete array ledgers. A separate exhaustive search proves Proposition 42 from the three-block series (113); its certificate also checks the exact fourth-block equation counts and the rank-feasibility inequality $s \leq 3m_1$. No production three-block quotient or linear-completion rate is claimed. At tractable sizes, the artifact builds 128 four-block Macaulay matrices: 64 random instances have full column rank and 64 planted-solvable instances have the model's strongest $D = 1$ behavior, with the planted evaluation vector spanning the right kernel. A separate boundary campaign constructs 48 planted four-block systems across five negative-Hilbert profiles having at least one degree-one block; every matrix has exact right corank one. This confirms that degree one still retains all variables and behaves normally in small profiles, but it is not a production-degree theorem. A further reproducible campaign keeps the *production-selected multidegrees* $(3, 3, 3, 4)$, $(1, 2, 3, 5)$, and $(2, 2, 2, 3)$ and the corresponding values $m_1 = 5, 7, 9$, while reducing the block size to $s = 2$ so exact sparse elimination is feasible. The three planted matrices have $(M, R) = (15{,}000, 268{,}500)$, $(3{,}780, 69{,}678)$, and $(2{,}160, 47{,}628)$ and all have exact right corank one. At reduced size only the first Hilbert coefficient is negative, so this check is evidence about the selected degree shapes, not an argument that the production solving degree is proved. A 48-matrix two-block control contains one planted case with a two-dimensional right kernel. The latter is no longer a conceptual failure—finite quotients of dimension larger than one are covered by Theorem 37—but we still do not promote the $\ell = 2$ special-XL estimates because the control does not establish border completeness or production degree. These are generic small-profile experiments, not production SNOVA timings.

The channel artifact independently constructs (14) in $A = \mathbb{F}_{19}[t]/(t^4 - t - 1)$; its 10×10 matrix has exact rank ten and determinant eight. A second exact script enumerates every whole-channel square core for all three levels, recomputes B, B^+, the SLP bound, the finite- cardinality extension degree, the Jacobian loss, both retry denominators, the packed output size, and the dense coefficient-volume ceiling. It proves the globality statement in Proposition 45. This is an exact arithmetic certificate, not a symbolic-homotopy implementation or a claim that the hidden multiplier κ_{hom} is small.

Exhaustive checks over $\mathbb{F}_3$ and $\mathbb{F}_5$ verify the exact first and factorial second moments, the compatible-collision formula, the rank-spectrum upper bound, and the singleton and planted-distribution inequalities on 110 complete quadratic maps. Seventeen maps have nonmaximal cross-polar rank; in twelve, compatibility makes η_0 strictly smaller than $\bar{\eta}$. Finally, on six actual cost-driving $\ell = 2$ cores, the homogeneous quadratic outputs are linearly independent, the common polar radical—the set of directions annihilated by every polar form—is zero, and all 384 sampled homogeneous Jacobians (matrices of first derivatives of the homogeneous quadratic parts) attain the maximum rank permitted by their dimensions. These are genericity diagnostics, not solving-degree measurements.

8.4 Scope of the estimates

Once a public preflight passes, the symmetry quotient, affine reduction, stable kernel, fixed-skew identity, and rejection handling are exact. For a structured slice, the load-bearing root quantity is $\bar{\eta}_L$. Theorem 27 proves the required aggregate bound with explicit failure probability in the random-XOF idealization, provided the candidate is constructed from the decoupled cross data. The complete projective rank histogram remains an exact deterministic certificate for a fixed key, and direct enumeration is dominated by the Level-III/V compact solves. Thus the structured rank spectrum is no longer an unproved sampled minimum; the model boundary is whether one accepts the random-XOF idealized-expansion theorem or asks for a fixed-seed certificate.

The remaining production assumption for the streamed-XL branch is solver behavior at the selected Macaulay degree. The compact scalar-Wiedemann rows use $\sigma = 1$ in Theorem 35; the robust explicit rows charge only $\sigma \geq 1/8$. A complete commuting border basis of dimension $D > 1$ still gives zero-error recovery by Theorem 37, but the extra nullspace/block work needed to obtain it must fit the displayed operator ledger. The D_{30} values in Table 9 price only FGLM-style postprocessing once the multiplication tables have been supplied.

The fixed-core-free branch also does not share that selected-degree premise. The same aggregate spectrum that controls collisions bounds full-Jacobian singularity by (61). The basis-complete rows run every possible multidegree-family count; the Level-III row solves the complete square residual system. Thus their root-completeness statement needs neither a production Macaulay degree nor a preselected nonzero Jacobian determinant. Their quantitative boundary is instead the explicit κ_{hom} and the potentially enormous zero-dimensional parametrization.

The Frobenius-channel branch does not share that selected-degree premise. Its exact preflight is a nonzero evaluation of the highest homogeneous Jacobian determinant. Conditional on that public certificate, Theorem 43 accounts for singular roots without independence, and Lemma 44 supplies all nonsingular core roots with constant success probability. Its remaining quantitative boundary is instead the explicit κ_{hom} multiplying a soft-O arithmetic bound, together with the fact that the cited theorem gives no low-memory implementation guarantee. The paper therefore treats the channel exponents as break-even sensitivity ledgers rather than production timings.

The state and operator accounting is no longer implicit. The streamed normal operator of Proposition 36 avoids Macaulay materialization. The compact four-block frontier charges its exact row count over $\mathbb{F}_{19^8}$ and has packed eight-array envelopes 17.7 GiB, 2.21 GiB, and 1.01 GiB. The three-plus-one route gives explicit Level-III/V time points $2^{197.02}$ and $2^{250.32}$ and a robust Level-V low-state point $2^{263.40}$ with 9.55 GiB. Taking the best joint-stress state point from either XL decomposition gives packed envelopes 1.53 TiB, 55.4 GiB, and 9.55 GiB. The larger Level-I field affects only random normal preconditioning. These are explicit state and arithmetic bounds, not production timings.

For $\ell = 2$, the headline uses the aggregate bounds $\Theta_{\mathrm{pol}} \le 19^{-1}$ and $\Psi_{\mathrm{pol}} \le 19^{-1}$; minimum ranks 49, 73, and 194 suffice, versus all-target values 96, 144, and 240. The sampled ranks and small-profile solves support these conditions but do not certify the complete spectra or the selected production solving degrees. No production-size end-to-end solve is claimed.

What would turn the estimates into end-to-end certificates. For a fixed slice, the exact target is the weighted sum $\bar{\eta}_L = \sum_{y \ne 0} 19^{-\operatorname{rank} T_y}$. The reserved-line theorem already controls this sum under the idealized key distribution. When a deterministic fixed-seed record is desired, Proposition 30 gives a finite exact certificate: disjoint normalized projective charts, finite-field equations, determinantal ideals, and quotient dimensions recover the complete rank histogram. Unit-ideal transcripts for the $(h+1) \times (h+1)$ minors give the stronger minimum-rank certificate. Direct enumeration is also exact and is already dominated by the Level-III and Level-V compact solves. The random-slice identity in Proposition 24 gives another route on the stable kernel. For the full-space $\ell = 2$ rows, the same projective construction gives Θ_{pol} and Ψ_{pol} exactly; the displayed minimum-rank bounds remain stronger sufficient certificates.

For the channel route, a fixed-key record need only add the exact 10×10 channel transform, the selected whole-channel core, one nonzero leading-Jacobian witness, and the output of a finite-field symbolic-homotopy implementation followed by all public filters. This would directly measure κ_{hom} and resident memory; it would not require any Macaulay selected-degree extrapolation.

For the fixed-core-free route, the corresponding record is even simpler logically: publish the ordinary or stable slice, the random within-family projection seeds for every pattern, the homotopy outputs, and the final public filter transcript. No leading-Jacobian witness is part of the premise; the successful pattern's nonsingularity is checked directly at the returned root.

For the streamed-XL solver, a full production run is sufficient but not logically necessary. Row-reduction certificates showing the border relations of Theorem 37 at the selected degree—or at a nearby degree still within the sensitivity budgets—would give a machine-checkable end-to-end root-extraction certificate. The natural production experiment is now sharply specified: use the streamed operator of Proposition 36, record the preconditioner seed, resident block size, black-box products, border relations, quotient dimension, and verifier-checked roots. For the three-plus-one route it should additionally record every partial root, the consistency and rank of (112), and the final diagonal checks. A complete circuit comparison would additionally need to price control flow, indexing, memory traffic, communication, and data movement.

9 Countermeasures

The design lesson is that security must be based on distinct quadratic polynomials, not on the number of ordered labels used to name them. A security analysis for any symmetric redesign should compute the actual quotient rank ρ_{quad} and use the unconditional cap

$$\rho_{\mathrm{quad}} \le K_d = \min\left\{ M, m_1 \binom{d+1}{2} \right\}, \qquad d = \dim_{\mathbb{F}_q} \mathbb{F}_q[S]. \tag{135}$$

Equality with K_d is a public rank condition, not a consequence of symmetry alone. Outer randomization does not restore the alternating directions removed before the outer map.

Proposition 49 (Necessary floor for restoring the ordered-label cap)**.** *Let an existing ordered-label analysis intend the quadratic-image cap*

$$K_{\mathrm{ord}} = \min\{M, m_1 d^2\}.$$

Any redesign that retains symmetric public forms and uses m_1' base-form groups and M' verifier outputs can restore even this cap only if

$$M' \geq K_{\mathrm{ord}}, \qquad m_1' \geq \left\lceil \frac{K_{\mathrm{ord}}}{\binom{d+1}{2}} \right\rceil. \tag{136}$$

If it also retains Version 2.3's coupling

$$x' = o'r', \qquad M' = dx', \qquad m_1' = \left\lceil \frac{x'}{d} \right\rceil,$$

then necessarily

$$x' \geq \max\left\{ \left\lceil \frac{K_{\mathrm{ord}}}{d} \right\rceil, d\left(\left\lceil \frac{K_{\mathrm{ord}}}{\binom{d+1}{2}} \right\rceil - 1 \right) + 1 \right\}. \tag{137}$$

These are necessary feature-count conditions only.

Proof. The quadratic image of any symmetric redesign has dimension at most $\min\{M', m_1'\binom{d+1}{2}\}$. Requiring this upper bound to reach K_{ord} forces both inequalities in (136). The least integer x' satisfying $\lceil x'/d \rceil \geq m_1'$ is $d(m_1' - 1) + 1$; combining this with $dx' \geq K_{\mathrm{ord}}$ gives (137). □

For every published row, $K_{\mathrm{ord}} = M$. Applying Proposition 49 gives the following unconditional minimum changes for a redesign that chooses to restore that cap while retaining the current coupling. A secure parameter set may require still larger changes, and the table does not prescribe how $x' = o'r'$ should be split.

Table 20: Necessary feature-count floor for restoring the ordered-label cap under $m_1 = \lceil or/\ell \rceil$. This restores only the quadratic-image cap implicit in the ordered-label analysis; it is not a security estimate.

Level	parameters	$m_1 \to m_1'$	$or \to o'r'$	$M \to M'$	increase
I	$(28, 5, 4, 4)$	$5 \to 8$	$20 \to 29$	$80 \to 116$	45.00%
I	$(48, 16, 2, 2)$	$16 \to 22$	$32 \to 43$	$64 \to 86$	34.38%
I	$(28, 4, 4, 5)$	$5 \to 8$	$20 \to 29$	$80 \to 116$	45.00%
III	$(40, 7, 4, 4)$	$7 \to 12$	$28 \to 45$	$112 \to 180$	60.71%
III	$(72, 24, 2, 2)$	$24 \to 32$	$48 \to 63$	$96 \to 126$	31.25%
III	$(38, 5, 4, 5)$	$7 \to 10$	$25 \to 37$	$100 \to 148$	48.00%
V	$(50, 9, 4, 4)$	$9 \to 15$	$36 \to 57$	$144 \to 228$	58.33%
V	$(96, 32, 2, 2)$	$32 \to 43$	$64 \to 85$	$128 \to 170$	32.81%
V	$(52, 6, 4, 6)$	$9 \to 15$	$36 \to 57$	$144 \to 228$	58.33%

Proposition 50 (Necessary output floor for removing the full fresh slice)**.** *Consider an $\ell = 4$ symmetric redesign that retains the reserved public vinegar line and the same public vinegar count v. If*

$$M \leq \ell(v - 1), \tag{138}$$

then every full-rank affine reduction satisfies $\dim_{\mathbb{F}_q}(V' \cap \ker C_{R,\mathbf{v}}) \geq K$ *and therefore retains the* K*-dimensional ordinary square-slice route of Corollary 29. Eliminating this route by dimensions alone requires*

$$M > \ell(v-1). \tag{139}$$

If M *remains a multiple of* ℓ*, then* $M \geq \ell v$*.*

Proof. By (77),

$$\dim D_0 \geq \ell(v-1) - (M-K) = K + \ell(v-1) - M.$$

Condition (138) therefore implies $\dim D_0 \geq K$. Negating that dimension guarantee gives (139); divisibility by ℓ rounds the strict inequality to $M \geq \ell v$. □

For the six public $\ell = 4$ rows, removing the route by this dimension test alone forces the following minimum output increases. These are necessary conditions only: satisfying the table does not prove security, and the actual intersection D_0 may exceed its lower bound.

Table 21: Necessary output increase to eliminate the guaranteed K-dimensional fresh ordinary slice by dimensions alone, while retaining v and $M \equiv 0 \pmod{\ell}$.

Level	parameters	current M	necessary M'	increase
I	$(28,5,4,4)$	80	112	40.00%
I	$(28,4,4,5)$	80	112	40.00%
III	$(40,7,4,4)$	112	160	42.86%
III	$(38,5,4,5)$	100	152	52.00%
V	$(50,9,4,4)$	144	200	38.89%
V	$(52,6,4,6)$	144	208	44.44%

Changing q or rejecting symmetric signature blocks does not change (138). A redesign can instead alter the affine geometry, remove the reserved-line separation, reduce the public vinegar space, or increase outputs beyond this floor; each such change must be reanalyzed rather than assumed secure.

This table does not claim that the team's $q = 19$ choice was intended to preserve K_{ord}: the 2025 presentation explicitly switched to the $o\binom{\ell}{2}$ alternating-form criterion [27]. It answers a narrower but unavoidable countermeasure question. A response that retains symmetric public forms and the current coupling cannot recover even the old ordered-label image cap with smaller changes; increasing q or rejecting symmetric signatures does not recreate the missing alternating directions.

The direct repairs are to increase parameters after accounting for (135), remove public-matrix symmetry, or redesign the left and right power actions so that exchanging their labels changes the quadratic polynomial. The symmetric-block rejection rule is not sufficient: the fixed-skew $\ell = 2$ lift makes every block nonsymmetric by construction, while an injective local skew map for the square $\ell = 4$ rows leaves accepted density at least $(1 - 19^{-4})^n$ and adds under 0.0014 bits to the priced retry factor. Returning unchanged to the submitted $q = 16$ design would instead restore the known characteristic-two attack surface.

A revised security analysis should publish the quotient rank, affine or filter rank, stable-kernel rank when applicable, the polar or cross-polar condition used for target availability, and solver logs for the actual cost-driving system.

10 Conclusion

The proposed odd-characteristic Version 2.3 construction has an unconditional quadratic-image defect. Its ordered power-pair labels do not represent independent polynomials: symmetry identifies (ξ, η) with (η, ξ) before the outer map, so every public base-form group factors through $\mathrm{Sym}^2(A)$. For $\ell = 4$, sixteen labels become ten; for $\ell = 2$, four become three. Exact affine elimination turns this identity into complete public-key forgery reductions, and fixed-skew, filtered, or injective-skew lifts defeat the proposed symmetric-signature rejection rule with exact acceptance bounds.

The former global-rank objection is no longer a sampled premise. A reserved public vinegar line fixes the attack geometry from cross data disjoint from fresh internal public coefficients. The random-XOF theorem bounds the full aggregate spectrum, and its excess form separates the unavoidable full-rank baseline from genuine rank defects. The same spectrum bounds the full-Jacobian singular locus projectively. Consequently, an accepted nonsingular root occurs with the second-moment probability in (60); no singleton assumption and no independence among rejection, singularity, and the MQ map is required. A fixed public seed can instead be accompanied by its exact finite projective rank histogram.

This yields a selected-degree-free, fixed-core-free frontier. Frobenius transport collapses the 55 near-square family-count patterns to 16 orbits. The robust per-certified-key costs are $2^{138.324}$, $2^{179.758}$, and $2^{220.832}$ AXN operations. An exact structural-preflight theorem proves success probability above 0.9423222679 for every official $\ell = 4$ shape in the code-faithful random-XOF model; charging the full inverse probability gives $2^{138.410}$, $2^{179.844}$, and $2^{220.918}$. This branch has neither a selected-degree nor a preferred-core premise, although its direct dense output can be enormous. The independent original-equation eigenblock-core route has much smaller output ceilings and robust per-certified-key exponents 134.077, 196.246, and 247.432, conditioned on an exact public nonzero Jacobian certificate. Both homotopy routes are operation-count theorems, not measured wall-clock claims.

The streamed-XL route supplies the complementary low-state result. Its state-minimizing four-block profiles cost $2^{141.70}$, $2^{203.63}$, and $2^{269.35}$ AXN operations with packed eight-array envelopes 17.7 GiB, 2.21 GiB, and 1.01 GiB. The exact three-plus-one decomposition gives a robust Level-V point $2^{263.40}$ with 9.55 GiB. This branch still depends on production selected-degree certificate behavior; correctness itself needs only a complete finite commuting border quotient, not exact corank one.

The countermeasure burden is therefore not a small rejection tweak. Merely restoring the old ordered-label quadratic-image cap while retaining symmetry and Version 2.3's coupling forces 31.25%–60.71% more verifier outputs across the nine rows. On the six $\ell = 4$ rows, eliminating the guaranteed full-dimensional fresh ordinary slice by dimensions alone forces 38.89%–52.00% more outputs. These are unconditional necessary floors, not sufficient repaired parameters. Removing public-matrix symmetry, changing the left/right actions or affine geometry, or paying a substantial parameter increase are the direct responses; increasing q or rejecting symmetric signature blocks does not recreate the missing alternating features.

The remaining empirical questions are sharply isolated. A homotopy implementation would measure κ_{hom} and resident output state; a streamed border-basis transcript would settle the production XL degree. Those questions affect the best concrete implementation claim, but not the symmetry-quotient theorem, the exact forgery reductions, the rejection bypass, the selected-degree-free root algorithms, or the necessary parameter floors.

A Proofs Deferred from the Main Text

The main text states the probability and structural results at the point where an attacker uses them. This appendix collects the longer proofs in the same order.

A.1 Stable-lift kernel

Proof of Theorem 6. Choose an $\mathbb{F}_q$-basis of A. The spanning family in (16) has $m_1 d^2$ elements, proving the dimension bound. For $\zeta \in A$,

$$(w^\top \Sigma_\xi P_i \Sigma_\eta)\Sigma_\zeta = w^\top \Sigma_\xi P_i \Sigma_{\eta\zeta},$$

so $\mathcal{U}_w$ is closed under right multiplication by A.

A cross term produced by substituting $u_j = \Sigma_{R_j} u + \Sigma_{\Gamma_j} w$ has, when w is on the left, a row coefficient $w^\top \Sigma_\xi P_i \Sigma_\eta$. When w is on the right, transpose the scalar and use $S^\top = S$, $P_i^\top = P_i$, and commutativity of A; the coefficient has the same form with the two algebra labels reversed. Thus every row of $\mathcal{F}_{R,\mathbf{v}}$ lies in $\mathcal{U}_w$, and right-A stability puts every row of $\mathcal{O}^{\mathcal{F}}_{R,\mathbf{v}}$ there as well.

Because (17) contains the complete right orbit of every row of $\mathcal{F}_{R,\mathbf{v}}$, its row space is closed under right multiplication by every element of A. Since $A \cong \mathbb{F}_{q^d}$ is a field, this row space is an A-vector space. Hence $d \mid \rho_{\mathcal{F}}$ and

$$\rho_{\mathcal{F}} \le \dim_{\mathbb{F}_q} \mathcal{U}_w \le m_1 d^2.$$

The first block of (17) is $\mathcal{F}_{R,\mathbf{v}}$ itself, so $\mathcal{F}_{R,\mathbf{v}} \ker \mathcal{O}^{\mathcal{F}}_{R,\mathbf{v}} = 0$. For x in this kernel and $\zeta \in A$, every product $S^a\zeta$ is an $\mathbb{F}_q$-linear combination of $1, S, \ldots, S^{d-1}$; therefore every row $f\Sigma_{S^a}$ of the orbit matrix also kills $\Sigma_\zeta x$. Thus the kernel is A-linear. Rank–nullity over A gives

$$\dim_A H^{\mathcal{F}}_{R,\mathbf{v}} = n - \rho_{\mathcal{F}}/d \ge n - m_1 d,$$

as claimed. □

A.2 Stable affine slices

Proof of Corollary 8. The first two ranks in (20) give the full-lift case of Proposition 2; in particular, every target determines a nonempty affine space $a(t) + \ker C_{R,\mathbf{v}}$. The orbit matrix contains $C_{R,\mathbf{v}}$ as its first block, so $H_{R,\mathbf{v}} \subseteq \ker C_{R,\mathbf{v}}$. Hence $a(t) + y + L \subseteq a(t) + \ker C_{R,\mathbf{v}}$ whenever $y \in \ker C_{R,\mathbf{v}}$. Because $L \subseteq H_{R,\mathbf{v}}$, Theorem 6 gives $\mathcal{F}_{R,\mathbf{v}} L = 0$, and therefore $\mathcal{F}_{R,\mathbf{v}} u$ is constant on that slice.

By Lemma 7, the unordered power-pair features can be changed invertibly to eigenblock coordinates. Because Q_R has full column rank, row reduction of the complete output system isolates K rows whose homogeneous quadratic parts are an invertible linear image of z_{sym}. Applying the inverse image change to those rows expresses the quadratic parts in eigenblock coordinates. Full rank of the quadratic feature map restricted to L ensures that this restriction loses no eigenblock component. Translating a diagonal feature introduces only quadratic, linear, and constant terms from one block; translating an off-diagonal feature introduces terms from its two blocks. If h_a is a homogenizer for block a, multiplication by the unique powers of the h_a turns these affine polynomials into exact multidegrees $2\mathbf{e}_a$ and $\mathbf{e}_a + \mathbf{e}_b$.

Distinct unordered multidegrees have disjoint monomial supports. Each component is generated by the m_1 base forms and therefore has rank at most m_1. The total restricted rank is the sum of these component ranks. Since there are $\binom{\ell+1}{2}$ components and the total is $K = m_1\binom{\ell+1}{2}$, every component must have rank exactly m_1. □

A.3 Accepted targets for the $\ell = 2$ filter

Lemma 51 (Quadratic fiber bound). *Let q be odd and let $g : \mathbb{F}_q^N \to \mathbb{F}_q^K$ be quadratic. For each $b \in \mathbb{F}_q^K \setminus \{0\}$, let $r(b)$ be the polar rank of the scalar quadratic $b \cdot g$, put $r_* = \min_{b \neq 0} r(b)$, and set $\alpha = 1 - q^{-K}$. Then every target $z \in \mathbb{F}_q^K$ satisfies*

$$\Big| |g^{-1}(z)| - q^{N-K} \Big| \le \alpha q^{N - r_*/2}. \tag{140}$$

More generally, if $W \subseteq \mathbb{F}_q^N$ is an affine subspace of codimension c, then

$$|\{x \in W : g(x) = z\}| \le q^{N-c-K} + \alpha q^{N-r_*/2}. \tag{141}$$

Proof. Fix a nontrivial additive character ψ of $\mathbb{F}_q$. Fourier inversion gives

$$|g^{-1}(z)| = q^{-K} \sum_{b \in \mathbb{F}_q^K} \psi(-b \cdot z) \sum_{x \in \mathbb{F}_q^N} \psi(b \cdot g(x)).$$

The $b = 0$ term is q^{N-K}. For $b \neq 0$, let $r = r(b)$. An invertible linear change of variables splits the polar form into a nondegenerate r-dimensional part and a radical of dimension $N - r$. If the linear part of $b \cdot g$ is nonzero on the radical, summing over the radical gives zero. Otherwise the radical contributes q^{N-r}, while diagonalizing the nondegenerate quadratic part reduces its sum to a product of one-dimensional Gauss sums of absolute value $q^{1/2}$. In either case the inner sum has absolute value at most $q^{N-r/2}$ [15]. Summing the $q^K - 1$ nonzero Fourier coefficients proves (140).

Write $W = a + U$, where U has codimension c. Translation does not change a quadratic polynomial's polar form. Restricting a bilinear form to $U \times U$ lowers its rank by at most $2c$: after extending a basis of U to one of $\mathbb{F}_q^N$, deleting the last c rows and columns changes matrix rank by at most $2c$. Thus every nonzero scalar combination of $g|_W$ has polar rank at least $r_* - 2c$. If $r_* \ge 2c$, the same Fourier calculation on the $(N-c)$-dimensional affine space W bounds each nonzero character sum by

$$q^{(N-c)-(r_*-2c)/2} = q^{N-r_*/2}.$$

If $r_* < 2c$, the trivial bound q^{N-c} is already at most $q^{N-r_*/2}$. This proves (141) in both cases. □

Proof of Proposition 17. By Lemma 51, every target fiber has between $q^{N-K} - \alpha q^{N-r_*/2}$ and $q^{N-K} + \alpha q^{N-r_*/2}$ roots. A rejected root lies on the intersection of some t_{rej} block-symmetry hyperplanes. Each such intersection is an affine subspace of codimension t_{rej}, so (141) bounds the number of roots it contains by $q^{N-t_{\mathrm{rej}}-K} + \alpha q^{N-r_*/2}$. Subtracting the union bound over the B intersections from the full-fiber lower bound gives

$$q^{N-K}(1 - Bq^{-t_{\mathrm{rej}}}) - \alpha q^{N-r_*/2}(1 + B).$$

This is positive exactly under the sufficient inequality (38).

Every fiber has size at most $q^{N-K}(1 + \alpha q^{K-r_*/2})$. Since $|\mathcal{X}_{\mathrm{acc}}| = a_n q^N$, double-counting accepted assignments by their images proves (39). Finally, let $T_z = |g^{-1}(z)|$ and let R_z be the number of rejected roots. The accepted fraction is $1 - R_z/T_z$. Applying the union-bound upper estimate to R_z and the character-sum lower estimate to T_z gives (40). □

A.4 Root collisions and the rank-spectrum bound

Proof of Theorem 18. Average over the $q^{\dim V-h}$ cosets and q^K targets. Every point of V contributes once to the first moment, which gives (45). For an ordered pair of distinct points in one coset, write the second point as $x+y$ with $y \in L \setminus \{0\}$. The collision equation is

$$g(x+y)=g(x) \quad \iff \quad T_y(x)=-(g(y)-g(0)).$$

It has $\kappa(y)q^{\dim V-r(y)}$ solutions. Dividing the total number of ordered collisions by the number of target–coset pairs proves (46).

For the rank-spectrum identity, note that $q^{-r(y)}=q^{-K}|\ker T_y^*|$ and double-count pairs (b,y) satisfying $b\cdot T_y=0$. The term $b=0$ contributes q^h-1. For $b\neq 0$, the number of nonzero $y\in L$ annihilated by $b\cdot T_y$ is $q^{h-e_L(b)}-1$. Therefore

$$\bar{\eta}=q^{-K}\left((q^h-1)+\sum_{b\neq 0}(q^{h-e_L(b)}-1)\right)=\mu(1+\Theta_L(g))-1.$$

The inequality $\eta_0\le\bar{\eta}$ is immediate.

Using $\mathbb{E}[Z^2]=\mu(1+\eta_0)$ gives

$$\mathrm{Var}(Z)=\mu(1+\eta_0-\mu)\le\mu^2\Theta_L(g),$$

and the second-moment inequality proves (49). Finally, $e_*\ge h-\Delta$ implies $\Theta_L(g)\le(q^K-1)q^{-h+\Delta}<q^{\tau+\Delta}$. ☐

A.5 Unique-root trials and the planted distribution

Proof of Corollary 19. For every nonnegative integer z,

$$\mathbf{1}_{z=1}\ge z-z(z-1), \qquad \mathbf{1}_{z>0}\ge z-\binom{z}{2}, \qquad \mathbf{1}_{z\ge 2}\le\binom{z}{2}.$$

Apply these inequalities to Z and use (46) together with $\eta_0\le\bar{\eta}$. Dividing the upper bound on $\Pr[Z\ge 2]$ by (52) proves (53).

The planted law size-biases a target–coset pair by Z. Hence

$$\Pr_{\mathsf{P}}[Z\ge 2]=\frac{\mathbb{E}[Z\mathbf{1}_{Z\ge 2}]}{\mu}\le\frac{\mathbb{E}[Z(Z-1)]}{\mu}=\eta_0.$$

Conditioned on $Z=1$, both U and P are uniform on the same set of target–coset pairs. If two distributions have the same conditional law on an event, their total variation distance is at most the larger mass they assign to its complement. Here the complement is $\{Z\ge 2\}$. The same indicator inequality gives the sharper exact-parameter lower bound $\Pr[Z>0]\ge\mu(1-\eta_0/2)$. Combining it with $\Pr[Z\ge 2]\le\mathbb{E}[Z(Z-1)]/2=\mu\eta_0/2$ gives

$$\Pr_{\mathsf{U}}[Z\ge 2]\le\frac{\eta_0}{2-\eta_0}\le\eta_0 \qquad (\eta_0<1).$$

The displayed size-bias bound gives $\Pr_{\mathsf{P}}[Z\ge 2]\le\eta_0$. Both are at most $\bar{\eta}$, proving the total-variation claim. ☐

A.6 Full-space target availability

Proof of Corollary 32. Take $L = V$ in Theorem 18. Then $e_L(b)$ is the polar rank and $\Theta_L(g) \le (q^K - 1)q^{-r_*} = \alpha q^{K-r_*}$. For the last statement, (140) gives

$$|g^{-1}(z)| \ge q^{N-K} - \alpha q^{N-r_*/2} > 0$$

when $r_* \ge 2K$. □

B Dimension Formulas

For a Version 2.3 row,

$$n = v + o, \qquad m_1 = \left\lceil \frac{or}{\ell} \right\rceil, \qquad M = or\ell, \qquad N_0 = n\ell, \qquad N_{\mathrm{raw}} = m_1 \ell^2 r^2, \qquad K = m_1 \binom{\ell+1}{2}.$$

For a full structured lift, $\operatorname{rank} C_{R,\mathbf{v}} = M - K$ and

$$\dim \ker C_{R,\mathbf{v}} = N_0 - M + K.$$

If $\operatorname{rank} \mathcal{O}_{R,\mathbf{v}} = m_1 \ell^2$, then

$$\dim_{\mathbb{F}_q} H_{R,\mathbf{v}} = N_0 - m_1 \ell^2, \qquad \dim_A H_{R,\mathbf{v}} = n - m_1 \ell.$$

The selected special-XL slice has A-dimension s, base-field dimension $h = \ell s$, and retry exponent $\tau = K - h$.

Table 22: Structured-lift dimensions for the six Version 2.3 $\ell = 4$ rows.

Parameters	M	K	$\dim_A H$	s	τ
$(28, 5, 4, 4)$	80	50	13	10	10
$(28, 4, 4, 5)$	80	50	12	10	10
$(40, 7, 4, 4)$	112	70	19	15	10
$(38, 5, 4, 5)$	100	70	15	15	10
$(50, 9, 4, 4)$	144	90	23	19	14
$(52, 6, 4, 6)$	144	90	22	19	14

C Cost-Search Certificate

For each $m_1 \in \{5, 7, 9\}$ and every admissible integer s, the artifact expands (98) exactly. Here admissible means $1 \le s \le \min\{\dim_A H, \lfloor K/4 \rfloor\}$. The $K/4$ bounds are $12, 17, 22$. They bind or tie for both $m_1 = 5$ rows and both $m_1 = 9$ rows. For $m_1 = 7$, the bounds are 17 for $(40, 7, 4, 4)$ and 15 for $(38, 5, 4, 5)$, the latter coming from $\dim_A H$. The artifact searches to the larger group bound and then filters the certificate separately for each parameter set; both $m_1 = 7$ rows attain the same optimum at $s = 15$. The monomial count $M_s(\mathbf{d})$ is coordinatewise increasing, so any multidegree that could match or beat the incumbent lies in a finite rectangular box. By symmetry it suffices to enumerate sorted multidegrees. The search checks 3,690, 11,217, and 27,449 candidates for $m_1 = 5, 7, 9$, respectively.

The optima, unique up to block permutation, are those in Table 10. No zero signed coefficient occurs anywhere in these bounded searches, so using a strictly negative rather than a non-positive trigger does not change the result. The optimal coefficients are

$$-14{,}677{,}784, \qquad -124{,}983{,}335{,}190, \qquad -1{,}164{,}044{,}975{,}088{,}360.$$

The corresponding Macaulay column counts are

$$23{,}417{,}049{,}656, \quad 44{,}237{,}557{,}981{,}725{,}696, \quad 236{,}094{,}291{,}515{,}544{,}000{,}000.$$

Using the sharper full-rank retry bound would give published-model exponents 128.81713, 171.68562, and 214.12301. The strengthened headline instead uses Retry_τ^+, adding $\log_2(19/18)$ and giving 128.89513, 171.76362, and 214.20101. Under the validated signed AXN circuit, the corresponding headline exponents are 132.24987, 175.11835, and 217.55575. The NAND sensitivity gives 133.11831, 175.98679, and 218.42418.

Low-state certificate. A second search allows every block degree $d_i \ge 1$ and minimizes $M_s(\mathbf{d})$ subject to a cost budget. Degree one is legitimate here: the Macaulay columns still contain the homogenizer and every degree-one variable in that block, while only diagonal equation multiples that exceed the target degree are omitted. For the published-model budgets 143/207/272, the exact global minima are

Level	(s, τ)	$\mathbf{d}$	$[\mathbf{t}^{\mathbf{d}}]H$	$M_s(\mathbf{d})$
I	$(8, 18)$	$(1, 2, 3, 3)$	$-70{,}875$	11,026,125
III	$(9, 34)$	$(1, 1, 2, 4)$	$-16{,}396$	3,932,500
V	$(11, 46)$	$(1, 2, 2, 3)$	$-31{,}536$	26,574,912.

Their published-model exponents are 140.12986, 205.46174, and 262.52898. Under the AXN budgets 143/207/272, the exact global minima are $(s, \tau, \mathbf{d}) = (9, 14, (2, 2, 3, 4))$ at Level I, $(10, 30, (1, 2, 3, 3))$ at Level III, and $(11, 46, (1, 2, 2, 3))$ at Level V. Their Macaulay column counts are 475,832,500, 59,383,896, and 26,574,912, and their AXN exponents are 137.69565, 199.96188, and 265.88371.

Here is the finite-tail proof of globality. For fixed s, $M_s(\mathbf{d})$ and each displayed cost are coordinatewise increasing. A sorted four-tuple with largest coordinate D has

$$M_s(\mathbf{d}) \ge \binom{s + D}{D} (s + 1)^3.$$

Thus, once the right-hand side exceeds the incumbent, every larger degree is irrelevant. At smaller s, even the degree-one floor lies above the relevant cost budget; at larger s, that floor or the first negative-Hilbert profile already exceeds the incumbent. Exact enumeration of every remaining finite box gives the displayed unique minima up to block permutation. The companion JSON records every per-s exclusion, every tested negative coefficient, and the exact row counts in (102). As a boundary check, the artifact also constructs 48 planted four-block systems across five negative-Hilbert profiles containing degree-one blocks; every exact Macaulay matrix has one-dimensional right kernel.

Three-plus-one certificate. For the series (113), the artifact performs an independent exact search over sorted triples $1 \le d_1 \le d_2 \le d_3$ and every admissible s. It charges (116), the exact square-row acceptance density, normal-preconditioner restarts over $\mathbb{F}_{19^8}$, and the extraction reserve. The compact time minima are

Level	(s,τ)	$\mathbf{d}$	$[\mathbf{t}^{\mathbf{d}}]H^{(3)}$	$M^{(3)}$	$R^{(3)}$
I	$(9,14)$	$(4,4,5)$	$-416{,}990$	1,023,472,450	2,958,813,000
III	$(11,26)$	$(3,3,6)$	$-1{,}321{,}334$	1,639,770,496	4,286,113,104
V	$(13,38)$	$(3,4,4)$	$-646{,}618$	3,172,064,000	7,332,242,400.

Their compact/robust AXN exponents are respectively 144.24760/148.16994, 197.02045/200.94294, and 250.31880/254.24141. The Level-V robust minimum-state row is $(s,\tau,\mathbf{d}) = (12, 42, (2, 3, 5))$, with coefficient $-416{,}615$, $M^{(3)} = 256{,}214{,}140$, $R^{(3)} = 559{,}191{,}087$, exponent 263.40011, and an eight-array state of 9.545 GiB.

For every $d_i \ge 1$, $R^{(3)}/M^{(3)} \ge 3m_1/(s+1)^2$, so any profile within a given work incumbent has an explicit finite cap on $M^{(3)}$. Sorted-degree monotonicity then gives $M^{(3)} \ge (s+1)^2\binom{s+d_3}{d_3}$ and closes every omitted tail. The companion script records the per-s caps and checks 507, 455, and 558 sorted degree triples for Levels I, III, and V; 8, 35, and 60 have negative Hilbert coefficient in the certified boxes.

D Same-Basis Ordered-Label Estimator Baseline

The comparison with the official Version 2.3 table mixes cost conventions, so we also recost a reproducible *estimator baseline* under exactly the same AXN arithmetic basis as the selected attacks. This baseline feeds the full ordered-label equation/variable dimensions from the earlier affine-column accounting into the pinned generic-MQ optimizer and replaces each charged base-field operation by $g_1^{\mathrm{AXN}} = 441$. An attacker can indeed feed the unreduced affine-column equations to a generic solver. What is counterfactual is the estimator's treatment of the ordered labels as though they represented independent generic quadratic features: on every symmetric key, reversed labels define the same polynomial by Theorem 3. Thus the table is a controlled historical estimator comparison, not evidence for a second quadratic space of the stated ordered-label rank. It reports every row rather than only the range quoted in the introduction.

Table 23: Historical ordered-label generic-MQ estimator and selected attack, both recosted in the same AXN arithmetic basis. “Estimator input” gives the equation/variable dimensions of the unreduced affine-column equations. The system is executable, but treating its reversed labels as independent generic quadratic features is counterfactual on a symmetric key.

Level	Parameters	Estimator input	Ordered-label AXN	Selected AXN	Saving
I	$(28, 5, 4, 4)$	80/132	206.69	132.25	74.44
I	$(48, 16, 2, 2)$	64/128	171.77	133.82	37.95
I	$(28, 4, 4, 5)$	80/128	206.69	132.25	74.44
III	$(40, 7, 4, 4)$	112/188	277.52	175.12	102.40
III	$(72, 24, 2, 2)$	96/192	242.15	187.82	54.33
III	$(38, 5, 4, 5)$	100/172	250.29	175.12	75.17
V	$(50, 9, 4, 4)$	144/236	345.49	217.56	127.93
V	$(96, 32, 2, 2)$	128/256	311.08	239.57	71.51
V	$(52, 6, 4, 6)$	144/232	347.13	217.56	129.58

The exact unrounded savings are 37.94724–129.57738 bits. This comparison still inherits the generic-MQ and special-XL solving models, but it removes the field-operation and gate-basis mismatch from the comparison itself.

E Fixed-Skew Certificate

For the official $\ell = 2$ matrix, direct arithmetic gives

$$9I + 10S = \begin{pmatrix} 0 & 1 \\ 1 & 7 \end{pmatrix}, \qquad e_0^{\mathsf{T}}(9I + 10S) = e_1^{\mathsf{T}}.$$

Together with (35), this proves (36) without a rank or genericity assumption. The remaining public preflight checks $\operatorname{rank} Q_R = K$ and $\operatorname{rank} C_{R,\mathbf{v}} = M - K$.

The resulting residual dimensions are 48/112, 72/168, and 96/224. The reoptimized generic-MQ convention gives per-solve exponents 129.44030, 183.43803, and 237.84310. The last optimum uses Hashimoto's $a = 1$ boundary and the direct $\mathrm{MQ}(19, 1, 1)$ solver described in the main text. Replacing the coarse factor b_1 by the validated signed AXN circuit g_1^{AXN} gives per-solve arithmetic exponents 133.74643, 187.74416, and 242.14922. By Corollary 33, the headline condition $\Theta_{\mathrm{pol}} \le 19^{-1}$ adds less than $\log_2(20/19)$ bits, giving Model/AXN exponents 129.51430/133.82043, 183.51203/187.81816, and 237.91710/242.22322. The minimum ranks $K + 1 = 49, 73, 97$ suffice for the aggregate condition. Positive AXN nominal arithmetic gap only requires ranks 46, 68, and 89; the all-target condition $r \ge 2K$ gives 96, 144, and 192. These are the values in Table 16 and Table 19.

F Artifact and Reproduction

The companion artifact, distributed separately from this manuscript package, contains a self-contained evidence directory with:

- the public-key generator, direct verifier, and official Level-I known-answer public key and signature;
- the quotient, affine-feature, structured-offset, stable-kernel, and rejection checks;
- all 121 structured preflight records, all 13 structured-slice records, and 60 fixed-skew preflights;

- 102,796 structured cross-polar ranks, 21,992 dual directional-derivative ranks, 3,888 fixed-skew polar ranks, and 29,424 zero-offset filter ranks, including the exceptional-direction campaigns;
- exhaustive $\ell = 2$ relation enumeration, both rejection certificates, accepted-target bounds, and generic-MQ costs;
- the complete special-XL and two-block cost searches, 128 exact $\ell = 4$ Macaulay experiments, 48 additional degree-one boundary experiments, 48 two-block controls, and the target-generation bound;
- 110 exhaustive toy checks of the exact collision and rank-spectrum formulas, including rank-defective and compatibility-defective cases, plus genericity diagnostics on six actual $\ell = 2$ headline cores;
- an independent rank implementation checked on 2,036 matrices, a projective-distinctness audit of all 124,788 structured rank-test directions, and independent scalar evaluators for the released gate circuits and headline cost conversions;
- the certified compact and joint-stress four-block time–state frontiers, including exact Hilbert coefficients, row counts, preconditioner bounds, monotone tail certificates, and the $\mathbb{F}_{19^{12}}$ Level-I extension ledger;
- the exact three-plus-one Hilbert search, streamed operator ledger, linear-completion theorem, and compact/robust time–state certificate underlying Table 13;
- the explicit streamed normal-operator ledgers and global state-minimality certificates underlying Tables 11 and 12, including the $16R(s+1)^2M$ and $32R(s+1)^2M$ work bounds and eight-array memory reserves;
- the exact random-XOF seeded-spectrum and excess-spectrum evaluations of Theorem 27 and Corollary 29, retaining the modulo-19 atom 14/256;
- the complete 55-pattern enumerator, exact reduction to 16 Frobenius orbits, multihomogeneous Bézout and companion homotopy sums, per-orbit extension/SLP ledgers, candidate filters, structural-preflight density certificate, and the normalized fixed-core-free frontier of Proposition 48;
- the complete-square homotopy ledger for $K = 50, 70, 90$, including field thresholds, root-success denominators, output ceilings, and break-even factors;
- the exact necessary repair-floor calculations of Propositions 49 and 50 and tables 20 and 21;
- three exact reduced-size Macaulay checks at the production-selected multidegrees, with full row and column counts and right-corank transcripts; and
- the exact projective-spectrum enumeration cost calculation of Proposition 31.

All finite-field ranks are computed exactly over $\mathbb{F}_{19}$. The companion artifact records hashes of the scripts, public keys, stable kernels, selected slices, and polar bases, together with a manifest for every file. None of the finite sampling in that artifact is presented as a proof of a production aggregate spectrum.

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